MILITARY SPECIFICATION

POWER SUPPLIES, SHIPBOARD, ELECTRONIC

28 VOLTS, KEY CODE CC3A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this associated detail specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-P-24764.

1. SCOPE

1.1 Scope. This document along with MIL-P-24764, establishes the requirements and test procedures for an electronic switching mode power supply with a programmable 28 volt output.

1.2 Classification. This specification covers family S1 power supplies in accordance with MIL-P-24764.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Crane Division, Code 603A, SHARP Project Office, Bldg 2917, Naval Surface Warfare Center, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-P-24764 - Power Supplies, Shipboard, Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

MIL-C-83421/01 - Capacitors, Fixed, Specification for

HANDBOOK

MILITARY

MIL-HDBK-241 - Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS Standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Power supplies shall comply with the requirements of MIL-P-24764, except as specified herein.

3.2 Correlation requirements. All testers and test equipment shall be correlated in accordance with MIL-P-24764.

3.3 Reliability requirements. All power supplies shall have a minimum mean time between failures (MTBF) of 100,000 hours when predicted in accordance with MIL-P-24764. The prediction shall be made with a mounting surface temperature of 50 degrees Celsius (°C).

3.4 Performance requirements. Power supplies shall conform to the electrical requirements specified in table I and shall operate within the environment specified in MIL-P-24764.

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TABLE I. Power supply electrical requirements.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C TO 60°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	IV	Isolation - input to chassis and output 800 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A2	IV	Isolation - output to chassis, 50 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A3	IV	Isolation - excessive temperature, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A4	IV	Isolation - unsafe temperature, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A5	IV	Isolation - output status, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A6	IV	Isolation - input connector used to unused contacts, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A7	IV	Isolation - output connector used to unused contacts, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A8	IV	Isolation - input to chassis and output, 800 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A9	IV	Isolation - output to chassis, 50 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A10	IV	Isolation - excessive temperature, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A11	IV	Isolation - unsafe temperature, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A12	IV	Isolation - output status, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A13	IV	Isolation - input connector used to unused contacts, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A14	IV	Isolation - output connector used to unused contacts, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm

-See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C TO 60°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
B1	V	Chassis contact to chassis continuity	X	0	1.0	0	1.0	0	1.0	0	1.0	Ohm
B2	V	28V return to remote ON/OFF return continuity	X	0	1.0	0	1.0	0	1.0	0	1.0	Ohm
B3, B4	V	28V return to programming return continuity	X	0	1.0	0	1.0	0	1.0	0	1.0	Ohm
B5-B40	V	Parallel connector contact continuity	X	0	1.0	0	1.0	0	1.0	0	1.0	Ohm
B41-B45	V	Chassis contact to chassis continuity	Q	0	0.1	0	0.1	0	0.1	0	0.1	Ohm
C1	VI	Load regulation, 28V, 2A, direct	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
C2	VI	Load regulation, 28V, 20A, direct	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
C3	VI	Load regulation, 30.5V, 2A	X	30.20	30.80	30.04	30.96	29.89	31.11	29.74	31.26	VDC
C4	VI	Load regulation,30.5V, 20A	X	30.20	30.80	30.04	30.96	29.89	31.11	29.74	31.26	VDC
C5	VI	Load regulation, 30.5V, 20A, direct	X	30.20	30.80	30.04	30.96	29.89	31.11	29.74	31.26	VDC
C6	VI	Load regulation, 28V, no load	X	27.44	28.56	27.30	28.70	27.16	28.84	27.02	28.98	VDC
C7	VI	Load regulation, 28V no load, output status	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC
C8	VI	Load regulation, 30.5V, no load	X	29.89	31.11	29.74	31.26	29.59	31.42	29.43	31.57	VDC
C9	VI	Load regulation, 30.5V, no load, output status	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C TO 80°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
D1	VII	Line regulation, 86 VRMS, 2A	S	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D2	VII	Line regulation, 86 VRMS, 20A	S	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D3	VII	Line regulation, 144 VRMS, 2A	S	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D4	VII	Line regulation, 144 VRMS, 20A	S	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D5	VII	Line regulation, 108 VDC, 2A	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D6	VII	Line regulation, 108 VDC, 20A	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D7	VII	Line regulation, 202 VDC, 2A	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D8	VII	Line regulation, 202 VDC, 20A	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D9	VII	Line regulation, auxiliary input 108 VDC, 20A	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D10	VII	Line regulation, auxiliary input 202 VDC, 20A	X	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D11	VII	Line regulation, 20A 47 Hz, 115 VRMS	Q	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
D12	VII	Line regulation, 20A 63 Hz, 115 VRMS	Q	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
E	VIII	Low input voltage protection	Q	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
F1-F3	VIII	Loss of phase protection	Q	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
G1	IX	Efficiency, 20A	X	75	100	75	100	75	100	75	100	%

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C TO 60°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G2	IX	Efficiency, 2A	Q	60	100	60	100	60	100	60	100	%
G3	IX	Efficiency, 28V, 20A direct	Q	78	100	78	100	78	100	78	100	%
G4	IX	Efficiency, 30.5V, 20A	Q	75	100	75	100	75	100	75	100	%
H	IX	Power factor	Q	0.8	1.0	0.8	1.0	0.8	1.0	0.8	1.0	-
J1-J3	IX	Input current balance	Q	-5	5	-5	5	-5	5	-5	5	%
K1, K2	X	Inrush current	Q	-	50	-	50	-	50	-	50	Apk
L1	XI	Overload protection	X	20	25	20	25.4	20	25	20	25.4	ADC
L2	XI	Short circuit recovery	Q	0	100	0	100	0	100	0	100	ms
L3	XI	Short circuit overshoot	Q	-1.40	1.40	-1.40	1.40	-1.40	1.40	-1.40	1.40	V
L4	XI	Short circuit current	S	0	25	0	25.4	0	25	0	25.4	ADC
M1	XII	Turn-on transient	Q	-1.40	1.40	-1.40	1.40	-1.40	1.40	-1.40	1.40	V
M2	XII	Turn-on time	S	1	250	1	250	1	250	1	250	ms
M3	XII	Turn-off transient	Q	-1.40	1.40	-1.40	1.40	-1.40	1.40	-1.40	1.40	V
M4	XII	Control turn-on transient	Q	-1.40	1.40	-1.40	1.40	-1.40	1.40	-1.40	1.40	V
M5	XII	Control turn-on time	S	1	250	1	250	1	250	1	250	ms
M6	XII	Control turn-off transient	Q	-1.40	1.40	-1.40	1.40	-1.40	1.40	-1.40	1.40	V

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C TO 80°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1, N2	XIII	Ripple, total, mVp-p (115 VAC, 10 Hz - 1 MHz) 20A	X	0	364	0	420	0	364	0	420	mVp-p
N3, N4	XIII	Ripple, low frequency (115 VAC, 10 Hz - 10 kHz) 20A	S	0	280	0	280	0	280	0	280	mVp-p
N5, N6	XIII	Ripple, high frequency (115 VAC, 10 kHz - 1 MHz) 20A	S	0	280	0	280	0	280	0	280	mVp-p
N7, N8	XIII	Ripple, total (115 VAC, 10 Hz - 20 MHz) 20A	Q	0	1680	0	1680	0	1680	0	1680	mVp-p
N9, N10	XIII	Ripple, total (115 VAC, overload, 10 Hz - 5 MHz) 20A	Q	0	1100	0	1100	0	1100	0	1100	mVp-p
N11, N12	XIII	Ripple, total (115 VAC, 10 Hz - 5 MHz) no load	Q	0	728	0	840	0	728	0	840	mVp-p
N13, N14	XIII	Ripple, total (86 VAC, 10 Hz - 5 MHz) 20A	Q	0	728	0	840	0	728	0	840	mVp-p
N15, N16	XIII	Ripple, total (144 VAC, 10 Hz - 5 MHz) 20A	Q	0	728	0	840	0	728	0	840	mVp-p
P1	XIV	Dynamic line regulation, +21%	Q	0	840	0	840	0	840	0	840	mV
P2	XIV	Dynamic line ringing, +21% (2nd half cycle)	Q	0	420	0	420	0	420	0	420	mV
P3	XIV	Dynamic line regulation, -21%	Q	0	840	0	840	0	840	0	840	mV
P4	XIV	Dynamic line ringing, -21% (2nd half cycle)	Q	0	420	0	420	0	420	0	420	mV

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q1	XV	Dynamic load regulation, 80% to 90%	X	0	840	0	840	0	840	0	840	mV
Q2	XV	Dynamic load ringing (2nd half cycle)	S	0	420	0	420	0	420	0	420	mV
Q3	XV	Dynamic load regulation, 90% to 80%	X	0	840	0	840	0	840	0	840	mV
Q4	XV	Dynamic load ringing (2nd half cycle)	S	0	420	0	420	0	420	0	420	mV
Q5	XV	Dynamic load regulation, 80% to 90% full capacitor load	Q	0	840	0	840	0	840	0	840	mV
Q6	XV	Dynamic load ringing (2nd half cycle)	Q	0	420	0	420	0	420	0	420	mV
Q7	XV	Dynamic load regulation, 90% to 80% Full capacitor load	Q	0	840	0	840	0	840	0	840	mV
Q8	XV	Dynamic load ringing (2nd half cycle)	Q	0	420	0	420	0	420	0	420	mV
Q9	XV	Dynamic load regulation, 5% to 95%	S	0	10	0	10	0	10	0	10	VDC
Q10	XV	Dynamic load ringing, 5% to 95%	S	0	5	0	5	0	5	0	5	VDC
Q11	XV	Dynamic load regulation, 95% to 5%	S	0	5	0	5	0	5	0	5	VDC
Q12	XV	Dynamic load ringing, 95% to 5%	S	0	5	0	5	0	5	0	5	VDC
R	XVI	Turn-on time into capacitive load	Q	1	250	1	250	1	250	1	250	ms
S1	XVII	Undervoltage - output status LED off	X	24.36	26.60	24.36	26.60	24.36	26.60	24.36	26.60	VDC
S2	XVII	Overvoltage - output status LED off 28V	X (4/)	29.68	31.72	29.68	32.00	29.68	31.72	29.68	32.00	VDC
S3	XVII	Overvoltage - output status LED off 28.5V	X (4/)	30.21	32.29	30.21	32.58	30.21	32.29	30.21	32.58	VDC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 80°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S4	XVII	Overvoltage - output status LED off 29.5V	X (4/)	31.27	33.42	31.27	33.72	31.27	33.42	31.27	33.72	VDC
S5	XVII	Overvoltage - output status LED off 30.5V	X (4/)	32.33	34.22	32.33	34.52	32.33	34.22	32.33	34.52	VDC
S6	XVII	Status LED check	X	----------			Status LED on	----------				
T1	XVII	Output status signal, logic low (5mA)	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC
T2	XVII	Output status signal, logic high	X	29.0	30.5	29.0	30.5	29.0	30.5	28.0	30.5	VDC
T3	XVII	Undervoltage- output status signal logic high	X	24.36	26.60	24.36	26.60	24.36	26.60	24.36	26.60	VDC
T4	XVII	Overvoltage- output status signal logic high - 28V	X (4/)	29.68	31.72	29.68	32.00	29.68	31.72	29.68	32.00	VDC
T5	XVII	Overvoltage- output status signal logic high - 28.5V	X (4/)	30.21	32.29	30.21	32.58	30.21	32.29	30.21	32.58	VDC
T6	XVII	Overvoltage- output status signal logic high - 29.5V	X (4/)	31.27	33.42	31.27	33.72	31.27	33.42	31.27	33.72	VDC
T7	XVII	Overvoltage- output status signal logic high - 30.5V	X (4/)	32.33	34.22	32.33	34.52	32.33	34.22	32.33	34.52	VDC
T8	XVII	Output status signal leakage current	Q	0	4.4	0	4.4	0	50	0	200	uADC
T9	XVII	Output status signal settling time (high to low)	Q	0	2.0	0	2.0	0	2.0	0	2.0	ms
T10	XVII	Output status signal settling time (low to high)	Q	0	2.0	0	2.0	0	2.0	0	2.0	ms
T11	XVII	Output status signal, logic low (.5mA)	Q	0	0.4	0	0.4	0	0.4	0	0.4	VDC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U1	XVIII	Excessive temp indication (5mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
U2	XVIII	Excessive temp indication (500mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
U3	XVIII	Unsafe temp indication (5mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
U4	XVIII	Unsafe temp indication (500mA current)	Q	-	-	-	-	0	0.4	-	-	VDC
U5	XVIII	Unsafe - excessive differential (5mA)	Q	-	-	-	-	1.0	25	-	-	C
U6	XVIII	Indicator reset (5mA)	Q	-	-	-	-	60	-	-	-	C
U7	XVIII	Unsafe - excessive differential (500mA)	Q	-	-	-	-	1.0	25	-	-	C
U8	XVIII	Indicate reset (500mA)	Q	-	-	-	-	60	-	-	-	C
V	XIX	Output voltage hold-up time, 108 VDC input	Q	50	-	50	-	50	-	50	-	us
W1	XX	AC input to chassis capacitance	S	0.0	0.03	0.0	0.03	0.0	0.03	0.0	0.03	uF
W2	XX	Auxiliary input to chassis capaciture	S	0.0	0.05	0.0	0.05	0.0	0.05	0.0	0.05	uF
X	XXI	Output impedance, 10Hz to 25 kHz 75% rated load	Q	0.0	300	0.0	300	0.0	300	0.0	300	mOhm
Y1	XXII	Spike voltage protection, AC input, initial transient	Q	-1.40	1.40	-1.40	1.40	-1.40	1.40	-1.40	1.40	V
Y2	XXII	Spike voltage protection, AC input, secondary transient	Q	-0.70	0.70	-0.70	0.70	-0.70	0.70	-0.70	0.70	V
Y3	XXII	Spike voltage protection, auxiliary input, initial transient	Q	-1.40	1.40	-1.40	1.40	-1.40	1.40	-1.40	1.40	V

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				Units 6/
				Initial		End of life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Y4	XXII	Spike voltage protection, auxiliary input, secondary transient	Q	-0.70	0.70	-0.70	0.70	-0.70	0.70	-0.70	0.70	V
			Q	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
Z	XXIII	Reverse voltage protection on auxiliary input	S	0	1.6	0	1.6	0	1.6	0	1.6	VDC
AA1	XXIII	Reverse voltage protection on output at 50 ADC	Q	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
AA2	XXIII	Reverse voltage protection on output at 200 Apk	S	0	20	0	20	0	20	0	20	mADC
BB1	XXIV	Forward voltage protection, on output at 300V	Q	27.72	28.28	27.58	28.42	27.44	28.56	27.30	28.70	VDC
BB2	XXIV	ON/OFF parallel outputs, OFF, direct	Q	2.4	5.5	2.4	5.5	2.4	5.5	2.4	5.5	VDC
BB3	XXIV	ON/OFF parallel outputs, OS high	Q		Output	status	LED	is	off.			--
BB4	XXIV	ON/OFF parallel outputs, OS LED off	S	28.91	30.09	28.76	30.24	28.61	30.39	28.47	30.53	VDC
BB5	XXIV	ON/ON Parallel outputs	S	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC
BB6	XXIV	ON/ON parallel outputs, OS on	S	27.44	28.56	27.30	28.70	27.16	28.84	27.02	28.98	VDC
BB7	XXIV	Positive pre-bias test	X	28.21	28.79	28.07	28.93	27.93	29.07	27.79	29.21	VDC
CC1	XXV	Output voltage, 28.5V, 2A	X	29.20	29.80	29.09	29.94	28.91	30.09	28.76	30.24	VDC
CC2	XXV	Output voltage, 29.5V, 2A	S	28.6	29.4	28.6	29.4	28.6	29.4	28.6	29.4	VDC
DD	XXVI	Open sense lead voltage direct output, 28V										

. See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
EE1	XXVII	Common mode current, broadband, 28V, 20A	Q	0	200	0	200	0	200	0	200	mAp-p
EE2	XXVII	Common mode current, broadband, 28V, 10A	Q	0	200	0	200	0	200	0	200	mAp-p
EE3	XXVII	Common mode current (50 Hz - 10 MHz)	Q	-INF	-60	-INF	-60	-INF	-60	-INF	-60	dBA \√Hz
EE4	XXVII	Common mode current (10 kHz - 10 kHz)	Q	-INF	-36.5	-INF	-36.5	-INF	-36.5	-INF	-36.5	dBA \√Hz
EE5	XXVII	Common mode current largest amplitude signal	Q	-INF	-26.0	-INF	-26.0	-INF	-26.0	-INF	-26.0	dBA \√Hz
FF1-FF3	XXVIII	Voltage programming contact signal rejection	Q	6	-	6	-	6	-	6	-	dBV
GG1	XXIX	Input voltage turn-on threshold	S	100	135	100	135	100	135	100	135	VDC
GG2	XXIX	Input voltage turn-off threshold	S	-	100	-	100	-	100	-	100	VDC
GG3	XXIX	Input voltage turn-on/turn-off hysteresis	S	18	-	18	-	18	-	18	-	VDC
GG4	XXIX	Input voltage turn-on threshold, auxiliary input	S	100	135	100	135	100	135	100	135	VDC
GG5	XXIX	Input voltage turn-off threshold, auxiliary input	S	-	100	-	100	-	100	-	100	VDC
GG6	XXIX	Input voltage turn-on/turn-off hysteresis, auxiliary input	S	18	-	18	-	18	-	18	-	VDC
GG7	XXIX	Maximum steady state input current	Q	-	13.5	-	13.5	-	13.5	-	13.5	ADC

- See footnotes at end of table.

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TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter 5/	Test type 1/ - 3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 80°C mounting surface temperature				Units 6/
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
GG8	XXIX	Slowly rising input peak current	Q	-	13.5	-	13.5	-	13.5	-	13.5	Apk
GG9	XXIX	Slowly rising input rate of current rise	Q	-	8.4	-	8.4	-	8.4	-	8.4	A/ms
GG10	XXIX	Input peak currents, remote turn-on	Q	-	13.5	-	13.5	-	13.5	-	13.5	Apk
GG11	XXIX	Slowly rising input rate of current rise, remote turn-on	Q	-	8.4	-	8.4	-	8.4	-	8.4	A/ms
HH1	XXX	Emergency condition input voltage protection	Q	28.98	32.03	29.74	31.26	29.74	31.26	29.74	31.26	VDC
HH2	XXX	Emergency condition input voltage protection	Q	30.20	30.80	30.04	30.96	29.89	31.11	29.74	31.26	VDC
JJ1	XXXI	Output status noise, positive	Q	0	500	0	500	0	500	0	500	mVpk
JJ2	XXXI	Output status noise, negative	Q	-500	0	-500	0	-500	0	-500	0	mVpk

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during power supply quality conformance testing.

2/ "S" parameters shall be sample tested (one percent or two power supplies from each inspection lot, whichever is greater) during quality conformance testing.

3/ "Q" parameters shall be tested during qualification testing. It is not required that these parameters be tested during quality conformance testing, however, the vendor is responsible for ensuring that the power supplies meet these parameters.

4/ Proper operation of the overvoltage protection circuitry shall be verified as an "X" test by the power supply vendor during manufacture of the supplies.

5/ Unless otherwise specified, the parameters apply to the 28V protected output configuration.

6/ The symbols used in this table and not defined elsewhere are as follows:

A - amperes
ADC - amperes direct current
Apk - amperes peak
cond - condition
INF - infinity
kOhm - kilOhm
mAp-p - milliamperes peak-to-peak
max - maximum
min - minimum
mOhm - milliohm
MOhm - megohm
mV - millivolts
mVpk - millivolts peak
mVp-p - millivolts peak-to-peak
% - percent
uADC - microamperes direct current
V - volts
VDC - volts direct current
VRMS - volts root-mean-square
dBA - decibel amperes
MHz - megahertz

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3.5 Power supply configuration. The configuration of the power supplies shall be in accordance with MIL-P-24764 and as specified herein.

3.5.1 Size. Power supplies shall be style 3.

3.5.2 Weight. Power supplies shall have a maximum weight of 14 pounds.

3.5.3 Visual indicator. The visual indicator shall be located as specified on figure 1. A green light emitting diode (LED) shall indicate status of the output voltage.

3.5.4 Marking. The power supply marking shall contain the information and shall be located in the relative location and orientation shown on figure 1. The marking "OUTPUT STATUS" shall be immediately adjacent to the output status LED. When authorization has been granted by the Shipboard Power Supply Products Quality Assurance Activity (SPSP-QAA), the qualification certification mark "QUAL" shall be marked as shown. The size of the letters, information per line, and spacing shall, within the limits of MIL-P-24764, be allowed to vary.

3.5.5 Keying. Keying shall be CC3A.

3.5.6 Mounting fastener torque. The power supply shall operate as specified when it is mounted to a cooling plate with the mounting fasteners torqued to 32 ± 3 inch-pounds.

3.6 Function diagram and contact assignments. The power supplies shall conform to the function diagram shown on figure 2 and contact assignment shall be as shown in table II. When multiple connector contacts are specified for an interface function (such as output) for current capacity, redundancy, and so forth, all multiple contacts for each function shall be connected together before being used to interface the power supply to external sources and loads. A summary of power supply functions is shown in table III.

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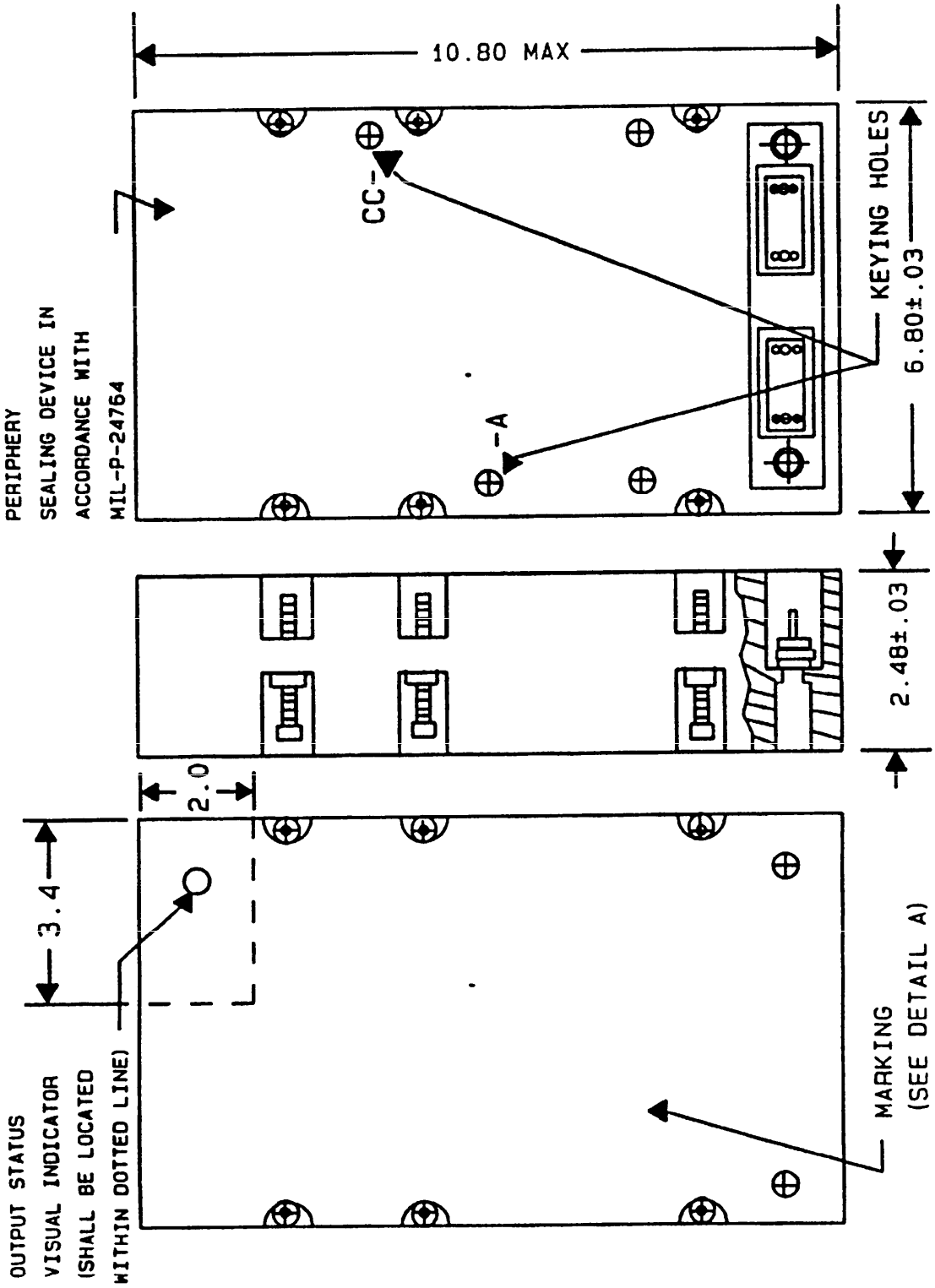
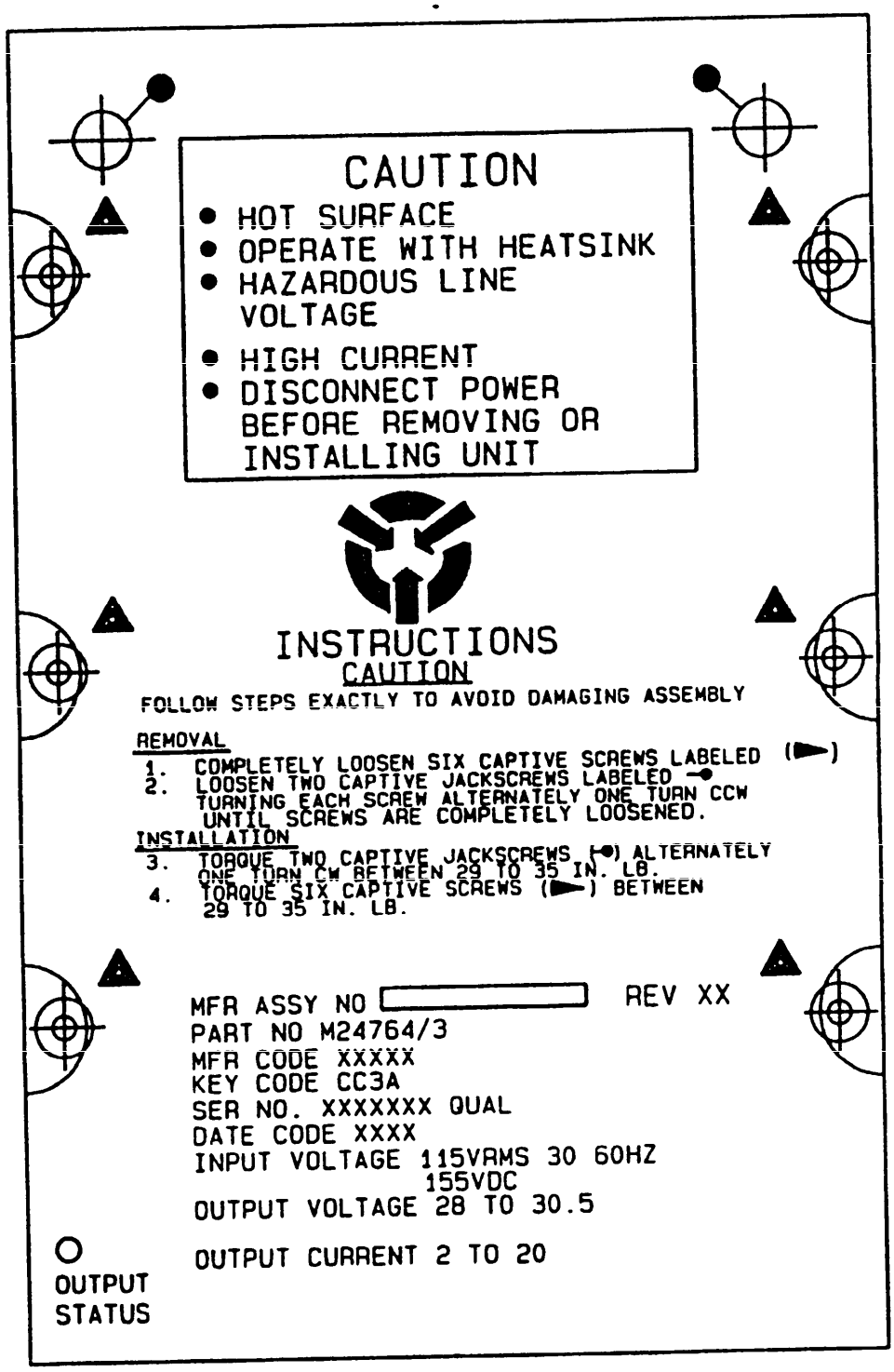


FIGURE 1. Power supply marking and visual indicator location.

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DETAIL A

FIGURE 1. Power supply marking and visual indicator location - Continued.

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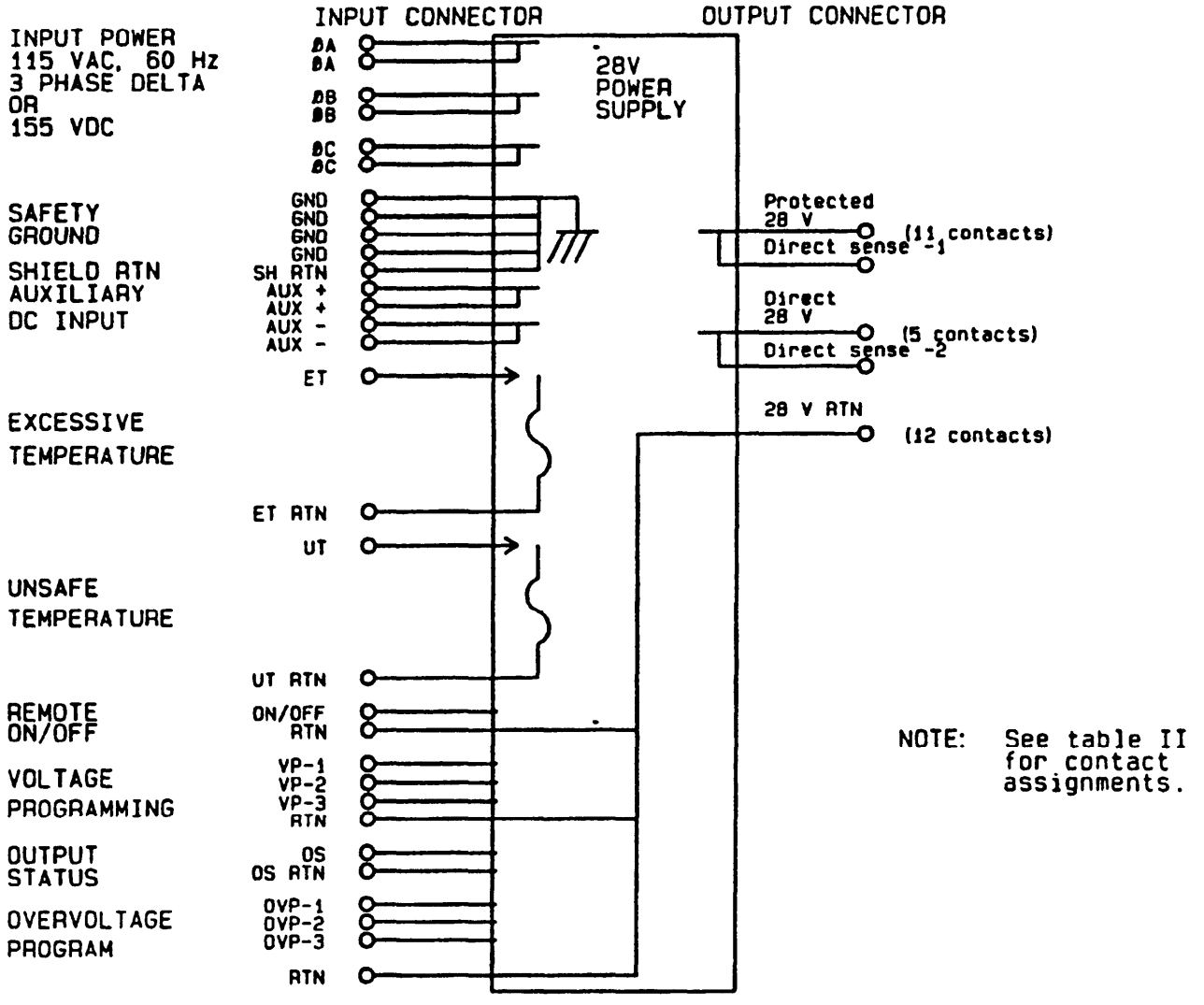


FIGURE 2. Power supply function diagram.

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TABLE II. Connector contact assignment.

INPUT CONNECTOR		OUTPUT CONNECTOR	
A	Phase A	A	Unused
B	Unused	B	Unused
C	Phase C	C	28 V Protected
D	Phase B	D	Unused
E	Phase A	E	28 V Protected
F	Unused	F	Unused
H	Phase C	H	28 V Protected
J	Phase B	J	Unused
K	Auxiliary DC (+)	K	28 V Protected
L	Unused	L	Unused
M	Auxiliary DC (-)	M	28 V Protected
N	Shield RTN	N	Unused
P	Auxiliary DC (+)	P	28 V Protected
R	Safety Ground	R	Unused
S	Auxiliary DC (-)	S	28 V Protected
T	Safety Ground	T	28 V RTN
U	Safety Ground	U	28 V Protected
V	Overvoltage Program -1	V	28 V RTN
W	Safety Ground	W	28 V Protected
X	Overvoltage Program -2	X	28 V RTN
Y	Excessive Temperature RTN	Y	28 V Protected
Z	Overvoltage Program -3	Z	28 V RTN
a	Excessive Temperature	a	28 V Direct Sense-1
b	Overvoltage Program RTN	b	28 V RTN
c	Unsafe Temperature RTN	c	28 V Protected
d	Voltage Program-1	d	28 V RTN
f	Unsafe Temperature	e	Unused
g	VP RTN	f	28 V RTN
h	Output Status RTN	h	Unused
i	Voltage Program -2	j	28 V RTN
j	Output Status	k	28 V Direct Sense-2
k	Remote ON/OFF RTN	m	28 V RTN
m	Remote ON/OFF	n	28 V Direct
n	Voltage Program -3	p	28 V RTN
		r	28 V Direct
		s	28 V RTN
		t	28 V Direct
		u	28 V RTN
		v	28 V Direct
		w	Unused
		x	28 V Direct
		y	Unused
		z	Unused

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TABLE II. Connector contact assignment - Continued.

INPUT CONNECTOR	OUTPUT CONNECTOR	
	AA	Unused
	BB	Unused
	CC	Unused
	DD	Unused
	EE	Unused
	FF	Unused
	HH	Unused

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TABLE III. Power supply functions.

Function	Logic "0" or closed circuit	Logic "1" or open circuit	Reference
Remote ON/OFF (Input contacts m & k)	Output ON	Output OFF	MIL-P-24764

Function	Direct output	Protected output	Reference
Output connection	Direct (tied to protected)	Protected contacts	3.10

Function	Logic "0"	Logic "1"	Reference
Output status signal (Input contacts j & h)	Voltage within rating and remote On/Off = On	Voltage out of tolerance or remote On/Off = Off	3.12

Function	On	Off	Reference
Output status LED	Voltage within rating and remote On/Off = On	Voltage out of tolerance or remote On/Off = Off	MIL-P-24764

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TABLE III. Power supply functions - Continued.

Function	Open 1/	Closure 1/	Reference
Excessive temperature (Input contacts a & Y)	Any Tj prior to an excessive closure	When Tj is from 1 to 25°C prior to the unsafe closure	MIL-P-24764
Unsafe temperature (Input contacts f & c)	Any Tj prior to an unsafe closure	When Tj exceeds the 80°C Tj by a minimum of 5°C.	MIL-P-24764

Output voltage programming (Input contacts d,i,n & g)	Output voltage	Overvoltage programming (Input contacts V,X,Z & b)	Overvoltage range
None	28.0 VDC	None	29.68 - 32.00 VDC
VP-1 to VP RTN	28.5 VDC	OVP-1 to RTN	30.21 - 32.58 VDC
VP-1, VP-2, to VP RTN	29.5 VDC	OVP-1, OVP-2, to RTN	31.27 - 33.72 VDC
VP-1, VP-2, VP-3 to VP RTN	30.5 VDC	OVP-1, OVP-2, OVP-3, to RTN	32.33 - 34.53 VDC

1/ The symbols used in this table and not defined elsewhere are as follows:

Tj - semiconductor junction temperature

3.7 Quality conformance tests. All production power supplies shall be tested in accordance with 4.2 and footnotes 1 and 2 in table I at a mounting surface temperature of 25°C ± 5°C.

3.8 Qualification. Initial qualification and periodic check shall be in accordance with MIL-P-24764 (see 4.5).

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3.9 End-of-life limits. Power supplies which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.3).

3.10 Output voltage polarity. The output of the power supply shall only be required to operate as a positive supply (28V RTN always connected to chassis or other reference point). However, the supply shall not be damaged by connecting the positive contact to chassis with the return floating.

3.11 Protected output. The power supply shall not be damaged when the output is connected indefinitely to a 143 VRMS, 60 Hz power source having a one Ohm source resistance or a 200 VRMS, 400 Hz, 1.5 kilovolt-amperes (kVA) power source. The power supply shall be capable of withstanding peak currents of 200 amperes and peak voltages of 300 volts. Semiconductor junction temperatures shall not exceed 125°C during the first 100 ms of the AC shorting condition and shall not exceed the manufacturer's maximum rated junction temperatures for the remainder of the shorted condition. Use of the protected output only requires connection of the load to the output terminals marked "28 V Protected." Figure 3 is a functional representation of the protection feature.

3.12 Direct output. Use of the direct output requires that the load be connected to the output terminals marked "28 V Direct." In addition, there must be a connection between the protected and direct outputs for proper output voltage sensing. Two connector contacts marked "Direct Sense-1" and "Direct Sense-2" have been designated for this purpose, however, any of the direct and protected terminals may be used in the interconnection.

3.13 Voltage programming. The power supply shall be programmable to any one of four discrete levels: 28 VDC, 28.5 VDC, 29.5 VDC, or 30.5 VDC. The programming shall be accomplished by connecting the appropriate combination of voltage programming contacts to the voltage programming return contact. The only guaranteed combinations of voltage programming contacts and the associated output voltages are tabulated in table III. No other combination of output voltage programming contacts shall result in nominal voltages less than 28 VDC or greater than 30.5 VDC.

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SUPPLY

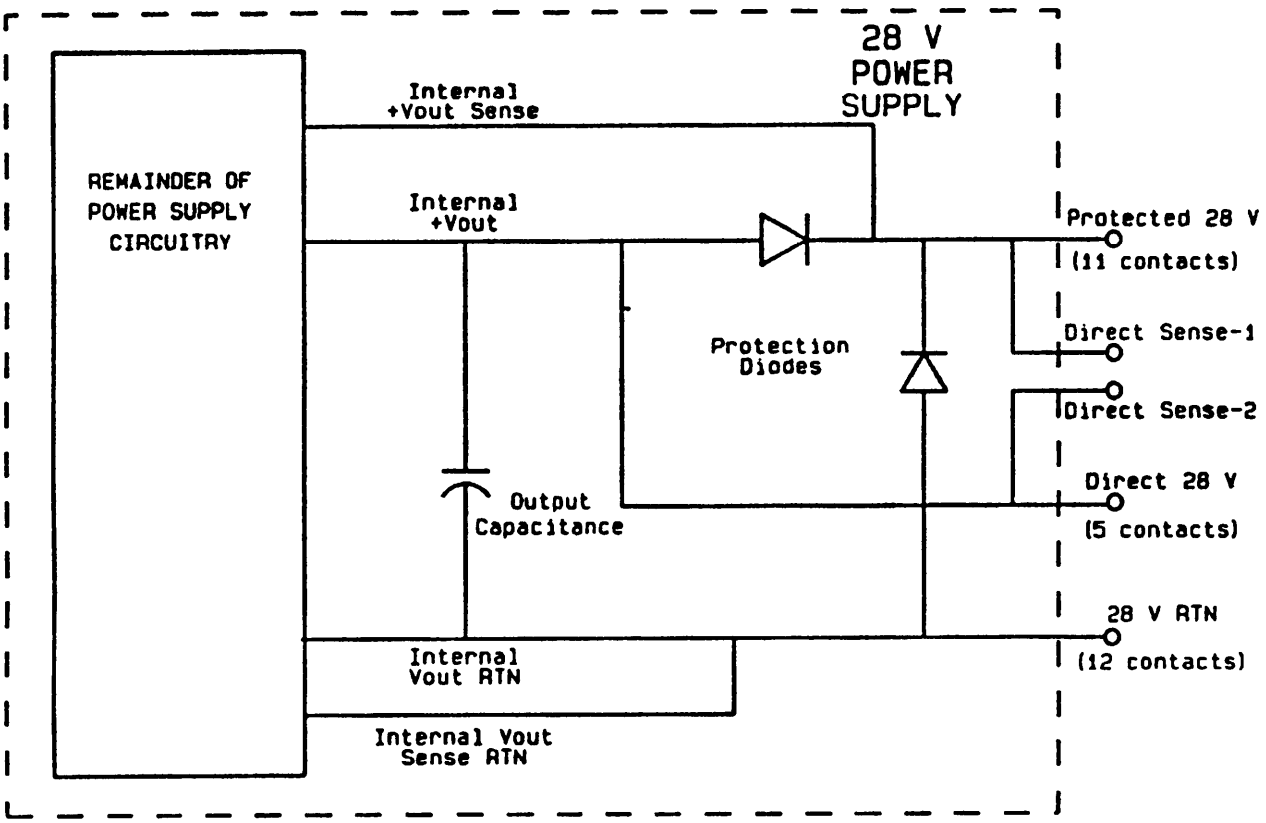


FIGURE 3. Functional diagram of outputs.

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3.14 Overvoltage programming. The power supply shall be programmable for overvoltage limits by connecting certain combinations of the overvoltage programming contacts to the overvoltage programming return contact. The only guaranteed combinations of overvoltage programming contacts and the associated overvoltage limits are tabulated in table III. No other combinations of overvoltage programming contacts shall result in overvoltage limits greater than 34.53 VDC. The overvoltage limits shall be programmable independent of the output voltage programming. Selection of nominal output voltages which exceed the selected overvoltage limits shall result in normal activation of the overvoltage protection circuitry without damage to the power supply. Selection of any of the overvoltage limits which exceed the selected nominal output voltage shall result in overvoltage protection within the selected limits.

3.15 Output status signal. The output status signal shall operate as defined in MIL-P-24764 for output voltage status. An output out of tolerance shall cause an appropriate indication on the status signal output. The output status signal shall be at a logic "1" whenever the remote ON/OFF equals OFF.

3.16 Parallel power supplies. The output of a CC3A power supply shall be capable of being connected in parallel with the output of one additional CC3A power supply under the conditions specified herein. When paralleling two power supplies, the same outputs (direct or protected) shall be used on each supply. Either direct or protected outputs are permissible for the ON/OFF configuration, but only the protected outputs shall be used for the ON/ON configuration. Two parallel configurations are possible.

(a) ON/OFF paralleling. One power supply shall be energized and the other unenergized. The remote ON/OFF control of the unenergized supply shall be programmed for the OFF condition as long as the other power supply is operating. In the ON/OFF parallel configuration, either power supply shall be voltage programmed to any permissible output voltage (see 3.13). Both power supplies shall be programmed for the same overvoltage limits. The overvoltage limits (see 3.14) shall be equal to or greater than 106 percent of the highest programmed output voltage. ORing diodes are not required.

(b) ON/ON paralleling. Both power supplies shall be energized at the same time. There shall be a minimum of 1.0 volts difference between the programmed output voltages (see 3.13) of the two power supplies. The overvoltage limit of the higher voltage output shall be programmed for any permissible level (see

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3.14) equal to or greater than 106 percent of the programmed output voltage. The overvoltage limit of the lower output voltage power supply shall be equal to or greater than the programmed limit of the higher output voltage power supply.

3.17 Output capacitive loading. When two CC3A power supplies are paralleled as in 3.16(a), the energized power supply shall provide specified output voltage and current into the maximum allowable capacitance of the external load and the output capacitance of the paralleled, unenergized power supply. The capacitance of the external load shall not exceed 30 microfarads (uF) per rated output ampere. The output capacitance of the paralleled power supply shall not exceed 30 uF per rated output ampere. Therefore, the energized power supply shall drive a maximum capacitance of 60 uF per rated output ampere.

3.18 Electromagnetic interference requirements. The power supply shall meet the following EMI requirements of MIL-STD-461, class A4 and class A5, with modifications or extensions to the limits where indicated (see 4.5.7):

(a) CE01 direct current (DC) power leads - see figure 4.

(b) CE01 alternating current (AC) power leads - see figure 5.

(c) CE03 narrowband, DC power leads - see figure 6.

(d) CE03 broadband, DC power leads - see figure 7.

(e) CE03 narrowband, AC power leads - see figure 8.

(f) CE03 broadband, AC power leads - see figure 9.

(g) CS01.

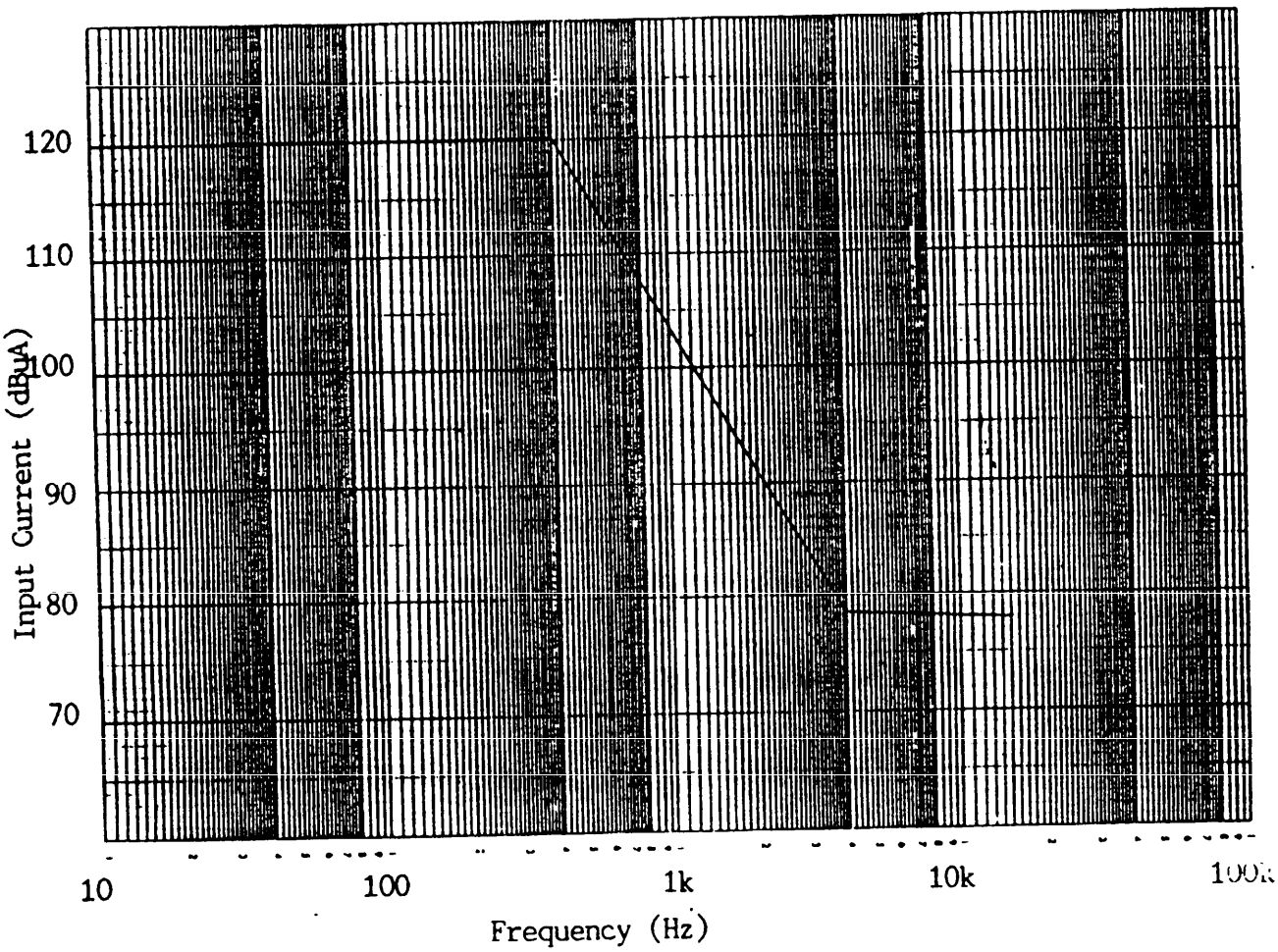
(h) CS02.

(i) RE01 - see figure 10.

(j) RE02 installed configuration, see 4.5.7.2(a). The maximum test frequency shall be 1 gigahertz (GHz).

(k) RE02 connector opening, see 4.5.7.2(b). The maximum test frequency shall be 1 GHz.

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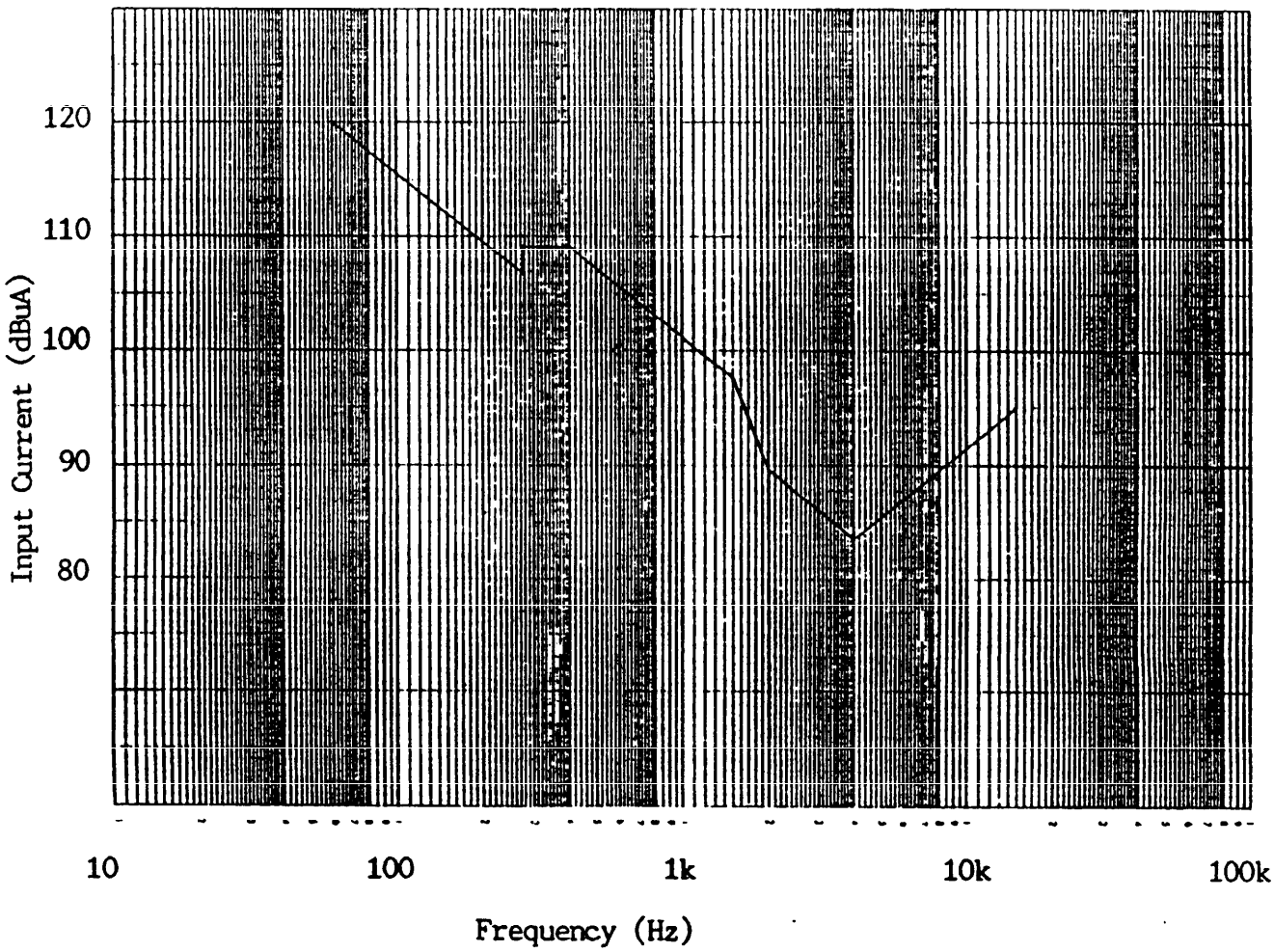
For power supplies with input current less than 1 ampere, the limits shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log (input current),

where input current is in amperes. The unit dBuA is decibels with respect to 1 microampere.

FIGURE 4. Powerline harmonic current limitation (CE01) on DC leads.

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NOTE:

For power supplies with input current less than 1 ampere, the limit shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log (input current),

where input current is in amperes. The unit dBuA is decibels with respect to 1 microampere.

FIGURE 5. Powerline harmonic current limitation (CE01) on AC leads.

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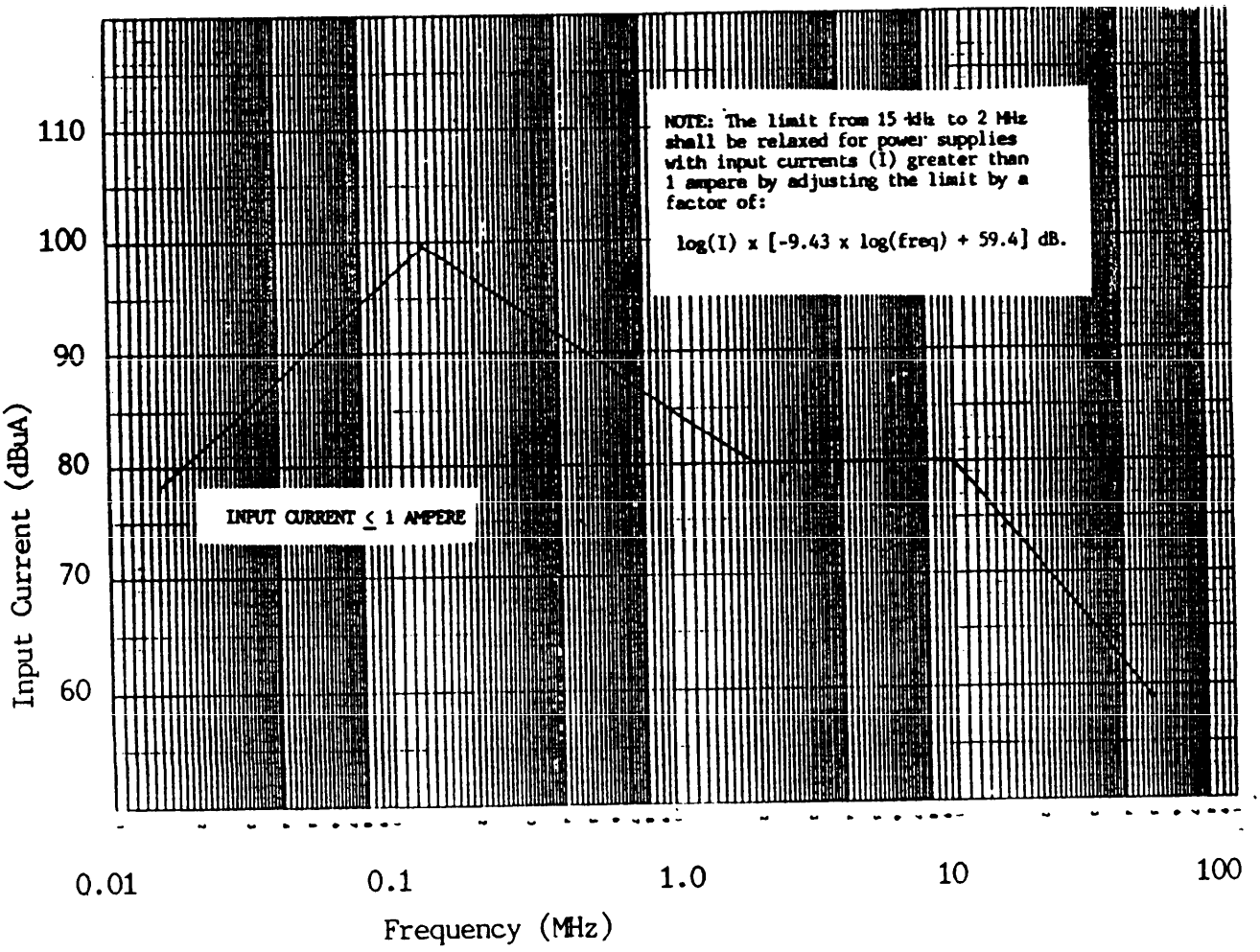


FIGURE 6. Narrowband conducted emissions on DC power leads (CE03).

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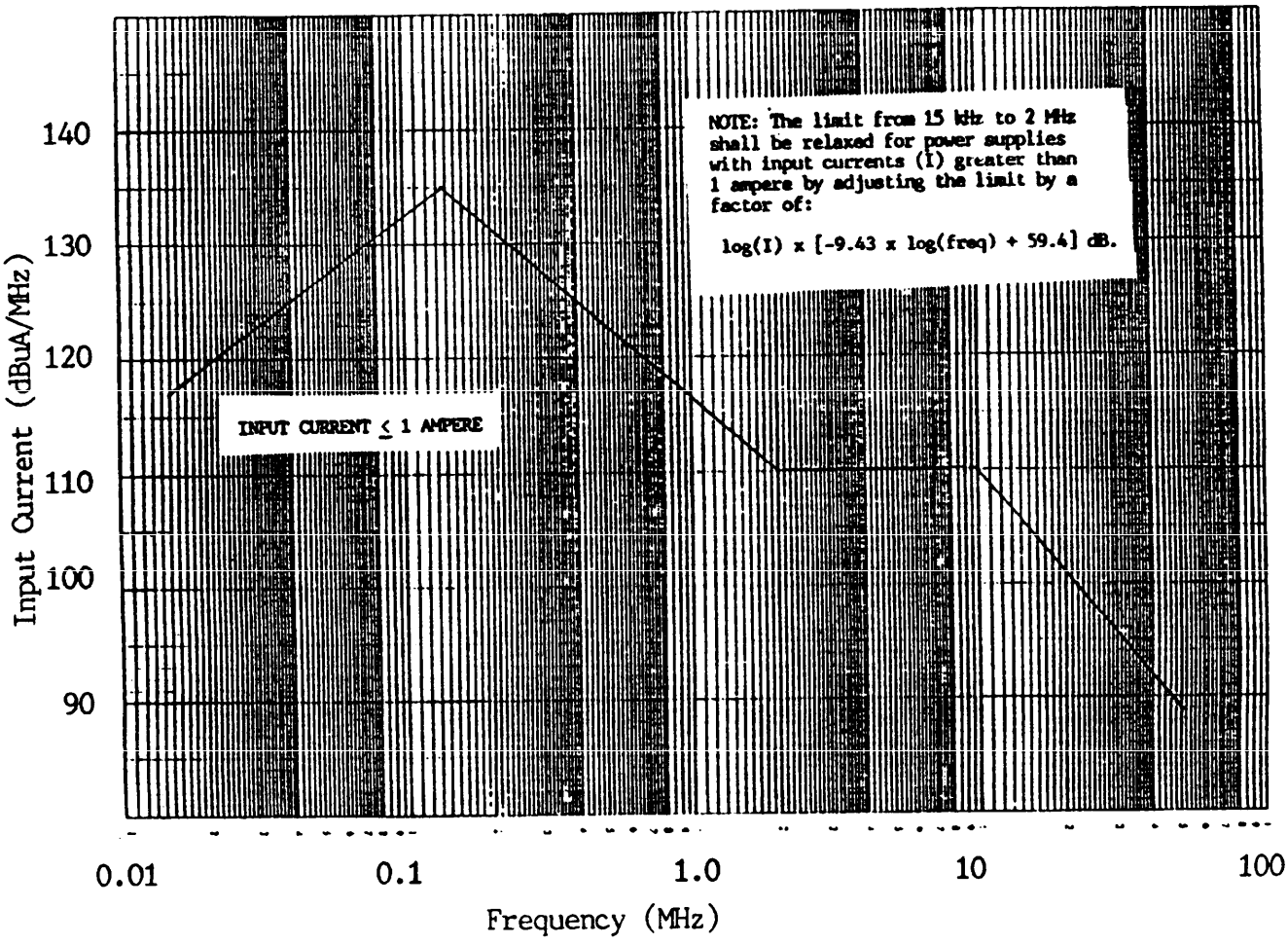


FIGURE 7. Broadband conducted emissions on DC power leads (CE03).

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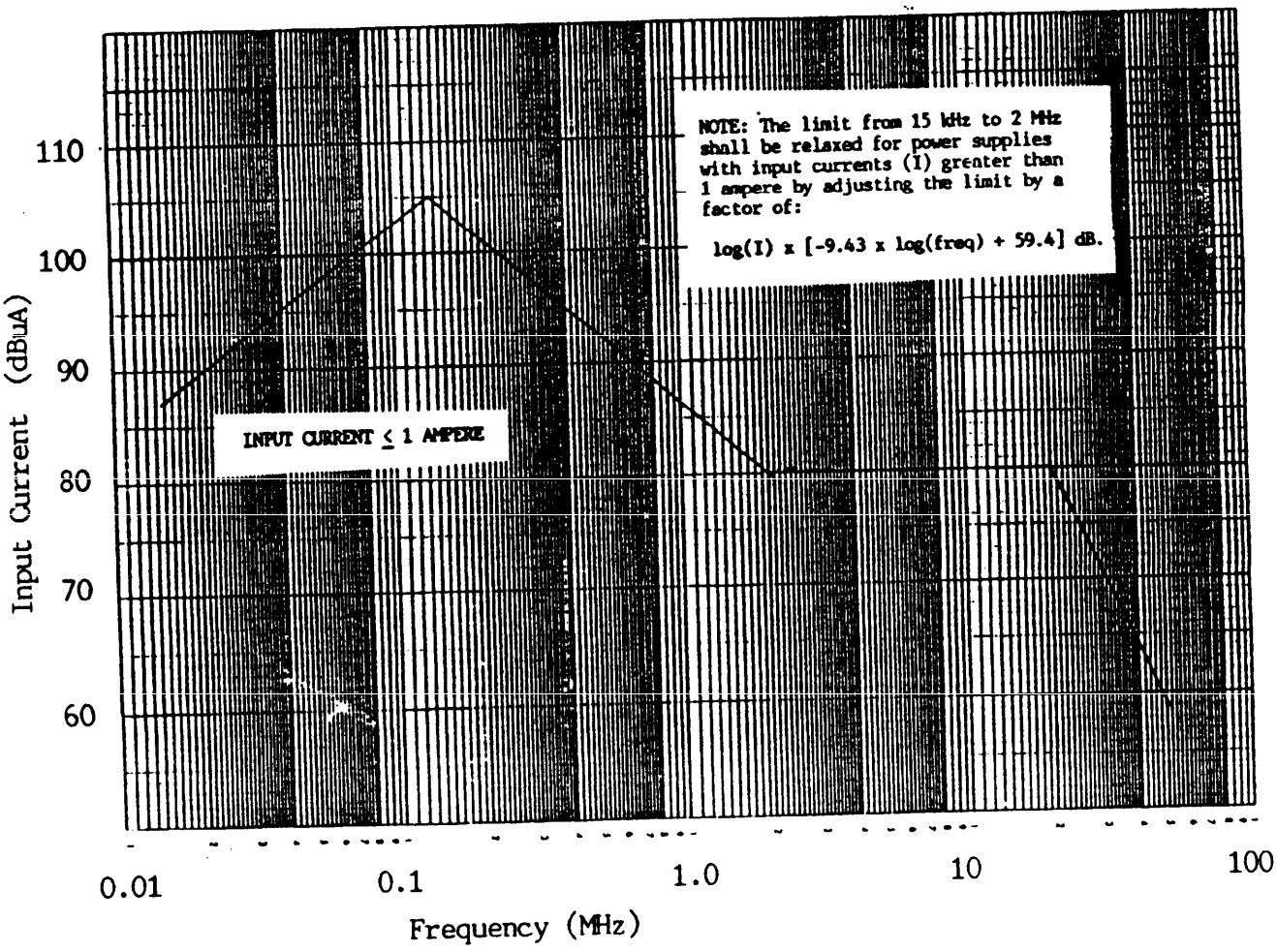


FIGURE 8. Narrowband conducted emissions on AC power leads (CE03).

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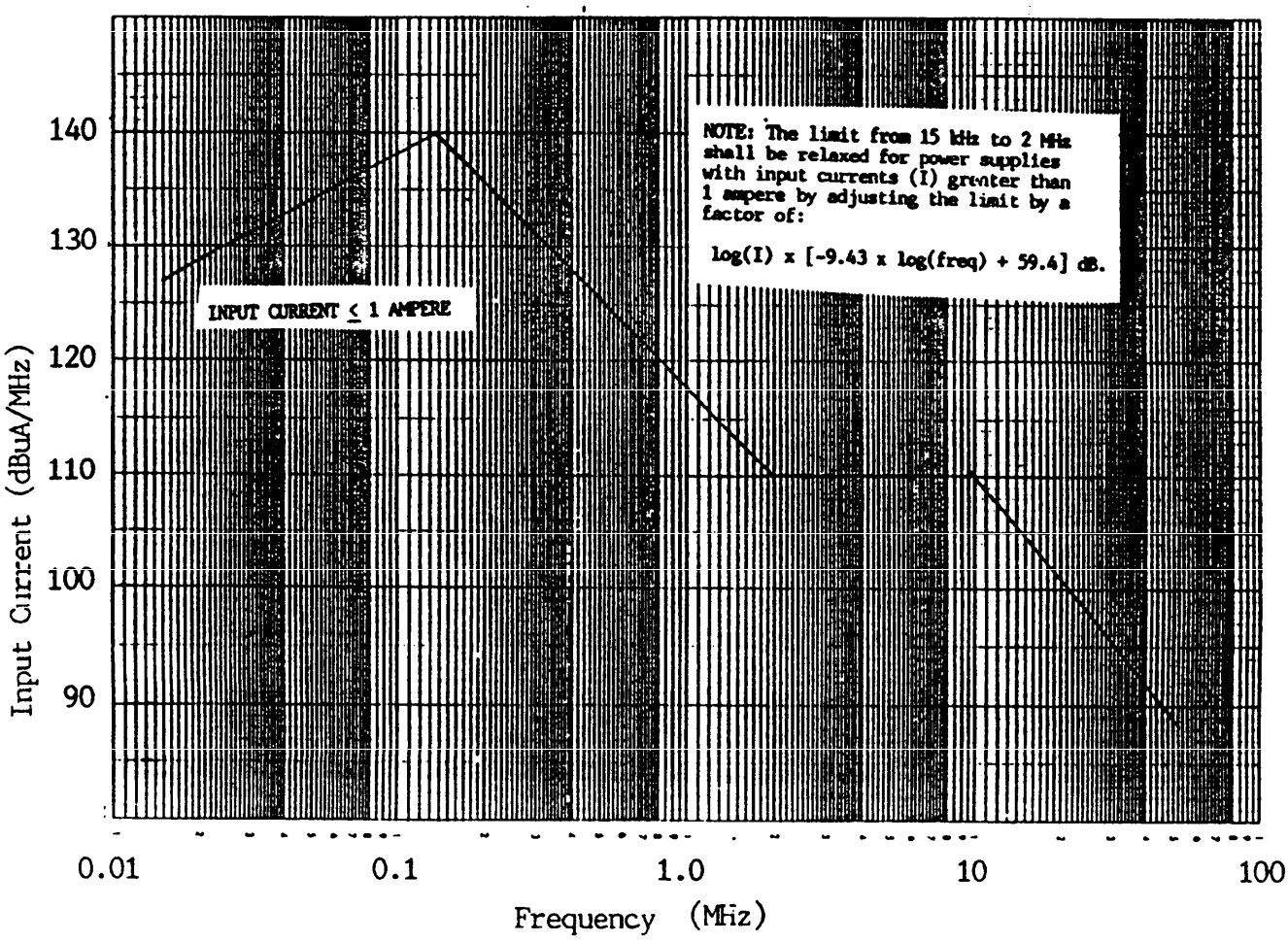


FIGURE 9. Broadband conducted emissions on AC power leads (CE03).

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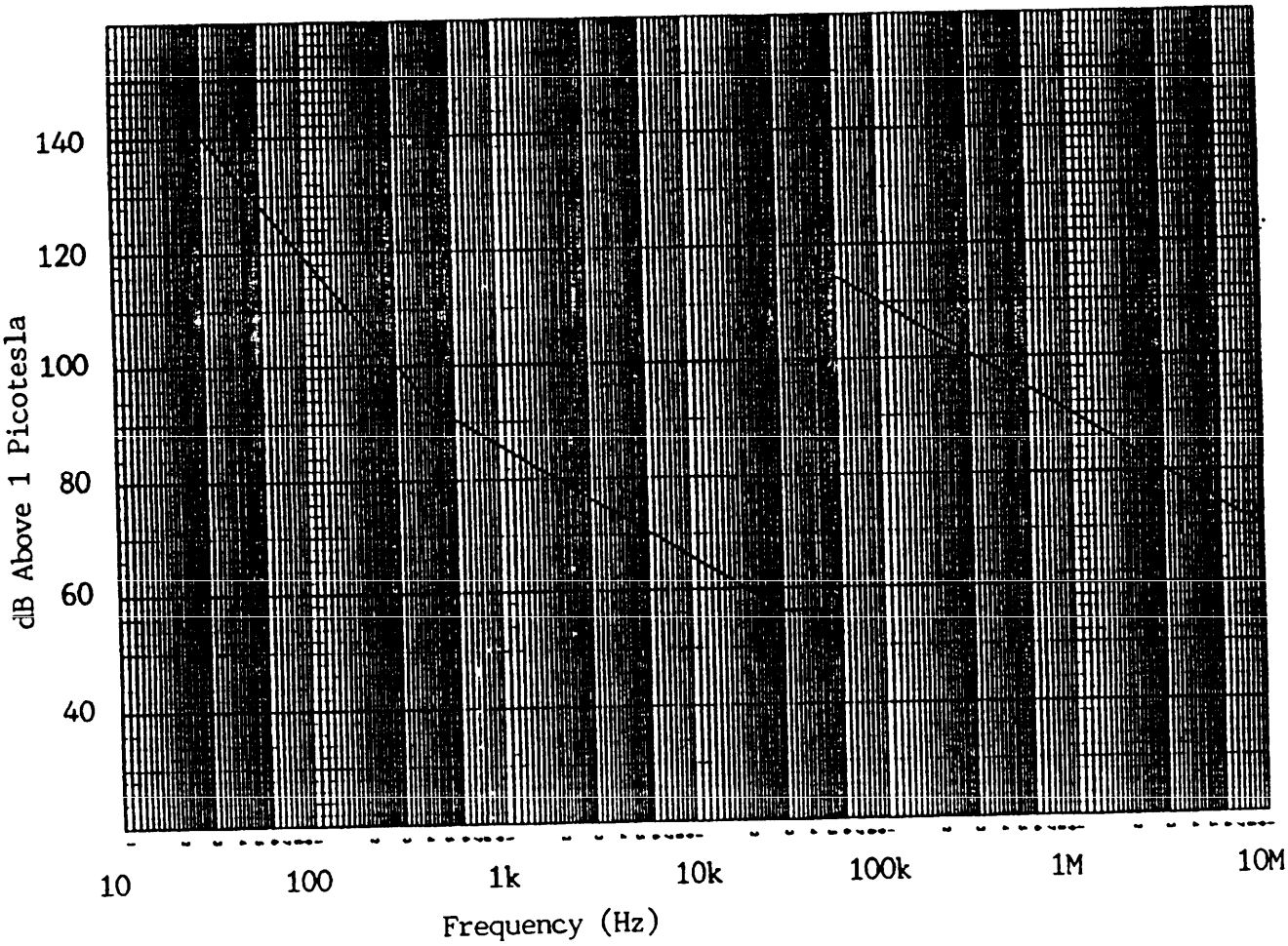


FIGURE 10. Magnetic field emissions (RE01).

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(l) RS01.

(m) RS02.

(n) RS03 class A4 for below deck metallic hull.

3.18.1 Criteria for susceptibility. When the power supply is being tested for susceptibility to a conducted or radiated signal, the output voltage shall be monitored for compliance to the table I output regulation limits.

3.19 EMI filter design. The power supply is not required to have an internal EMI filter. Therefore, for a power system to meet MIL-STD-461 conducted emissions requirement CE03, the system must have an EMI filter between the power source output and the power supply input. For one power supply operating at full load, the necessary EMI filter attenuation can be found by taking the difference between the allowed applicable power supply conducted emissions curves (see figures 4 through 9) and the system required conducted emissions. For reduced power supply loading and more than one power supply attached to an EMI filter, the determination of the required EMI filter attenuation becomes more complex. The power supply input impedance characteristics as specified in 3.19.1 are also needed in order to design an EMI filter. MIL-HDBK-241 should be referenced for precautions to be taken in the design of an EMI filter to prevent the filter/power supply combination from forming a negative impedance oscillator. The design of the power source system must ensure that damage cannot occur under the system's worst case minimum input voltage including brownout conditions. The power supply output voltage may continue within regulation as the input voltage drops to half of nominal 155 VDC (115 VRMS); however, the power source voltage may not be specified to go this low.

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3.19.1 Input impedance requirements. The power supply input impedance magnitude, at full load output, shall be greater than the figure 11 curve for the input voltages specified. The input impedance phase shall be more positive than the figure 12 curve for the input voltages specified. These curves are composites of the minimum input impedances from several different power supply designs. The actual model for each vendor's design may be obtained from the SPSP-QAA. During initial qualification testing, the SPSP-QAA will verify that each vendor's design meets the above impedance requirements.

3.19.2 Input impedance computer model. Each manufacturer of power supplies shall generate a computer model of the power supply input impedance (Zi) as observed from the input power contacts. The model shall diagram the input network and show both the AC and auxiliary inputs. It shall include elements simulating the negative impedance of the converter. Nominal component values together with their tolerances shall be given. To verify the accuracy of the model, each manufacturer shall measure the input impedance (magnitude and phase) of their power supply under the following conditions:

(a) From any two of the AC, phase A, B or C input contacts. Also from the auxiliary input contacts if the data taken would differ significantly from that taken at the AC input contacts.

(b) Over the frequency range of 100 Hertz (Hz) to 100 kilohertz (kHz) or up to the voltage feedback loop gain crossover frequency, whichever is greater.

(c) With input voltages of 108, 155 and 202 VDC. Also measurement at a minimum input voltage, just prior to the output voltage going out of regulation or turning off.

(d) At full load for all input voltages. For 108 VDC input voltage, also measure at 10 percent and 50 percent full load.

The Zi computer model shall be made to match the Zi measured data within ±1.0 decibel (dB) for gain and ± 6° for phase at frequencies up to 10 kHz. Above 10 kHz the required match shall be ± 2.0 dB for gain and ± 24° for phase. The measured data, the computer model, and curves obtained from the computer model shall be submitted to the SPSP-QAA with no restrictions on use by system manufacturers for stability analysis.

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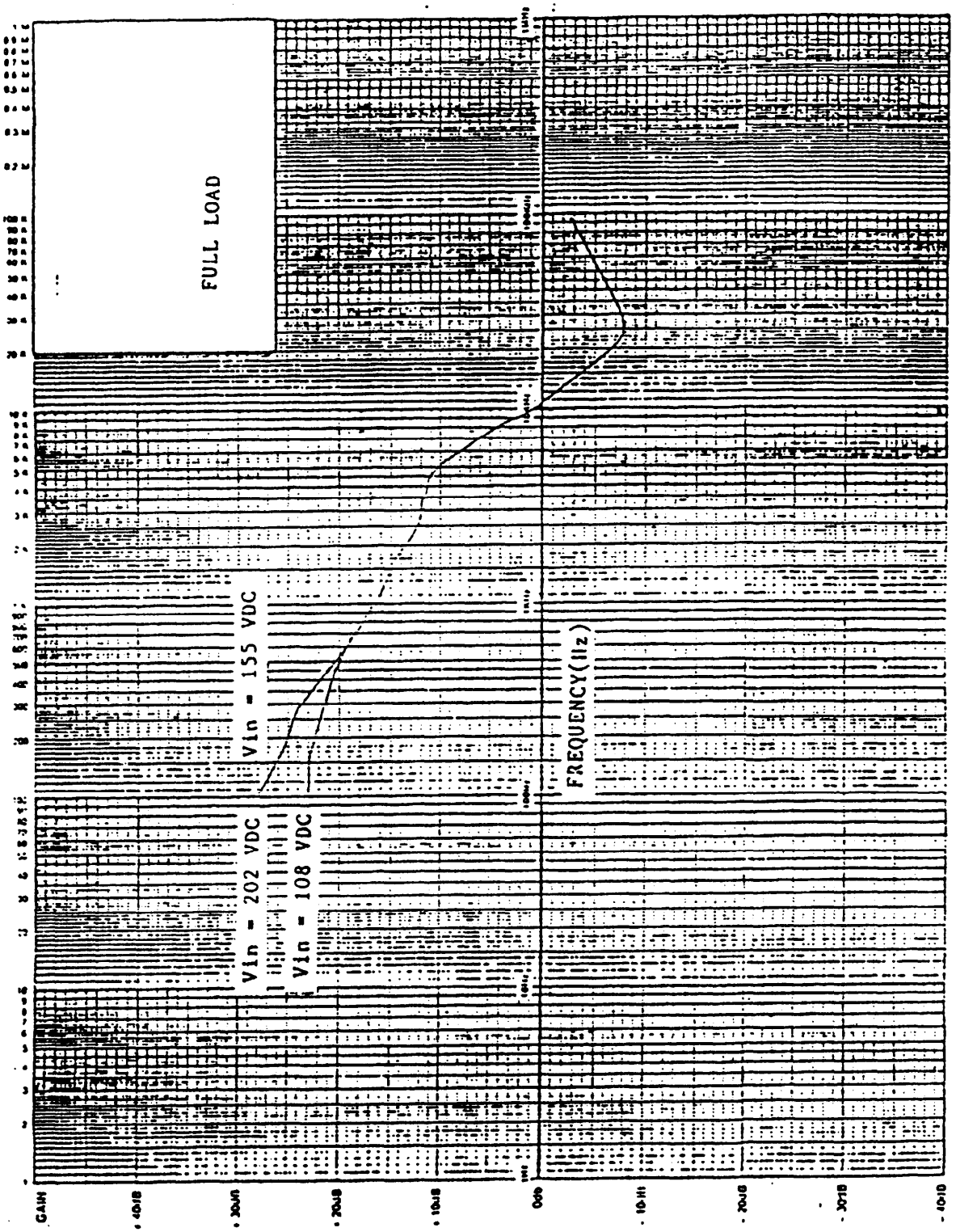


FIGURE 11. Minimum input impedance magnitude.

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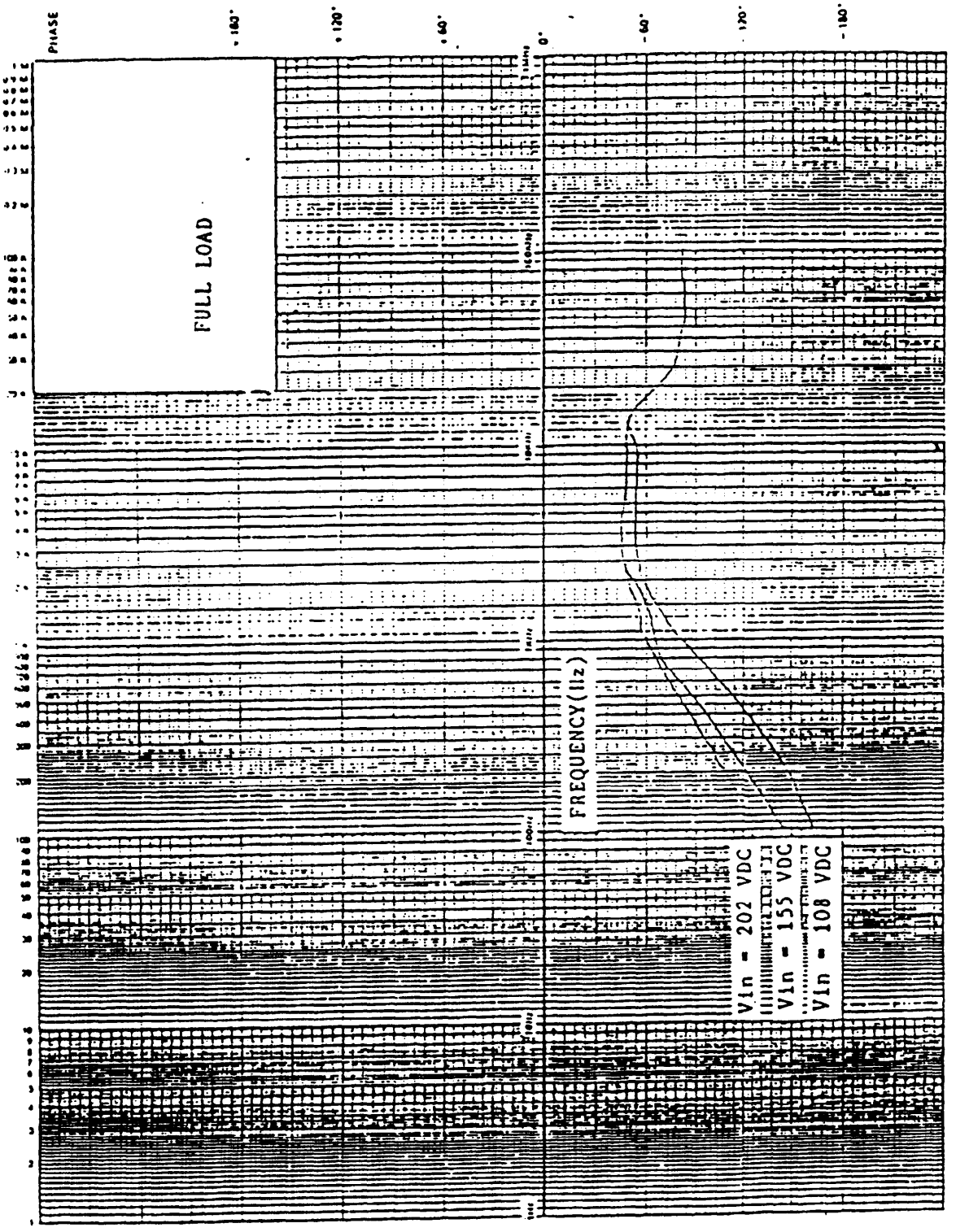


FIGURE 12. Minimum input impedance phase.

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3.20 Spike voltage. The power supply shall be designed to withstand, without damage, a 2500 volt spike clipped to a 250 volt amplitude by external system or equipment devices. The duration of the spike shall be 162 ± 16 microseconds (us) at the 250 volt level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-P-24764 shall apply.

4.2 Quality conformance tests. Quality conformance tests of all power supplies shall be in accordance with the procedures and equipment specified in MIL-P-24764.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Equivalent equipment may be used for quality conformance and qualification testing provided the correlation requirements of MIL-P-24764 are met.

4.4 Test procedures. The power supply shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SPSP-QAA and the test methods conform to the requirements of MLL-P-24764.

4.4.2 Order of testing. Test conditons A through JJ shall be performed in any order with the following exceptions. If test conditions A, Y, AA2, and HH are part of a sequence of tests, the order of testing shall be as follows: A, Y, AA2 and HH where applicable, followed by the remaining tests in the sequence. Test condition U shall be performed in accordance with 4.5.3.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in the MIL-P-24764 with the modifications or additions specified in 4.5.1 through 4.5.7.2.

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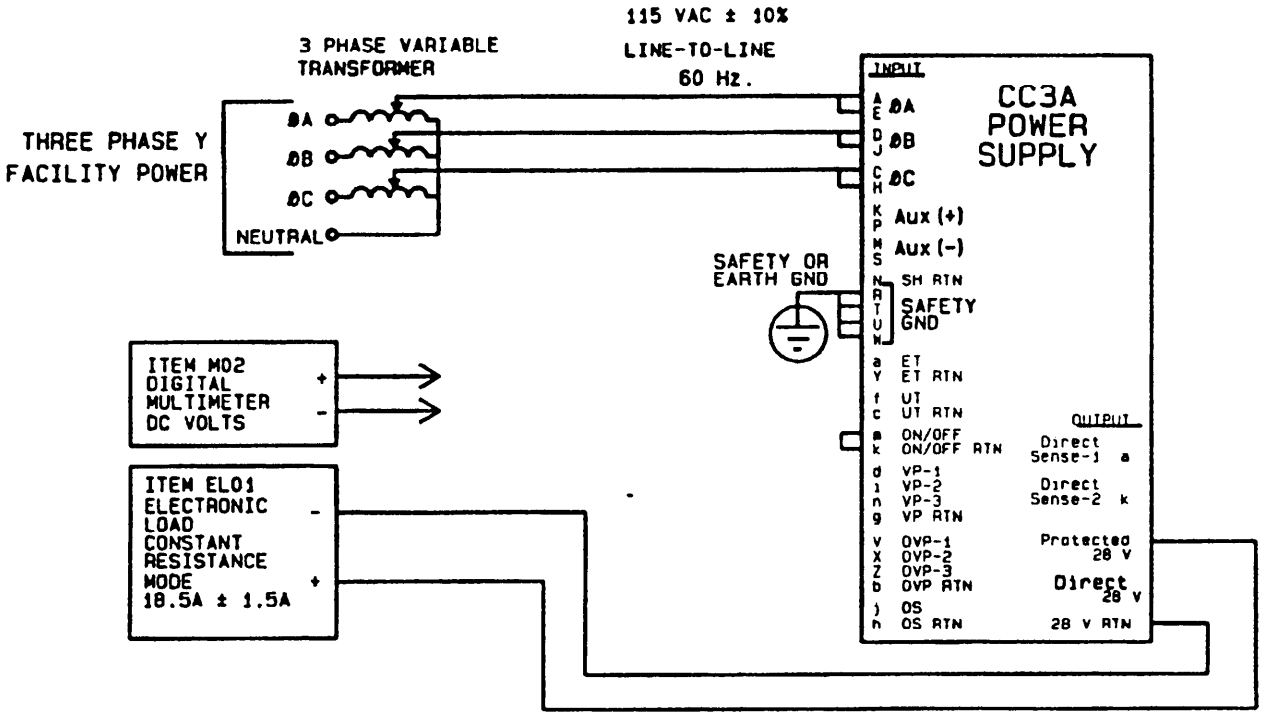
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-P-24764. Before starting these tests, test conditions A through T and V through JJ in table I shallbe performed in the order specified in 4.4.2 using the initial limits at 25 ± 5°C mounting surface temperature. At turn-on, following one hour stabilization at the low temperature, test conditions C3, C4, D2, D3, and M5 shall be performed. When electrical testing is called for at the end of each temperature stabilization period (at -10°C, 60°C, and 25°C post operational temperatures), test conditions C, D, G, H, L1, M1-M3, N1, N2, N7, N8, Q1-Q4, R, T, CC and GG1-GG3 shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. When electrical testing is called for at 80°C, test conditions C3-C4, D1-D4, L1, and N1-N2 shall be performed using the high temperature end-of-life limits. At all other times during the operating temperature tests, the power supply shall be operated in the test circuit shown on figure 13.

4.5.2 Component temperature tests. The coponent temperatures shall be measured in accordance with MIL-P-24764. The power supply shall be operated as shown on figure 13. These measurements shall be performed on each initial qualification submission that is the result of a new vendor or of a major redesign of a previously qualified vendor.

4.5.3 Over-temperature indicators. Operation of the excessive temperature indicator and unsafe temperature indicator shall be verified to be within the temperature liits specified in table I. Test conditions U1-U8 are non-operating tests designed to verify that the over-temperature indicators operate. Test condition U shall only be performed after all other electrical and environmental tests are completed.

4.5.4 Vibration tests. The vibration tests shall be conducted in accordance with MIL-P-24764. During the vibration tests, the power supply shall be operated in the circuit shown on figure 14. Following the vibration tests, the electrical tests in table I which are marked with an "X" in test type column shall be performed using the initial limits at 25 ± 5°C mounting surface temperature.

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NOTES:

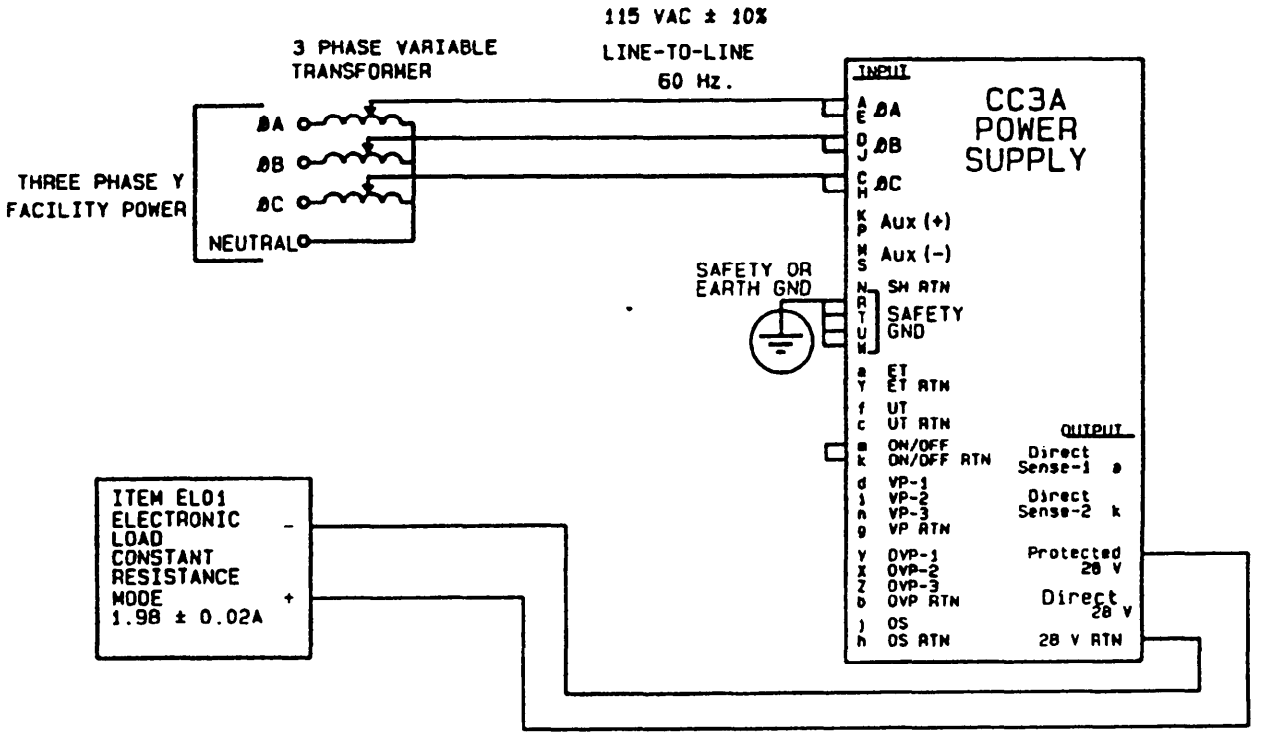
1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. For the long term stability test, the input voltage and output load current shall be within ± 1 percent when making output voltage measurements. Item M02 shall be used.

FIGURE 13. Operating temperature, component temperature, and long term stability test circuit.

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NOTES:

1. See table II for output connector contact assignment.

2. Output loading is specified at 15 percent of rated load to minimize the requirement for externally supplied cooling. The test may be performed at any load level between 10 percent and 100 percent rated load; however, the power supply mounting surface temperature shall not exceed 60°C.

FIGURE 14. Vibration and shock test circuit.

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4.5.5 Shock tests. The mechanical shock tests shall be conducted in accordance with MIL-P-24764. During the shock tests, the power supply shall be operated in the circuit shown on figure 14. Following the shock test, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C mounting surface temperature.

4.5.6 Long term stability test. The long term stability test shall be conducted in accordance with MIL-P-24764. The power supply shall be operated as shown on figure 13. After operating for 20 ± 2 hours, the baseline output voltage reading shall be taken. At the conclusion of the test, prior to disconnecting the power supply or reducing the temperature, the output voltage reading shall again be recorded. The end-of-test voltage shall not change by more than 0.2 percent from the baseline voltage. Following the return of the power supply to 25°C, electrical tests A through T, V, X, CC, EE, FF, and GG1-GG3 shall be performed in the order specified in 4.4.2 using the 25 ± 5°C mounting surface temperature end-of-life requirements in table I.

4.5.7 EMI test. The EMI tests shall be conducted in accordance with MIL-P-24764, the procedures of MIL-STD-462, and as modified herein. For requirements CE01 and CE03 the power supply shall be tested using 115 VRMS ± 2 percent, line-to-line, 3-phase 60 ± 1 Hz, input voltage and tested, using 155 VDC ± 2 percent input voltage. For requirement RE01, the supply shall be tested using the 115 VRMS ± 2 percent line-to-line input voltage. At the option of the testing facility, for the remainder of the requirements either the nominal AC or DC input voltage may be used. The loading shall be 90 to 100 percent of full load.

4.5.7.1 RE01 test modification. The RE01 upper frequency shall be 10 MHz as shown on figure 10.

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4.5.7.2 RE02 test modification. The test shall be performed for two conditions:

(a) The installed configuration shall have the power supply marking (see figure 1) facing the antenna, with the connector area and all leads completely shielded. The purpose of this configuration is to ensure that only the radiated emissions through the case and cover of the power supply are being measured.

(b) The connector opening configuration shall have the power supply connector opening facing the antenna and all leads up to the connector opening completely shielded. The purpose of this configuration is to ensure that only the radiated emissions from the power supply connector area are being measured.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-P-24764.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Power supplies specified herein are intended for use in military shipboard and subsurface ship systems and subsystems.

6.2 Acquisition requirements. Acquisition documents should specify the ordering data required in MIL-P-24764.

6.3 End-of-life. When any individual power supply parameter fails to conform to the end-of-life requirements in table I, the power supply is considered to have completed its useful life (see 3.9).

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6.4 Power supply description.

Keycode: CC3A

Case size: 2.48 inches (62.99 millimeters (mm)) by 6.80 inches (172.72 mm) by 10.8 inches (274.32 mm)

Mean time between failures (MTBF): 100,000 hours minimum

Input voltage: Accepts any of the following input voltages:

(a) 115 VRMS, line-to-line, 3-phase, 47-63 Hz, delta.

(b) 155 VDC to any two AC inputs.

(c) 155 VDC to the auxiliary input.

Output: Provides any one of the following output voltages and current:

28.0 volts, 28.5 volts, 29.5 volts and 30.5 volts at 2 to 20 amperes

6.5 Power supply function.

6.5.1 Voltage programming. The power supply output is programmable for a nominal voltage of 28.0 volts, 28.5 volts, 29.5 volts or 30.5 volts.

6.5.2 Overvoltage programming. The overvoltage limits are programmable and provide for adjustment of the limits to correspond to the selected output voltage.

6.5.3 Remote on/off control. The remote on/off control applies to the power supply output.

6.5.4 Direct and protected output. The protected output of the power supply provides protection for the power supply against shorting the output power distribution to an AC source. The direct output is not protected against AC shorts, but permits the power supply to operate at a higher efficiency.

6.5.5 Output current rating. The output is rated from 2 to 20 amperes. For operation less than 2 amperes, the 25°C initial load regulation and ripple tolerances are twice the tolerances for operation within the rated current limits.

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6.5.6 Recovery from overvoltage. The output voltage is not required to recover automatically from an output overvoltage fault and may require the input voltage to be taken to zero and then reapplied to reset the output. Cycling of the remote on/off control will not accomplish this reset function.

6.5.7 Exceptions to MIL-P-24764. This power supply shall have the following exceptions to the requirements of MIL-P-24764:

(a) The power input circuit can withstand a spike voltage of only 250 volts peak (see 3.20).

(b) The input voltage turn-on and turn-off thresholds are different (see table I, test conditions GG1 and GG2).

(c) Output remote sensing capability is not required.

(d) The overvoltage protection is not required to be a latching type.

(e) The EMI limits for requirements CE01, CE03, RE01 and RE02 of MIL-STD-461 are modified or extended (see 3.19).

(f) An external EMI filter is required to enable the power supply to meet requirement CE03 of MIL-STD-461.

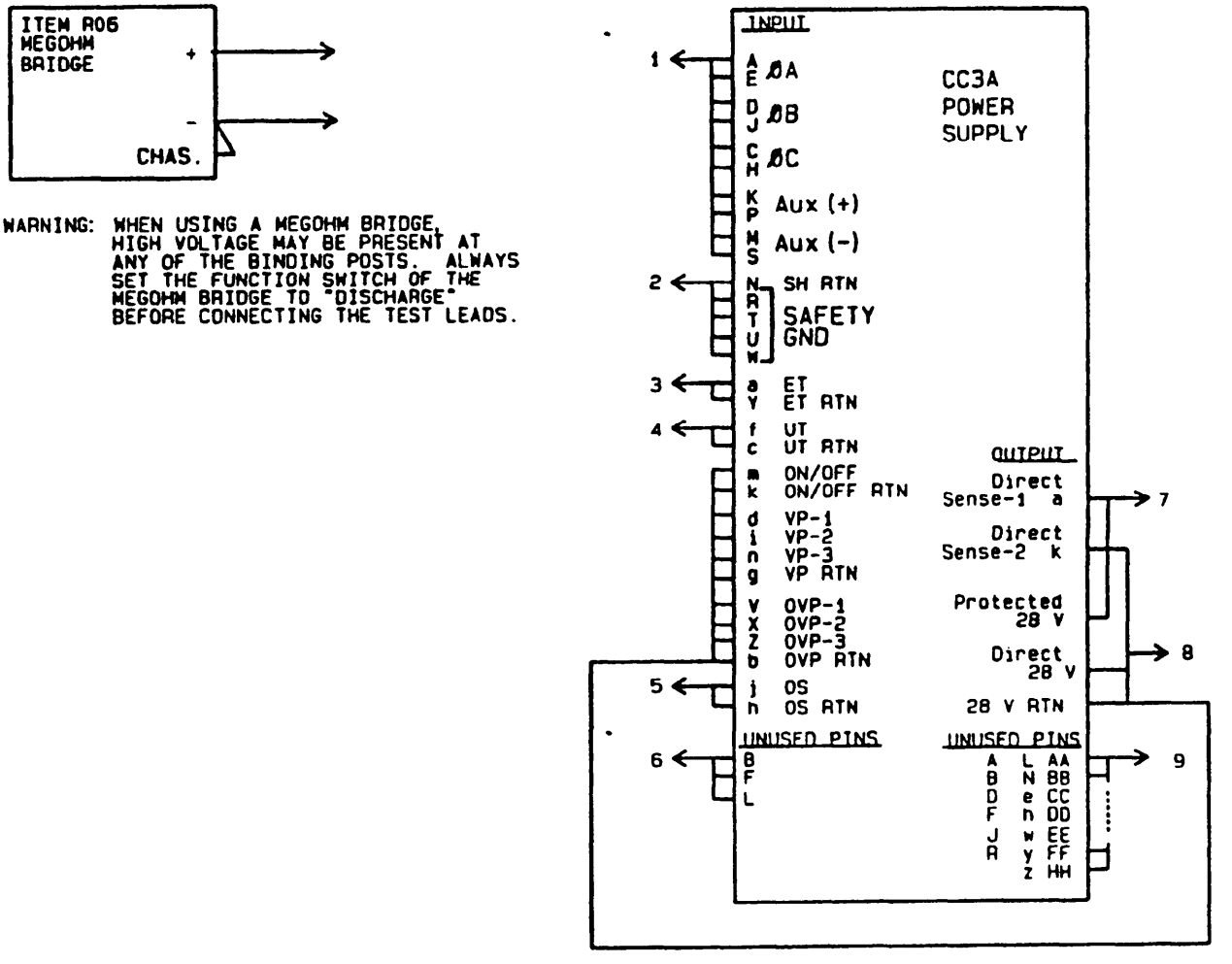
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TABLE IV. Isolation tests.

Test cond	Parameter	Item R06	Connect R06 to		Initial limits 25± 5°C mounting surface temperature		Unit	Test fig
		Voltage ± 10%	+ terminal points	- terminal points	Min	Max		
A1	Isolation - input to chassis and output	800 VDC	1	2,7,8	100	INF.	kOhm	15
A2	Isolation -output to chassis	50 VDC	7,8	2				
A3	Isolation - excessive temperature	100 VDC	3	1,2,4,5,7,8				
A4	Isolation - unsafe temperature	100 VDC	4	1-3,5,7,8				
A5	Isolation - output status	100 VDC	5	1-4,7,8				
A6	Isolation - input connector used to unused contacts	100 VDC	6	1-5	100		kOhm	
A7	Isolation - output connector used to unused contacts	100 VDC	9	7,8	10		MOhm	
A8	Isolation - input to chassis and output	800 VDC	1	2,7,8				
A9	Isolation - output to chassis	50 VDC	7,8	2				
A10	Isolation - excessive temperature	100 VDC	3	1,2,4,5,7,8				
A11	Isolation - unsafe temperature	100 VDC	4	1-3,5,7,8				
A12	Isolation - output status	100 VDC	5	1-4,7,8				
A13	Isolation - input connector used to unused contacts	100 VDC	6	1-5				
A14	Isolation - output connector used to unused contacts	100 VDC	9	7,8	10	INF.	MOhm	15

1/ Two or more numbers in one block indicate that these points shall be connected together in addition to being connected to the + terminal or - terminal (for example, "3,4" under column "+ terminal points" indicates point 3, connected to point 4, connected to the + terminal).

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NOTES:

1. See table II for the output connector contact assignment.
2. See table IV for item R06 connections and voltage setting.

FIGURE 15. Isolation test circuit for table IV.

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TABLE V. Continuity tests.

Test cond	Parameter	Connect to M02 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B1	Chassis contact to chassis continuity	Input R	Chassis	0	1.0	Ohm	-
B2	28 V RTN to remote ON/OFF RTN continuity	Output T	Input k				
B3	28 V RTN to voltage program RTN continuity	Output T	Input g				
B4	28 V RTN to overvoltage program RTN	Output T	Input b				
B5	Safety GND continuity	Input R	Input T				
B6	Safety GND continuity		Input U				
B7	Safety GND continuity		Input W				
B8	Safety GND to shield RTN continuity	Input R	Input N				
B9	Phase A contacts continuity	Input A	Input E				
B10	Phase B contacts continuity	Input D	Input J				
B11	Phase C contacts continuity	Input C	Input H				
B12	Auxiliary DC(+) contacts continuity	Input K	Input P				
B13	Auxiliary DC(-) contacts continuity	Input M	Input S	0	1.0	Ohm	-

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TABLE V. Continuity tests - Continued.

Test cond	Parameter	Connect to MO2 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B14	28 V protected output contacts continuity	Output C	Output E	0	1.0	Ohm	-
B15			Output H				
B16			Output K				
B17			Output M				
B18			Output P				
B19			Output S				
B20			Output U				
B21			Output W				
B22			Output Y				
B23	28 V protected output contacts continuity		Output c				
B24	28 V protected to direct sense-1	Output C	Output a				
B25	28 V direct output contacts continuity	Output n	Output r				
B26	28 V direct output contacts continuity	Output n	Output t				
B27	28 V direct output contacts continuity	Output n	Output v	0	1.0	Ohm	-

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TABLE V. Continuity tests - Continued.

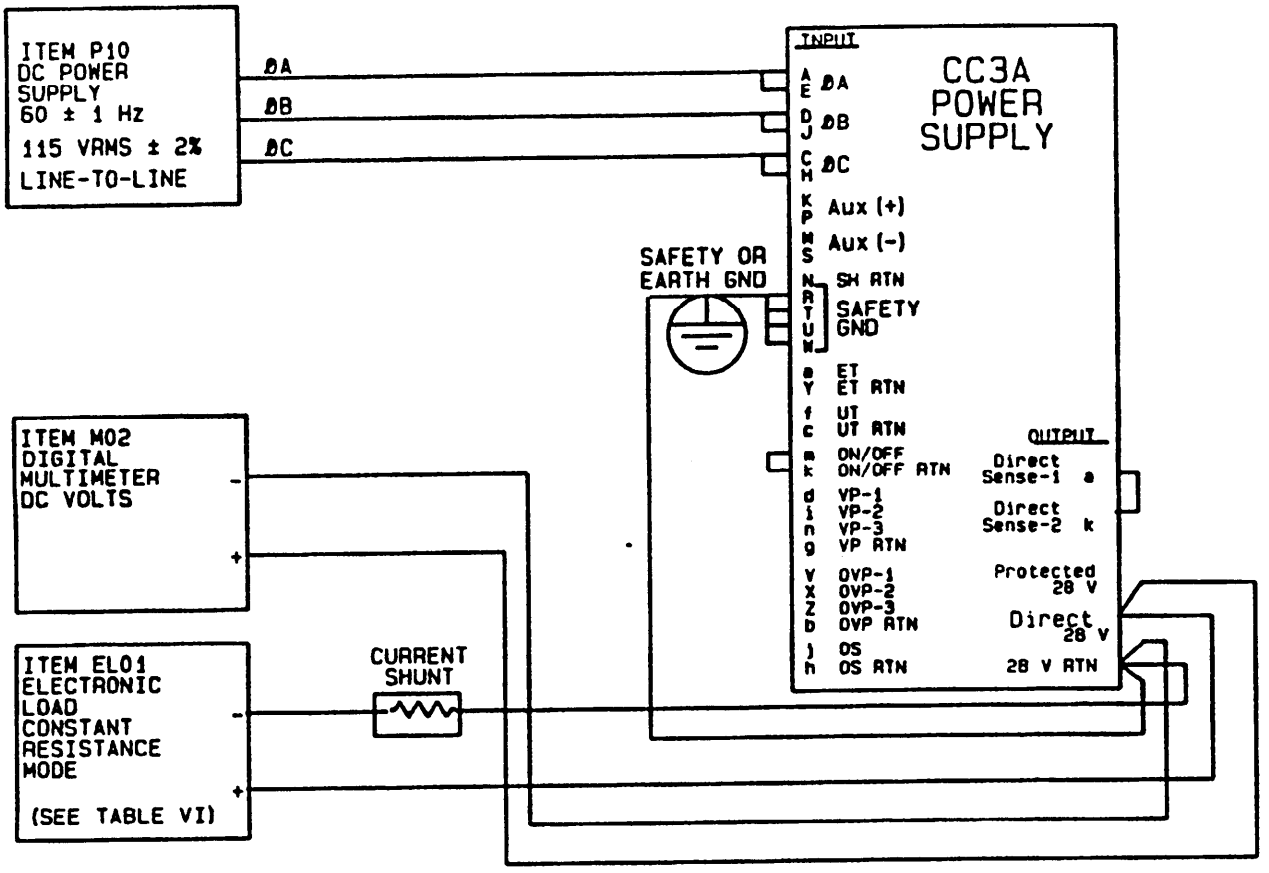
Test cond	Parameter	Connect to M02 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B28	28 V direct output contacts continuity	Output n	Output x	0	1.0	Ohm	-
B29	28 V direct to direct sense-2	Output n	Output k				
B30	28 V RTN contacts continuity	Output T	Output V				
B31			Output X				
B32			Output Z				
B33			Output b				
B34			Output d				
B35			Output f				
B36			Output j				
B37			Output m				
B38			Output p				
B39			Output s				
B40	28 V RTN contacts continuity	Output T	Output u	0	1.0		
B41	Chassis contact to chassis continuity	Input N	Chassis	0	0.1		
B42		Input R					
B43		Input T					
B44		Input U					
B45	Chassis contact to chassis continuity	Input W	Chassis	0	0.1	Ohm	-

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TABLE VI. Static load regulation tests.

Test cond	Parameter	Item EL01 Curr. +0, - 2%	Connect 1 to points	Connect 2 to points	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
C1	Load regulation, 28 V, direct	2A	-	-	27.72	28.28	VDC	16
C2	Load regulation, 28 V, direct	20A	-	-	27.72	28.28		16
C3	Load regulation, 30.5 V, protected	2A	3	4	30.20	30.80		17
C4	Load regulation, 30.5 V, protected	20A	3	4	30.20	30.80		17
C5	Load regulation, 30.5 V, direct	20A	-	-	30.20	30.80		16
C6	Load regulation, 28 V protected	0A	3	4	27.44	28.56		17
C7	Load regulation, 28 V output status	0A	5	6	0.0	0.4		17
C8	Load regulation, 30.5 V protected	0A	3	4	29.89	31.11		17
C9	Load regulation, 30.5 V output status	0A	5	6	0.0	0.4	VDC	17

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

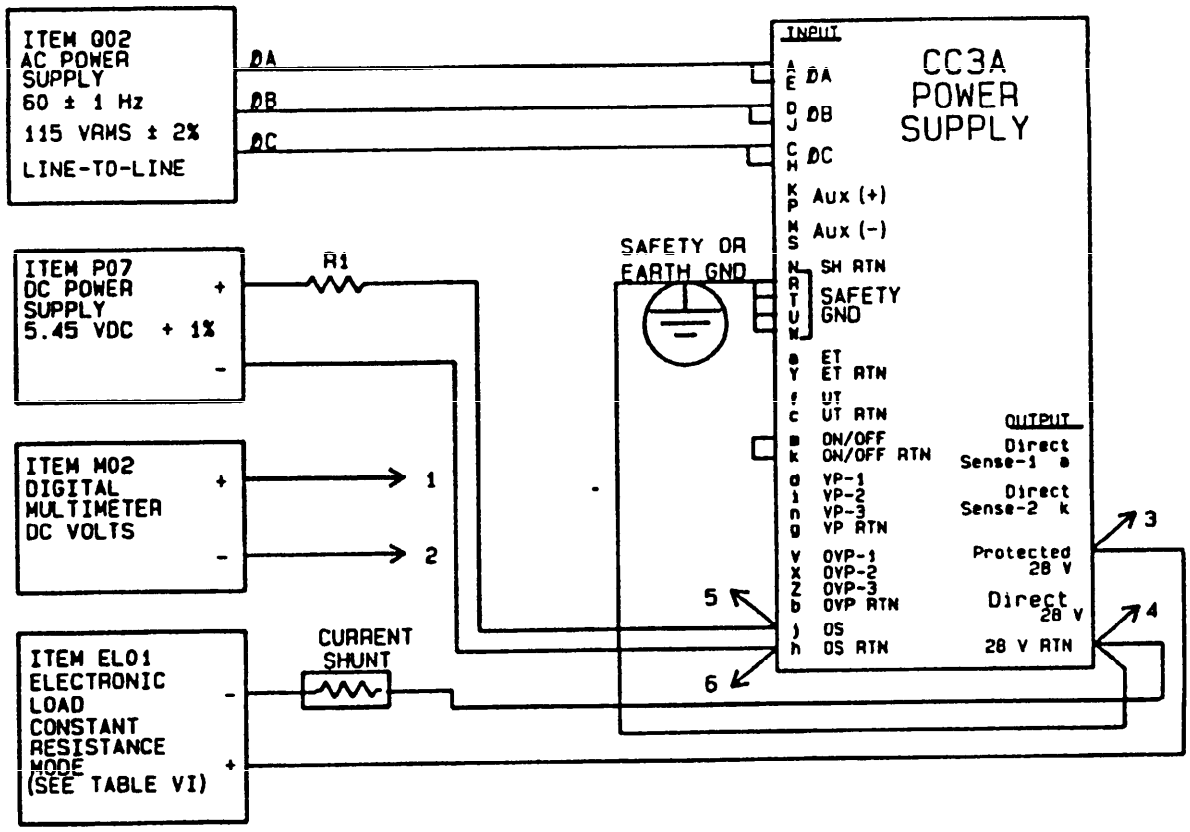
4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

6. For test condition C5, short input contacts d, i, n, and g together and short input contacts V, X, Z, and b together.

FIGURE 16. Static load regulation test circuit (direct output) for table VI.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 1.69 kOhm ± 1 percent, 1/8 watt, resistor.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. For test conditions C3, C4, C8, and C9, short input contacts d, i, n, and g together and short input contacts V, X, Z, and b together.

FIGURE 17. Static load regulation test circuit (protected output) for table VI.

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TABLE VII. Line regulation tests.

Test cond	Parameter	Input 1/ voltage ± 1%	Item EL01 curr. +0, - 2%	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
				Min	Max		
D1	Line reg., low AC input	86 VRMS	2A	27.72	28.28	VDC	18
D2	Line reg., low AC input, full load	86 VRMS	20A				
D3	Line reg., high AC input	144 VRMS	2A				
D4	Line reg., high AC input, full load	144 VRMS	20A				
D5	Line reg., low DC input	108 VDC	2A				
D6	Line reg., low DC input, full load	108 VDC	20A				
D7	Line reg., high DC input	202 VDC	2A				
D8	Line reg., high DC input, full load	202 VDC	20A				
D9	Line reg., low DC input, aux. input	108 VDC					
D10	Line reg., high DC input, aux. input	202 VDC					
D11	Line reg., low frequency	115 VRMS					
D12	Line reg., high frequency	115 VRMS	20A	27.72	28.28	VDC	18

1/ See figure 19, notes 5 and 6 for connection of input power. The root mean square (RMS) voltage is measured line-to-line.

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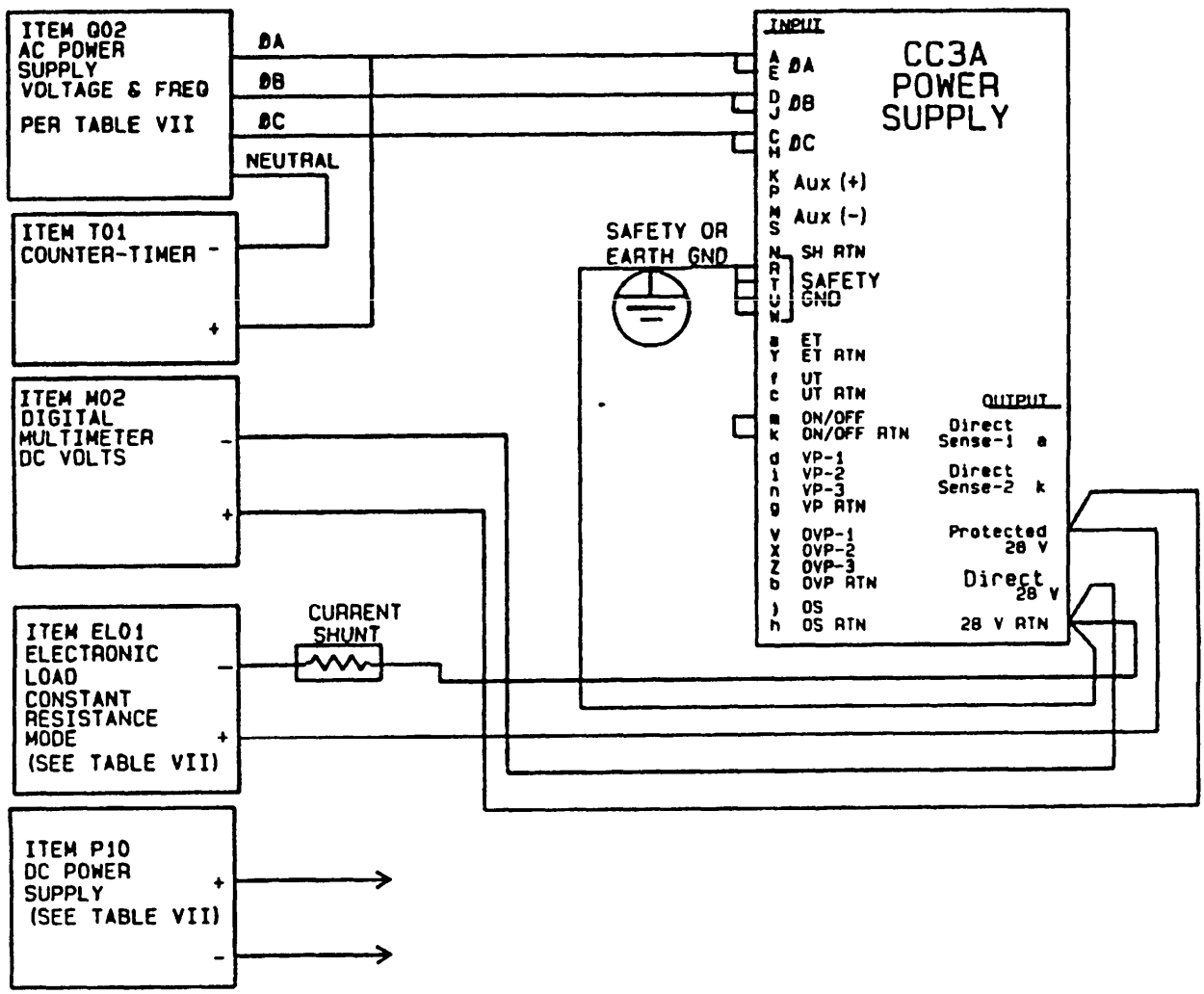


FIGURE 18. Line regulation test circuit for table VII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. The frequency of item Q02 shall be 60 ± 1 Hz for test conditions D1-D4, 47 ± 0.3 Hz for test condition D11 and 63 ± 0.3 Hz for test condition D12.

6. Where RMS voltage is specified (test conditions D1-D4, D11 and D12), item Q02 connected as shown shall be used. For test conditions D5-D8, disconnect item Q02 and connect item P10 + to input contacts A and E, and - to input contacts D and J. For test conditions D9 and D10, connect item P10 + to input contacts K and P and - to input contacts M and S. Only one power source shall be connected at any one time.

FIGURE 18. Line regulation test circuit for table VII - Continued.

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TABLE VIII. Low input voltage and loss of phase tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
E	Low input voltage protection	27.72	28.28	VDC	19
F1	Loss of phase A protection	27.72	28.28	VDC	19
F2	Loss of phase B protection				
F3	Loss of phase C protection				

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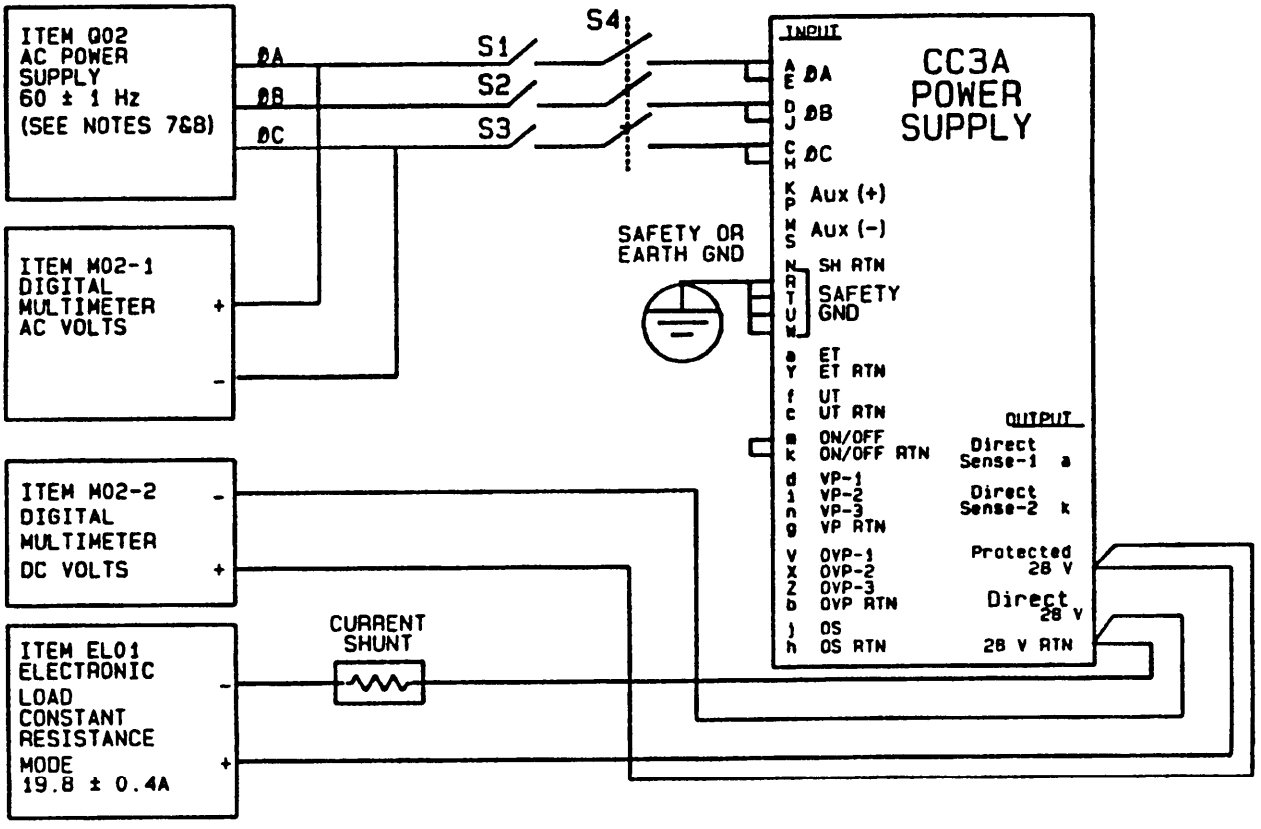


FIGURE 19. Low input voltage and loss of phase test circuit for table VIII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1=S2=S3= single pole single throw (spst), 160 VRMS, 15 amperes switch.

6. S4 = three pole single throw (3pst), 160 VRMS, 15 amperes switch.

7. For test condition E, connect the circuit shown with switches S1, S2, S3, and S4 closed. The item Q02 voltage shall be monitored with item M02-1. Open switch S4 and adjust the item Q02 voltage to 0 VRMS. Close switch S4 and slowly (less than 5 volts per second) increase the item Q02 voltage to 115 VRMS ± 2 percent line-to-line, then decrease it to 0 VRMS at the same rate. Open switch S4 and adjust the item Q02 voltage to 115 VRMS ± 2 percent line-to-line. Close switch S4. The item M02-2 voltage reading shall meet the test condition E limits.

8. Test conditions F1, F2 and F3 shall be performed as follows:

(a) For test condition F1, connect the circuit as shown with switches S1, S2, S3, and S4 closed. Open switch S1 for a minimum of 5 seconds, then close switch S1. The item M02-2 voltage reading shall meet the test condition F1 limits. The power supply is not required to meet the regulation limits during the loss of phase period.

(b) For test condition F2, repeat (a), except use switch S2 and test condition F2 limits.

(c) For test condition F3, repeat (a), except use switch S3 and test condition F3 limits.

FIGURE 19. Low input voltage and loss of phase test circuit for table VIII - Continued.

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TABLE IX. Efficiency, power, factor, and input current balance tests.

TEST COND	PARAMETER	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
G1	Efficiency, Protect 1/ 2/	75	100	%	20
G2	Efficiency, Protect 1/ 2/	60	100	%	
G3	Efficiency, Direct 1/ 2/	76	100	%	
G4	Efficiency, Protect 30.5V	76	100	%	
H	Power Factor 3/	0.8	1.0	-	
J1	Input Current Balance 4/	-5	5	%	
J2	Input Current Balance 4/	-5	5	%	
J3	Input Current Balance 4/	-5	5	%	20

- See footnotes on next page.

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1/ At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input shown for test conditions G1-G3.

2/ Efficiency is calculated as follows for test conditions G1, G2 and G3.

$$\text{Efficiency} = \frac{(\text{Vo})(\text{Io}) \times 100 \text{ percent}}{(\text{Item S03-1 watts}) + (\text{Item S03-2 watts})}$$

or for DC input

$$\text{Efficiency} = \frac{(\text{Vo})(\text{Io}) \times 100 \text{ percent}}{(\text{Vin})(\text{Iin})}$$

Where Vin is the input voltage and is measured with an item M02 multimeter and Iin is the input current and is measured with an item S03 wattmeter. The meters shall measure true RMS. Vo and Io are as defined on figure 20.

3/ Power factor is calculated as follows:

$$\text{power factor} = \frac{(\text{Item S03-1 watts}) + (\text{Item S03-2 watts})}{\left|\overline{\text{Va}^2 \times \text{Ia}^2 + \text{Vb}^2 \times \text{Ib}^2 + \text{Vc}^2 \times \text{Ic}^2}\right|^{1/2}}$$

Where Ia, Ib and Ic are measured with a Pearson 411 current transformer and an item M02 multimeter, and Va, Vb and Vc are line-to-line voltages measured with an item M02 multimeter as shown on figure 20. The meters shall measure true RMS.

4/ Current balance is calculated as follows:

$$J1 = \frac{(Ia-Ib) \times 100 \text{ percent}}{Iave}$$

$$J2 = \frac{(Ia-Ic) \times 100 \text{ percent}}{Iave}$$

$$J3 = \frac{(Ib-Ic) \times 100 \text{ percent}}{Iave}$$

where $Iave = \frac{Ia + Ib + Ic}{3}$.

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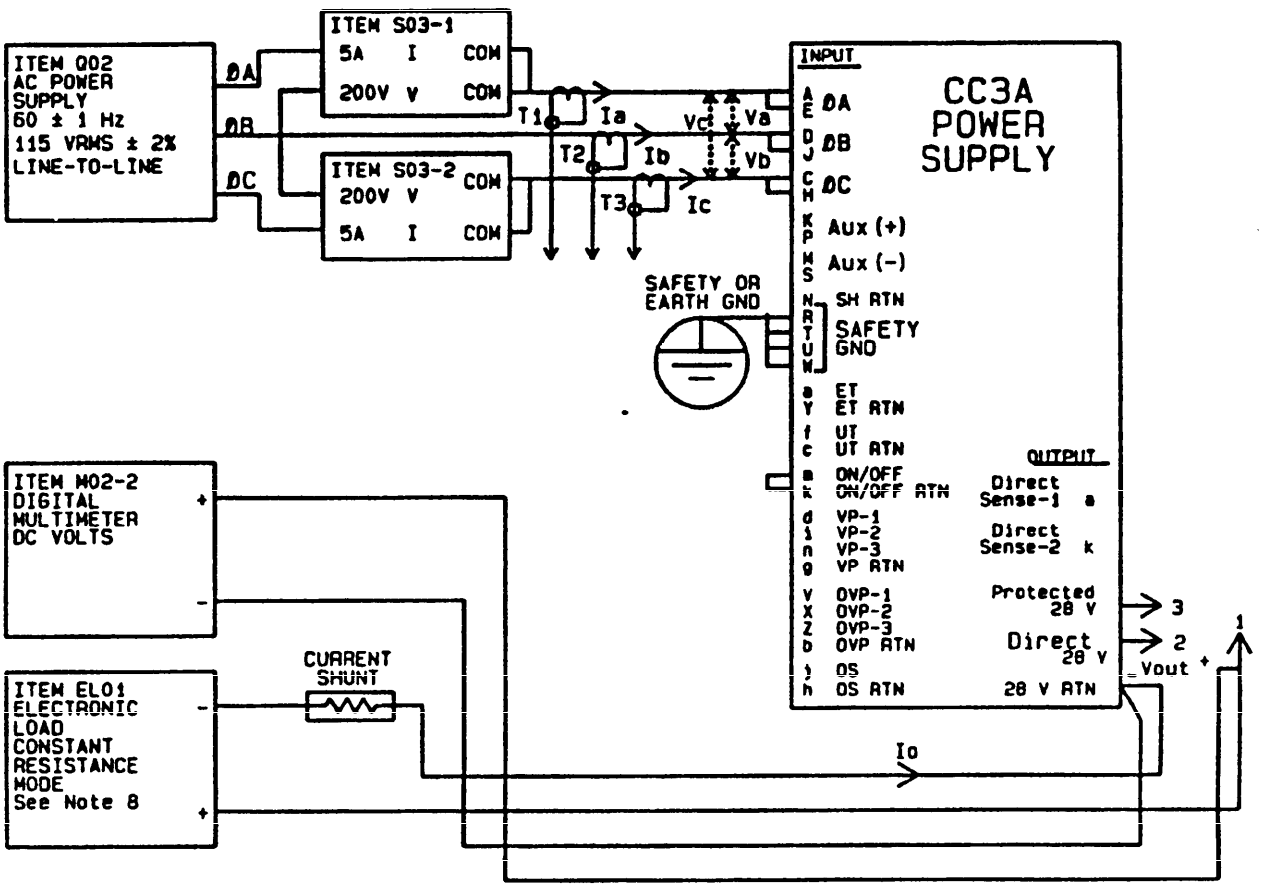


FIGURE 20. Efficiency, power factor, and input current balance test circuit for table IX.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. T1, T2 and T3 are Pearson Model 411 current transformers.

6. For test conditions G1, G2, G4, H, J1, J2, J3 connect point 1 to point 3 and, for test condition G3, connect point 1 to point 2 and output contacts a to k.

7. For test condition G4, connect the voltage programming contacts V, X, Z, and b together and the overvoltage programming contacts d, i, n and g together.

8. Item EL01 shall be set for 20 amperes +0, -2 percent, except for test condition G2, where it shall be set at 2 amperes ±1 percent.

FIGURE 20. Efficiency, power factor, and input current balance test circuit for table IX - Continued.

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TABLE X. Inrush current tests.

TEST COND	PARAMETER	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
K1	Inrush current-AC input	-	50	Apk	21,22
K2	Inrush current-auxiliary input	-	50	Apk	21,22

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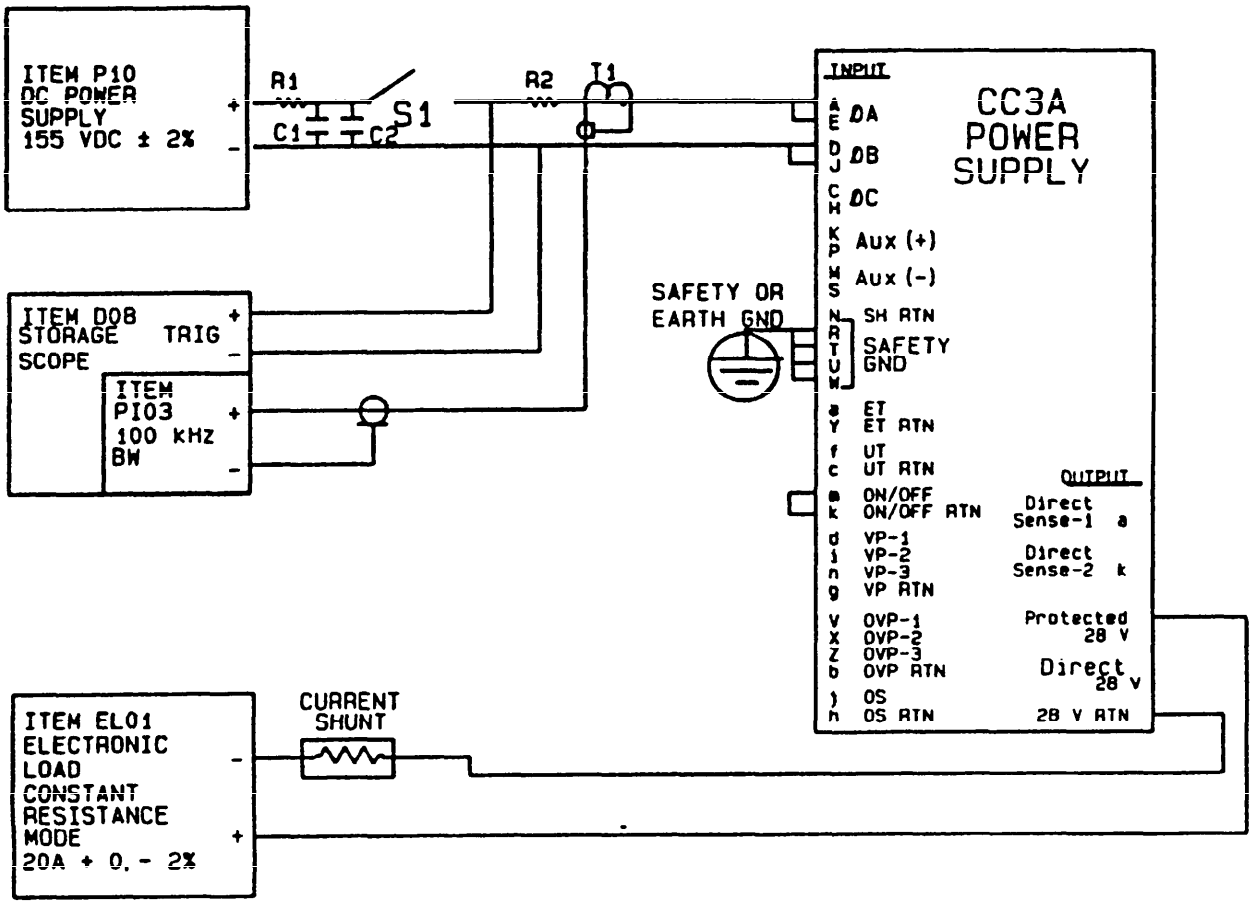


FIGURE 21. Inrush current test circuit for table X.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst, 225 VDC, 15 amperes switch.

6. T1 = Pearson Model 411 current transformer.

7. R1 = 1 Ohm ± 5 percent, 40 watt resistor.

8. R2 = 0.1 Ohm (represents maximum allowable series resistance).

9. C1 = C2 = 320 uF -15 percent, +50 percent, equivalent series resistance (ESR) of 0.4 Ohm max, 250 VDC capacitor (Sprague 80D331P250JE5D or equivalent).

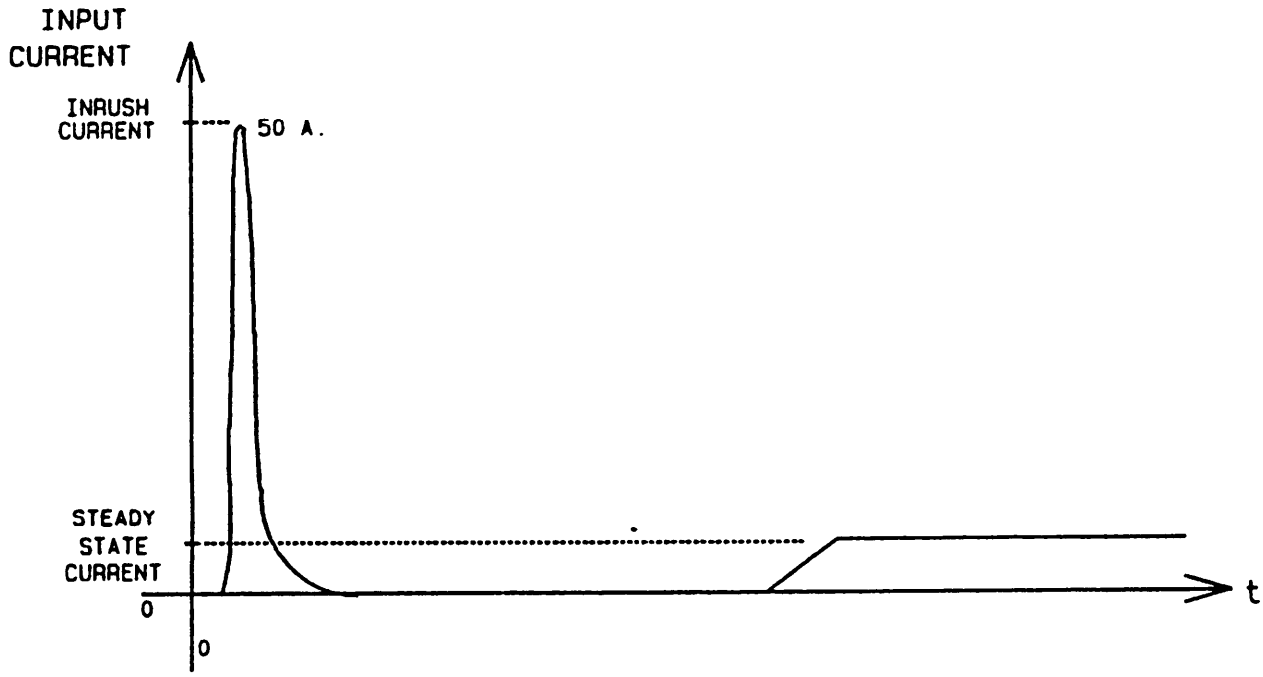
10. For test condition K1, connect item P10 to the AC inputs as shown. For test condition K2, connect item P10 to the auxiliary inputs: + to input contacts K and P, and - to input contacts M and S.

11. Apply power to the power supply under test by closing switch S1 with item P10 energized. Neither the power "OFF" period nor the power "ON" period shall be less than one second.

12. Keep the lead length between C1, C2, S1 and the power supply under test as short as possible.

FIGURE 21. Inrush current test circuit for table X - Continued.

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NOTE:

1. The steady state current is not observable using the figure 21 test circuit, but is shown to indicate its relationship to the inrush current.

FIGURE 22. Inrush current test waveform for table X.

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TABLE XI. Overload and short circuit protection tests.

Test cond	Parameter	Measurement device	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
L1	Overload protection	Shunt & item V04	20	25	ADC	23,24
L2	Short circuit recovery	Item D08	0	100	ms	
L3	Short circuit overshoot	Item D08	-1.40	1.40	V	
L4	Short circuit current	Shunt & item V04	0	25	ADC	23,24

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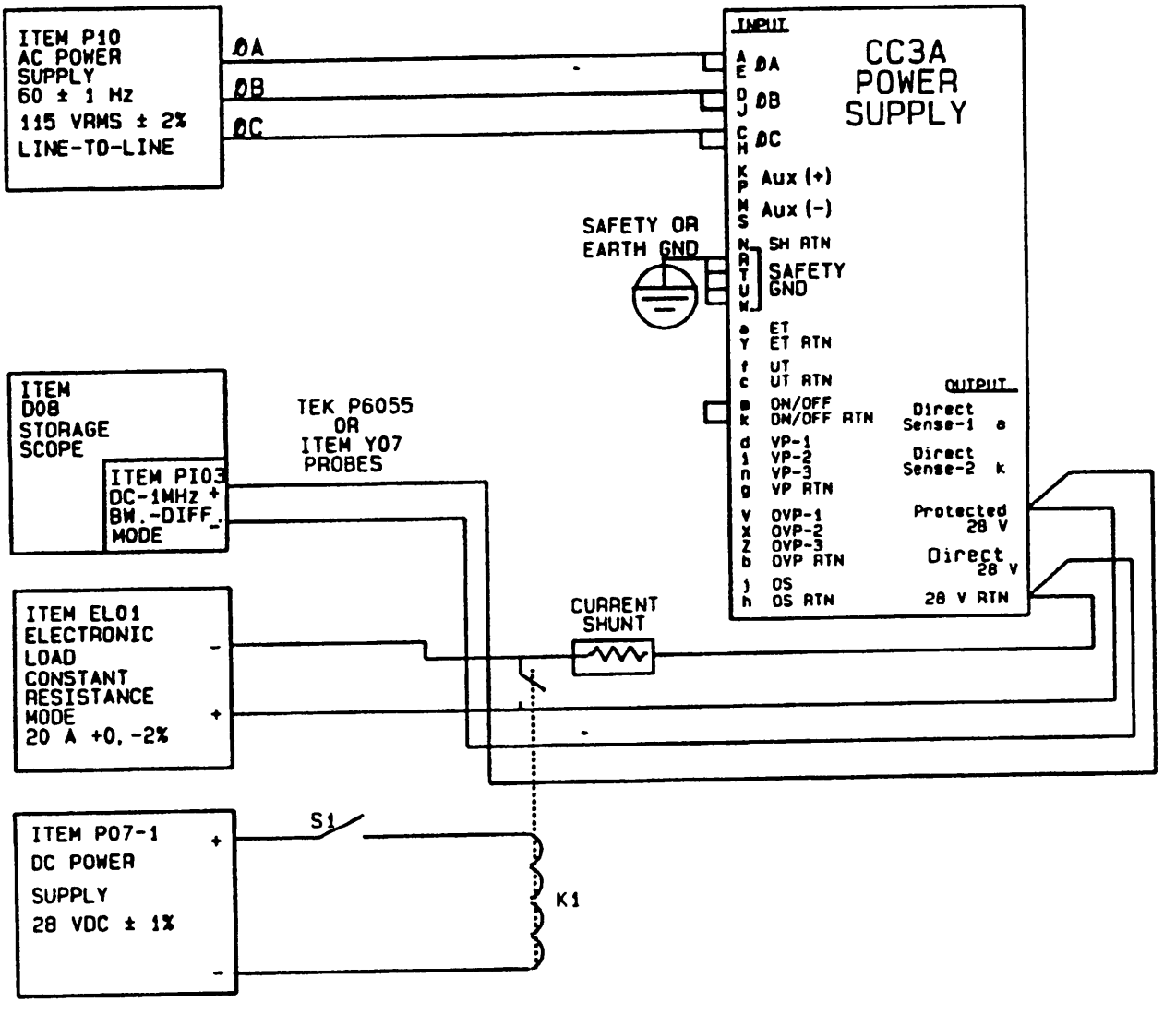


FIGURE 23. Overload and short circuit protection test circuit for table XI.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst switch.

6. K1 = Cutler Hammer MS24185D1 or equivalent contactor.

7. At the option of the test facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

8. For test condition L1: slowly increase load current, as indicated by the shunt voltage, until the maximum current indication is obtained (this may be current saturation or prior to current shutdown). This maximum current shall fall within the specified limits. If the current exceeds 25 amperes, stop this test and do not perform short circuit tests.

9. Keep the leads between the power supply under test and the electronic load as short as possible to minimize lead resistance.

10. Test conditions L2-L4 shall be performed as follows:

(a) Connect the circuit as shown and energize the power supply under test.

(b) Close the contactor K1 by closing S1. Short the output for a minimum of 10 seconds. The short circuit current as monitored by current shunt shall meet the limit of test condition L4.

(c) Remove the short and store the resulting transient on the storage scope (see figure 24). The power supply shall automatically recover within the regulation limits in the time limits of L2. The maximum overshoot after removal of the short shall meet the limits of L3.

FIGURE 23. Overload and short circuit protection test circuit for table XI - Continued.

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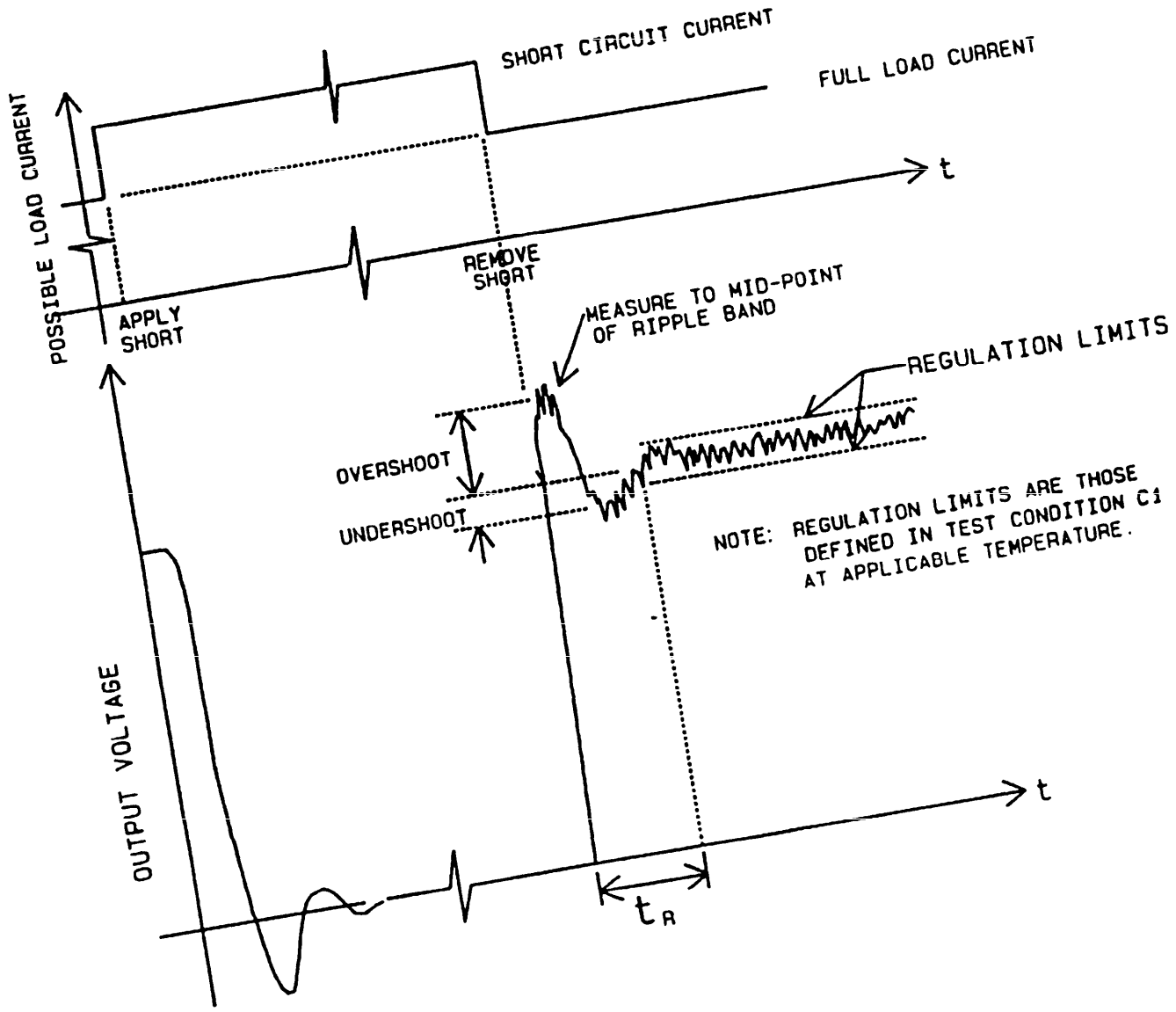


FIGURE 24. Short circuit recovery, overshoot definitions for table XI.

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TABLE XII. Turn-on/off transient and turn-on time tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
M1	Turn-on transient	-1.40	1.40	V	25, 27
M2	Turn-on time	1	250	ms	25, 27
M3	Turn-off transient	-1.40	1.40	V	25, 27
M4	Control turn-on transient	-1.40	1.40	V	26, 27
M5	Control turn-on time	1	250	ms	26, 27
M6	Control turn-off transient	-1.40	1.40	V	26, 27

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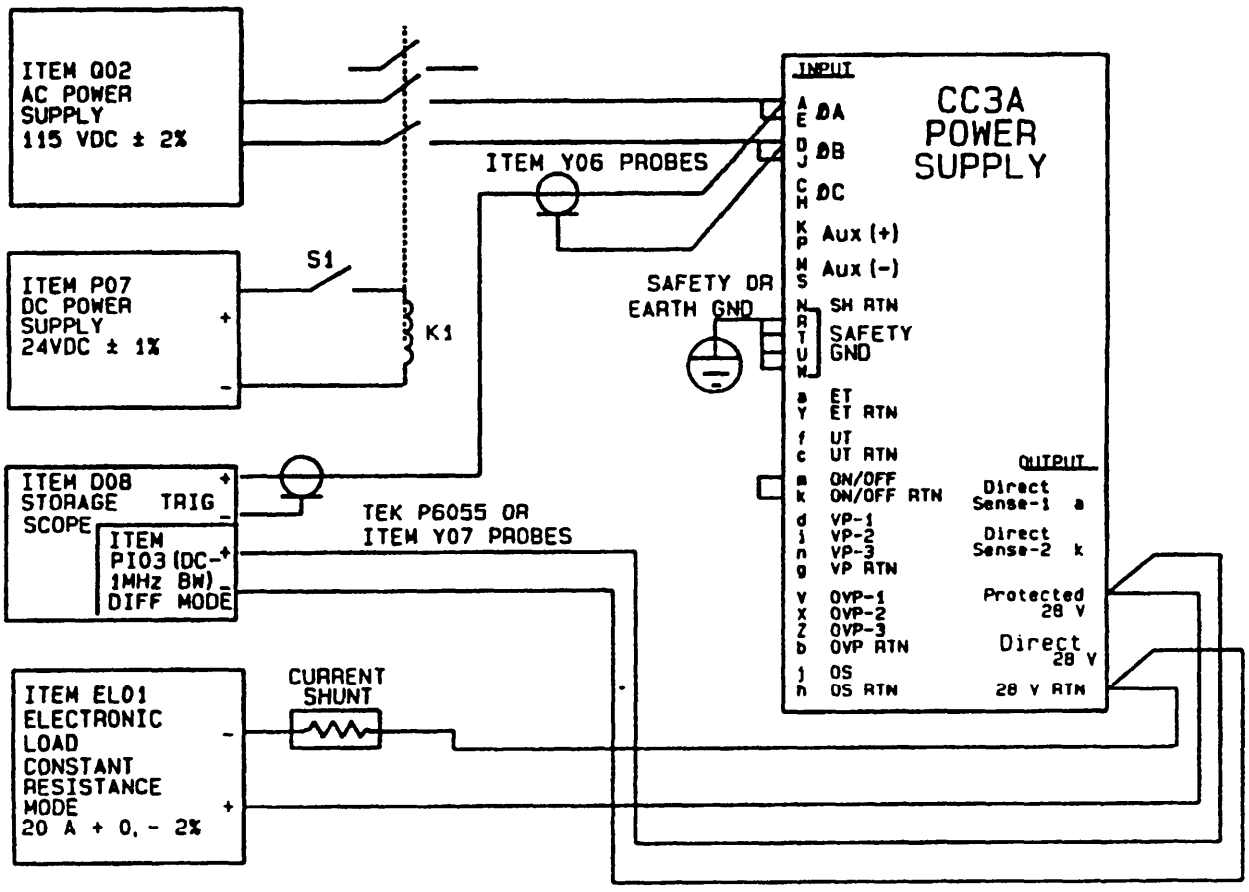


FIGURE 25. Turn-on/off transient and turn-on time test circuit for table XII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst switch.

6. K1 = 3pst Mercury relay, Magnecraft WM35AAA-24D or equivalent.

7. At the option of the testing facility, an AC input (instead of the DC specified) of 3-phase 115 VRMS ± 2 percent line-to-line, 60 ± 1 Hz can be applied at the phase A, B, and C inputs.

8. Each test shall be performed several times to ensure measurement of the highest transient and longest time.

FIGURE 25. Turn-on/off transient and turn-on time test circuit for table XII - Continued.

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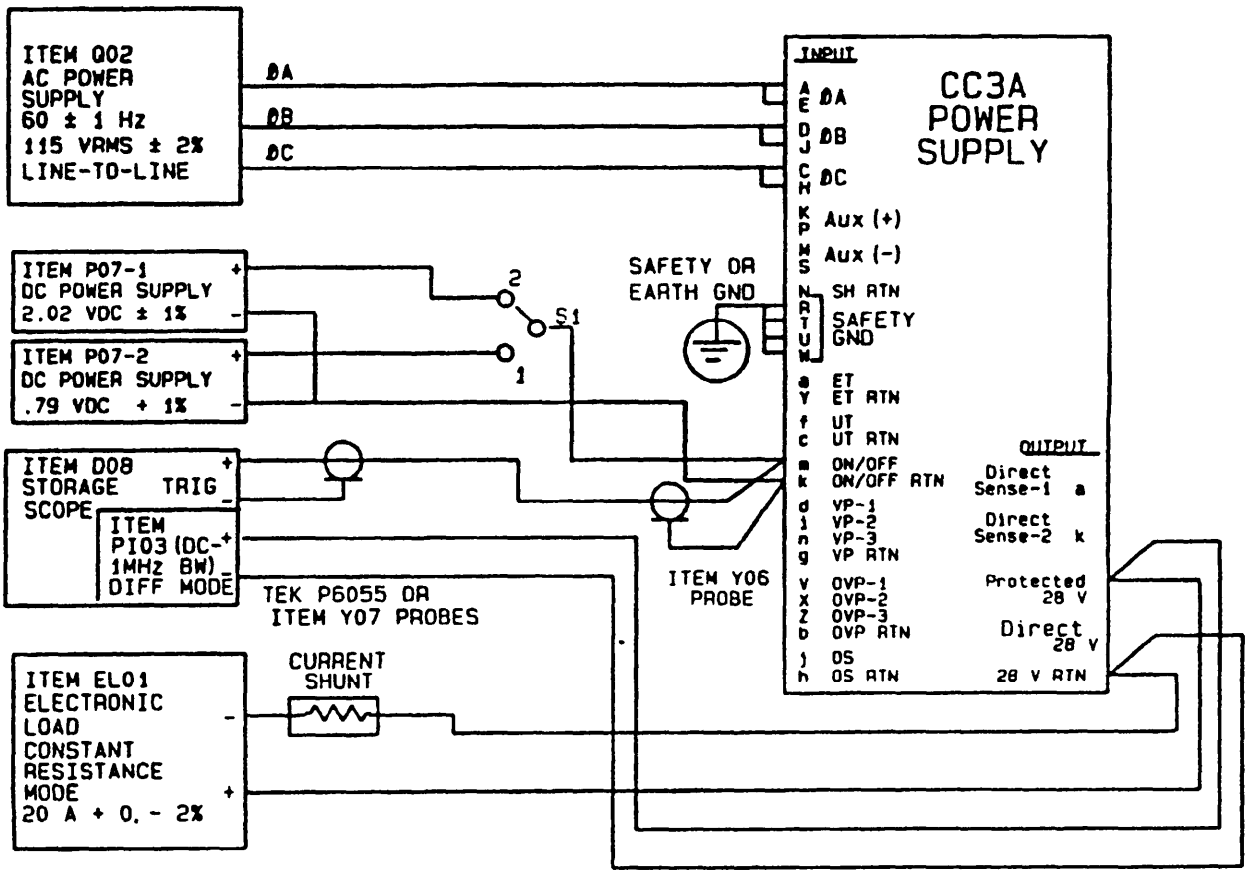


FIGURE 26. Control turn-on/off transient and turn on time test circuit for table XII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = single pole double throw (spdt) switch.

6. At the option of the testing facility, a DC input (item P10) of 155 volts ± 2 percent connected to any two AC inputs can be used instead of the AC input specified.

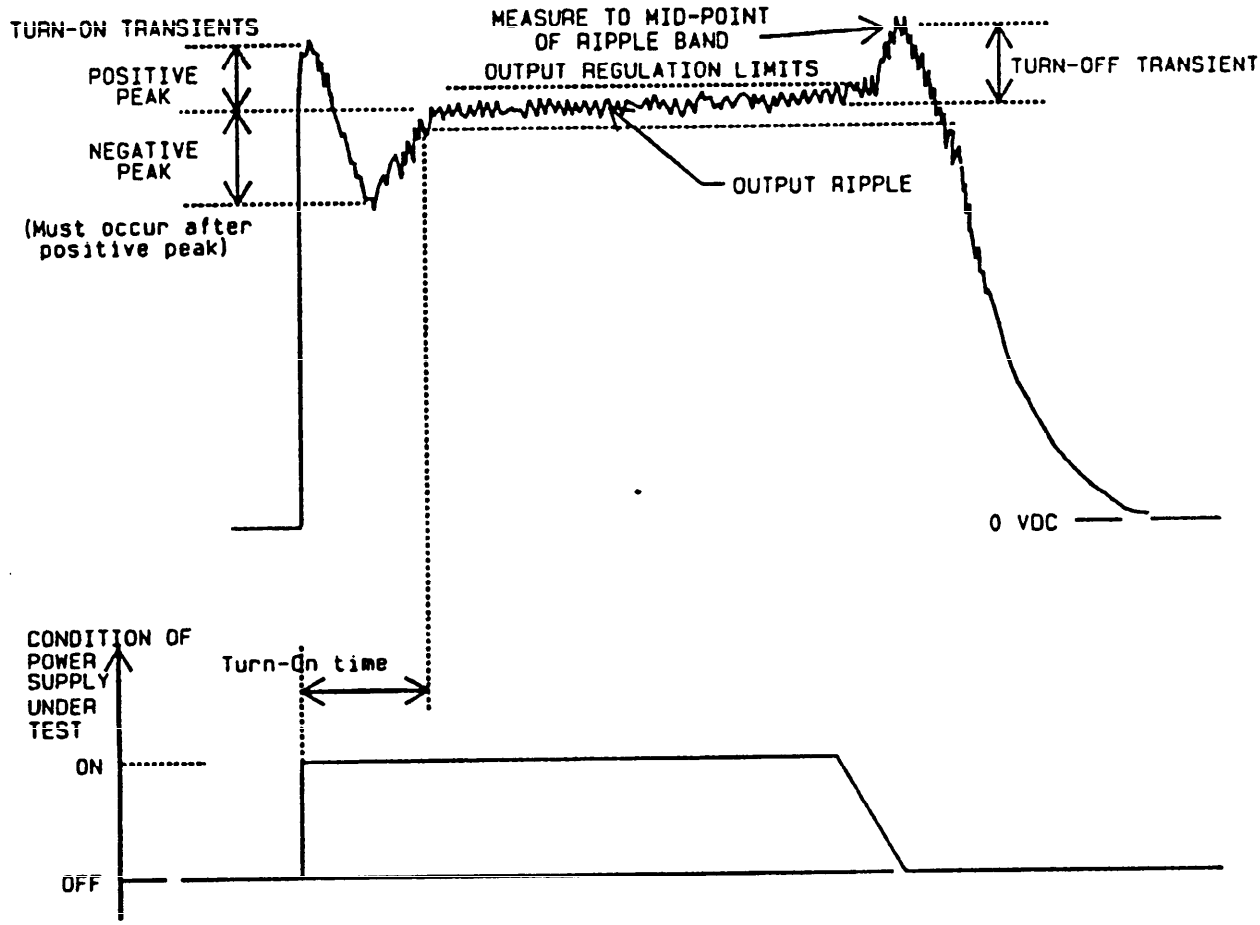
7. The test shall be performed several times to ensure measurement of the highest transient and longest time.

8. The power supply is turned on when S1 is switched from position 2 to 1. The power supply is turned off when S1 is switched from position 1 to 2.

FIGURE 26. Control turn-on/off transient and turn-on time test circuit for table XII - Continued.

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contact K1. Connect J3 pin 1 (input contact H) to safety GND (input



NOTES:

1. For turn-on transient, record the highest amplitude positive or negative peak.

2. The test condition M2 turn-on time is defined as the time from when the input power is applied to the power supply until the output recovers to and remains within the regulation limits. The regulation limits shall be test condition C1 limits (at the applicable temperature).

3. The test condition M5 control turn-on time is defined as the time from when voltage is applied to the control on/off input contacts until the output recovers to and remains within the regulation limits (see note 2).

FIGURE 27. Turn-on and turn-off waveforms for table XII.

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TABLE XIII. Ripple tests.

Test cond	Parameter	Measurement device (Item D08)	AC input line-to line voltage	Measurement frequency range (3dB pt) Low 1/	High	Connect 1 and 2 to	Initial limits 25 ± 5C mounting surface temperature Min	Max	Unit	Test fig
N1	Ripple, total, prot.	Item P103	115 VRMS ±2%	10 Hz	1 MHz	3	0	1680	mVp-p	28
N2	Ripple, total, dir.	Item P103			1 MHz	4		1680		
N3	Ripple, low frequency, prot.	Item P103			10 MHz	3		1100		
N4	Ripple, low frequency, dir.			10 Hz	10 MHz	4		1100		
N5	Ripple, high frequency, prot.			10 kHz	1 MHz	3		728		
N6	Ripple, high frequency, dir.	Item P103	115 VRMS ±2%	10 kHz	1 MHz	4	0	728	mVp-p	28

- See footnotes at end of table.

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TABLE XIII. Ripple tests - Continued.

Test cond	Parameter	Measurement device (item D08)	AC input line-to line voltage	Measurement frequency range(3dB pt) Low 1/	High	Connect 1 and 2 to	Initial limits 25 ± 5C mounting surface temperature Min	Max	Unit	Test fig
N7	Ripple, total, prot.	Item P101	115 VRMS ±2%	10 Hz	20MHz	3	0	1680	mVp-p	28
N8	Ripple, total, dir.	Item P101			20MHz	4		1680		
N9	Ripple, total, prot., overload	Item P102			5 MHz	3		1100		
N10	Ripple total, overload, dir.					4		1100		
N11	Ripple total, prot., no load					3		728		
N12	Ripple, total, dir., no load		115 VRMS ±2%			4		728		
N13	Ripple, total, low input, prot.		86 VRMS ±1%			3		728		
N14	Ripple, total, low input, dir.		86 VRMS ±1%			4				
N15	Ripple, total, high input, prot.		144 VRMS ±1%			3				
N16	Ripple, total, high input, dir.	Item P102	144 VRMS ±1%	10 Hz	5 MHz	4	0	728	mVp-p	28

1/ When using item P101 or P102, use AC coupling to obtain the 10Hz lower 3dB point setting.

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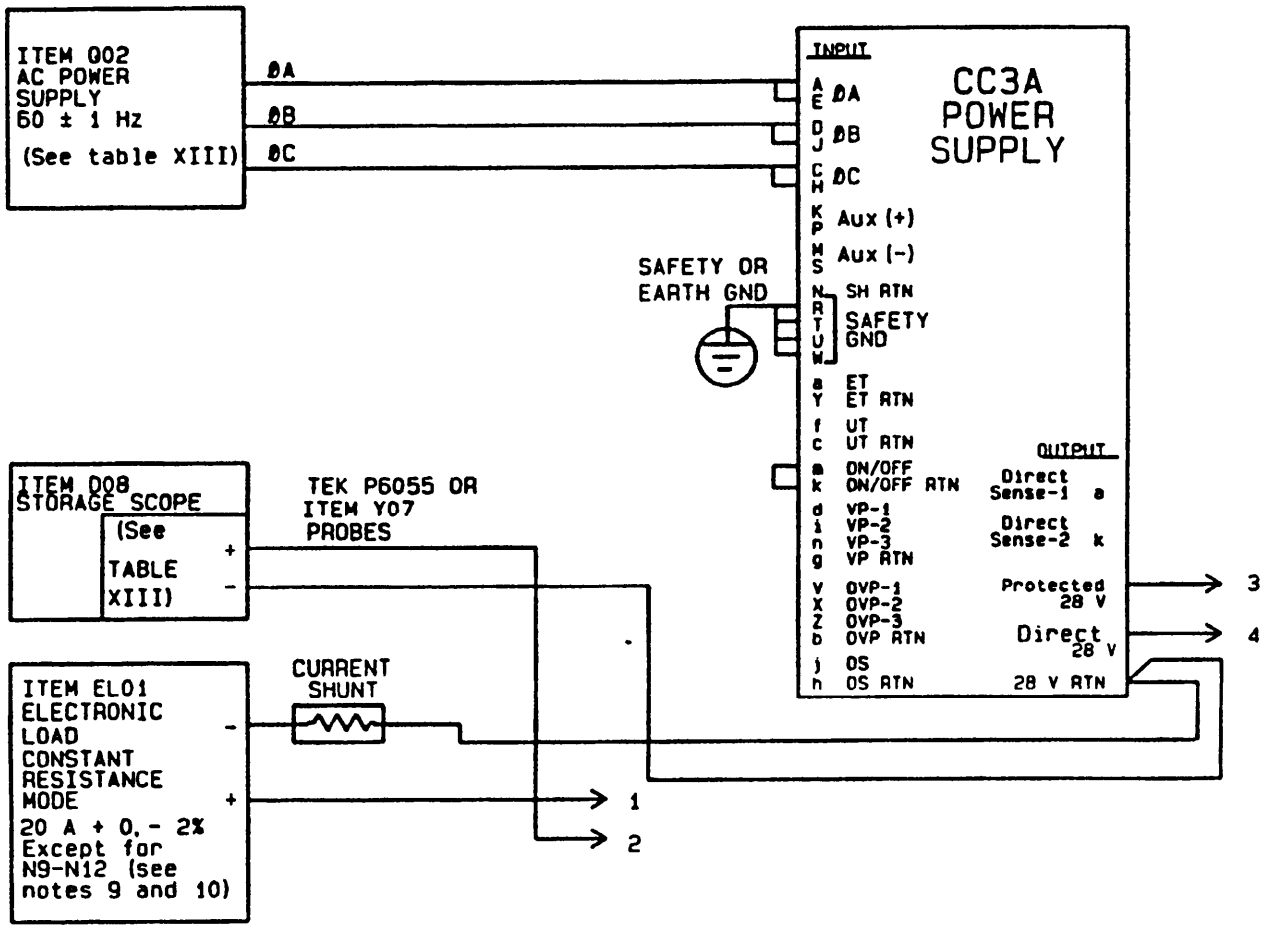


FIGURE 28. Ripple test circuit for table XIII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. The smaller block within item D08 represents a plug-in. See table XIII for the plug-ins used. Use items PI02 and PI03 in the differential mode.

6. For all test conditions, record the total peak-to-peak ripple and noise observed on the oscilloscope with the frequency settings per table XIII (including all spikes, hash, periodic and aperiodic components). Use the single sweep mode. The sweep rate shall be varied to obtain a maximum reading. The intensity must be turned up adequately to catch all the spikes. The scope trigger should be varied to assure observation of both positive and negative peaks.

7. For test condition N2, N4, N6, N8, N10, N12, N14 and N16, short output pins a to k (direct sense-1 to direct sense-2).

8. For test conditions N7 and N8, measurement shall be made with item D08 and item PI01 (bandwidth set to 20 MHz). The item Y07 shall be used so a differential measurement can be made.

9. For test conditions N9 and N10, the ripple measurements are made with the current protection circuit energized. The current on the output being measured shall be increased until one of the following things shall happen:

(a) to the point where the current saturates or begins to drop (in the case of current limiting or foldback),

(b) just prior to the point where the power supply shuts down (if the power shuts down on overcurrent),

FIGURE 28. Ripple test circuit for table XIII - Continued.

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NOTES: - Continued.

(c) until the output voltage decreases to 95 percent of the value recorded at the 20 amperes level.

The ripple measurements shall be made at the condition which occurs first. In no case shall the current be increased beyond 25 amperes.

10. For test conditions N11 and N12, adjust item EL01 to obtain 0 amperes load current.

FIGURE 28. Ripple test circuit for table XIII - Continued.

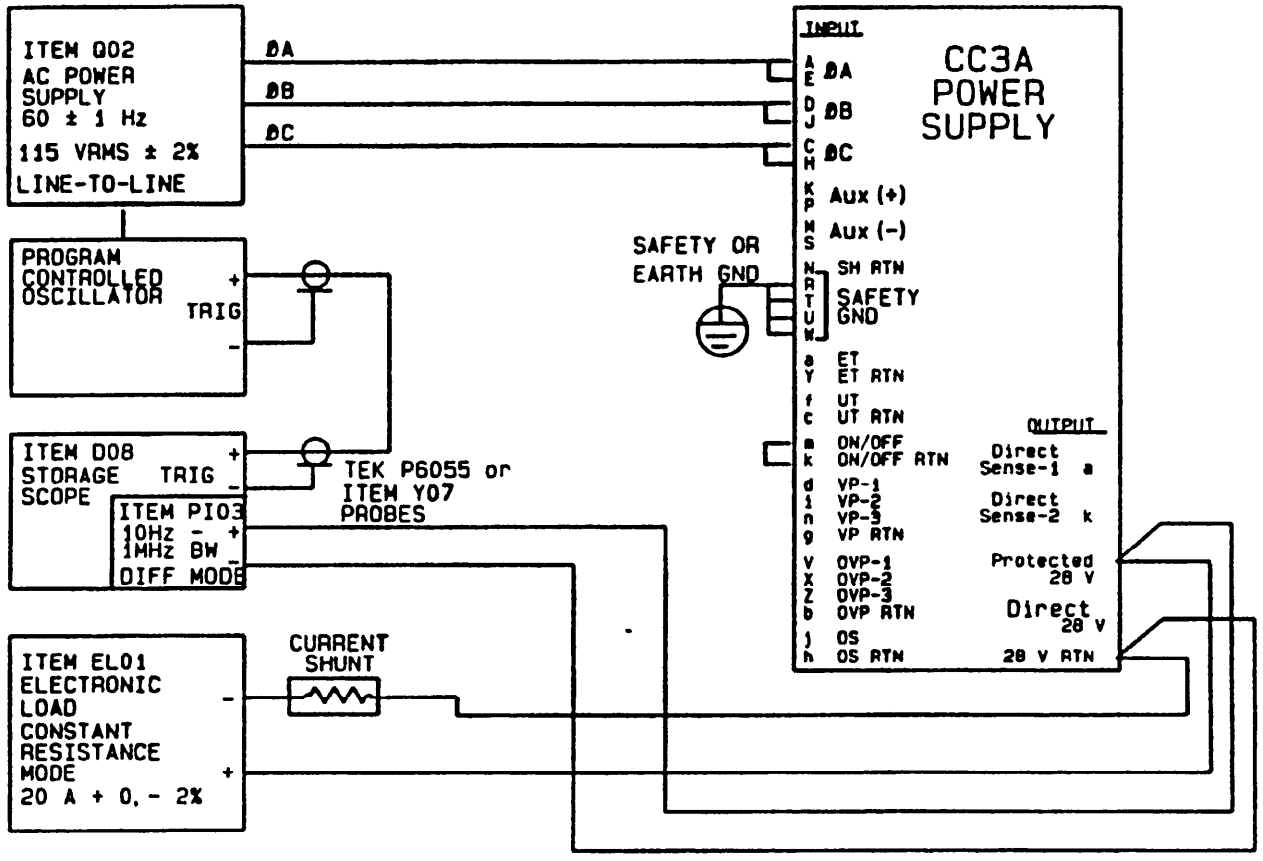
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TABLE XIV. Dynamic line regulation tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperture		Unit	Test fig
		Min	Max		
P1	Dynamic line regulation, +21%	0	840	mV	29,30
P2	Dynamic line ringing, +21% (2nd half cycle) 1/		420		
P3	Dynamic line regulation, -21%		840		
P4	Dynamic line ringing, -21% (2nd half cycle) 1/	0	420	mV	29,30

1/ +21 percent - from 115 VRMS to 139 VRMS: -21 percent - from 115 VRMS to 91 VRMS. See figure 30 for definitions.
2/ Any additional overshoot or ringing beyond the 2nd half cycle shall be within the ripple envelope of test condition N1 peak-to-peak limits.

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NOTES:

1. See table II for output connector contact assignment.

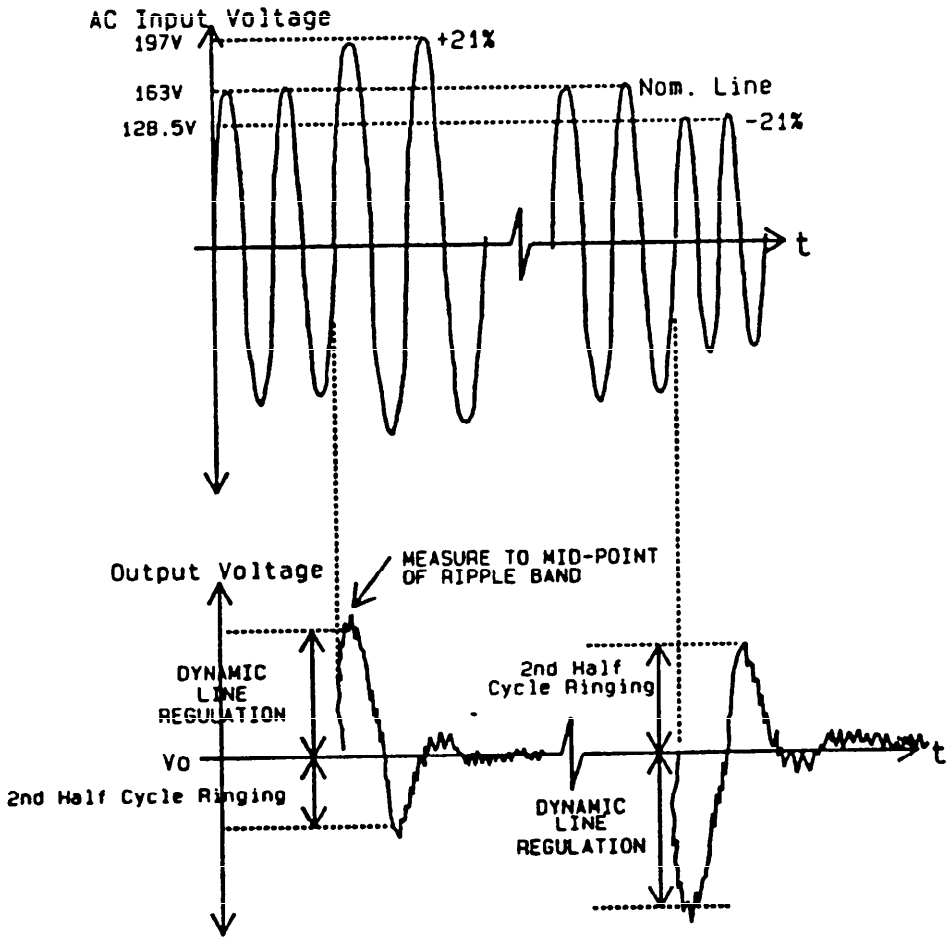
2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

FIGURE 29. Dynamic line regulation test circuit for table XIV.

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NOTES:

1. The initial overshoot may be positive or negative.

2. Transition time for the AC amplitude shall be between 50 and 500 us.

3. The AC input amplitude shall remain at each amplitude for a minimum of 2 seconds.

4. Nominal voltage is 115 VRMS (163 volts peak (Vpk)), + 21 percent = 139 VRMS (197 Vpk), - 21 percent = 91 VRMS (128.5 Vpk).

FIGURE 30. Dynamic line regulation test definitions for table XIV.

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TABLE XV. Dynamic load regulation tests.

Test cond	Parameter	Initial limits 25 ± 5C mounting surface temperature		Unit	Test fig
		Min	Max		
Q1	Dynamic load regulation, 80% to 90%	0	840	mV	31, 32
Q2	Dynamic load ringing, 80% to 90% 1/		420		
Q3	Dynamic load regulation, 90% to 80%		840		
Q4	Dynamic load ringing, 90% to 80% 1/		420		
Q5	Dynamic load regulation, full capacitance load, 80% to 90%		840		
Q6	Dynamic load ringing, full capacitance load, 80% to 90% 1/		420		
Q7	Dynamic load regulation, full capacitance load, 90% to 80%		840		
Q8	Dynamic load ringing, full capacitance load, 90% to 80% 1/		420	mV	
Q9	Dynamic load regulation, 5% to 95%		10	V	
Q10	Dynamic load ringing, 5% to 95%		5		
Q11	Dynamic load regulation, 95% to 5%		5		
Q12	Dynamic load ringing, 95% to 5%	0	5	V	31, 32

1/ Additional ringing beyond the 2nd half cycle shall be within the regulation limits of test condition C2 and the peak-to-peak noise envelope of test condition N1.

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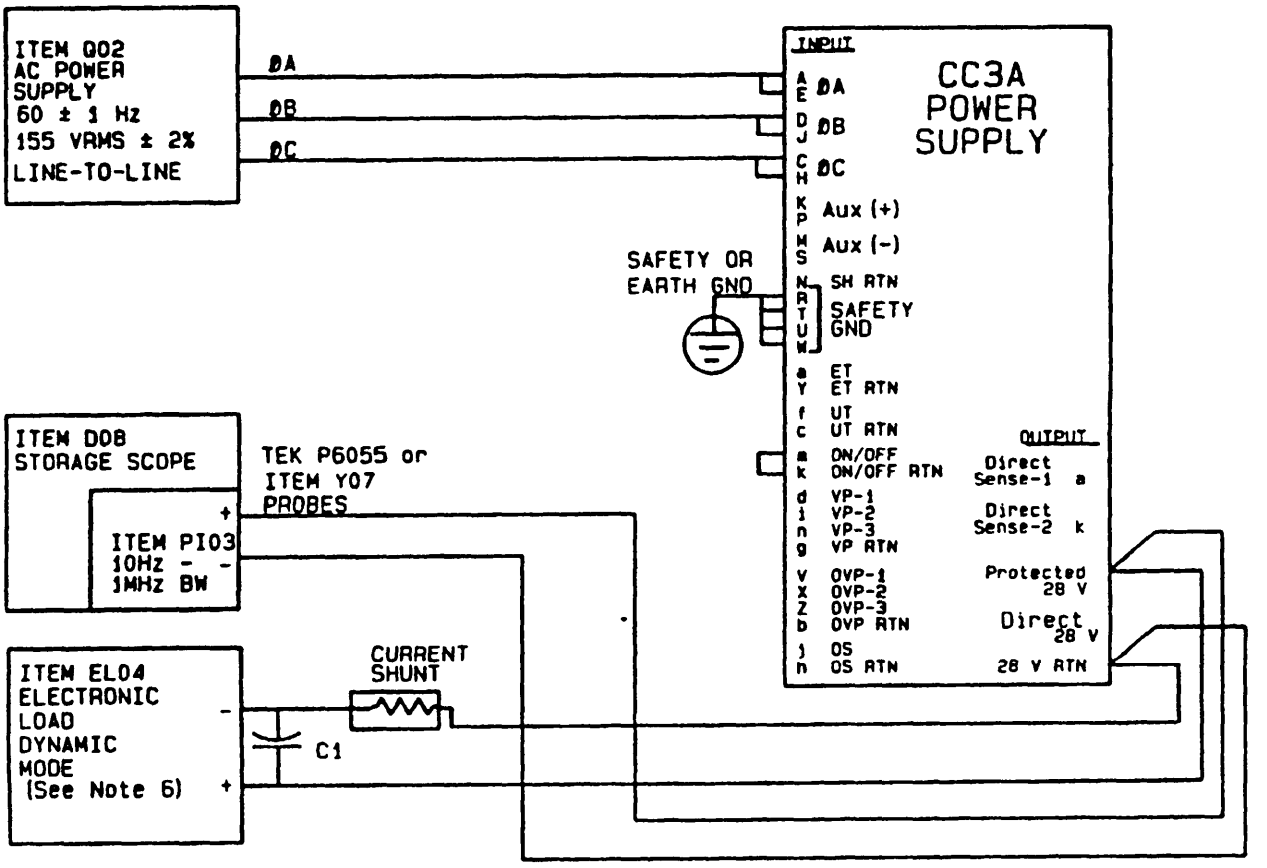


FIGURE 31. Dynamic load regulation test circuit for table XV.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

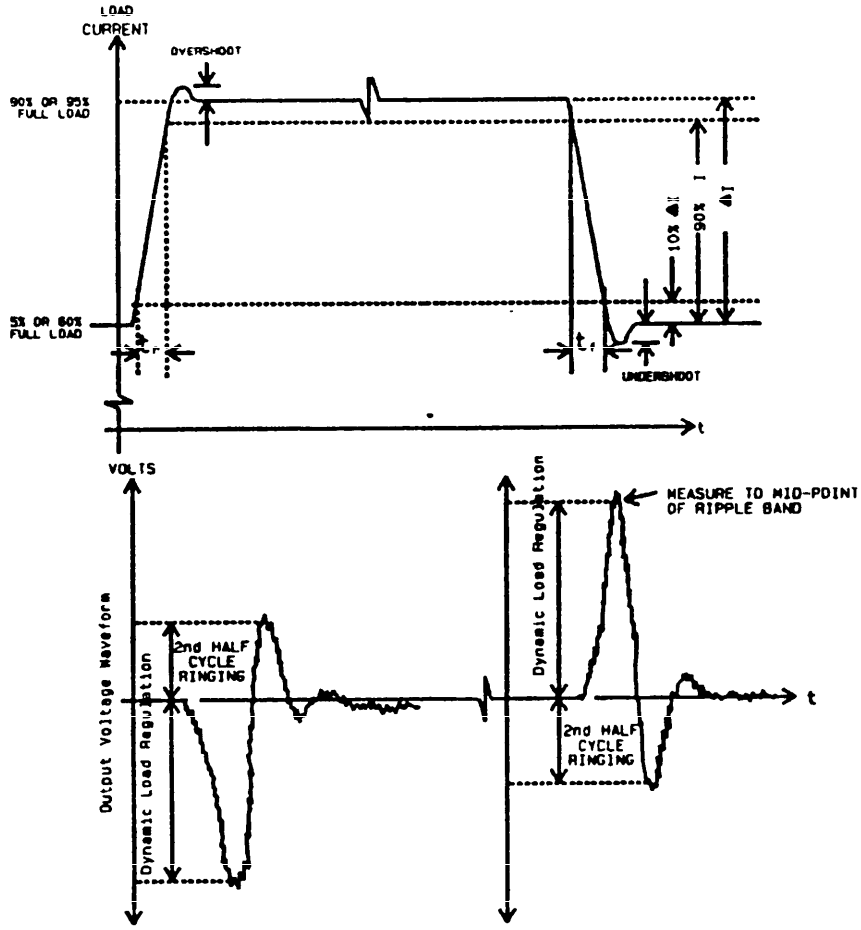
5. C1 = 7,200 uF, + 20, - 0 percent, 50 VDC capacitor. Used in test setup for test conditions Q5-Q8 only.

6. Item EL04 electronic load shall be in the dynamic mode, switching between 18 amperes and 16 amperes for test conditions Q1-Q8. It shall be in the dynamic mode, switching between 19 amperes and 1 ampere for test conditions Q9-Q12.

7. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

FIGURE 31. Dynamic load regulation test circuit for table XV - Continued.

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NOTES:

1. Rise time, tr, and fall time, tf, shall both be 60 us maximum.

2. The dynamic load may be a single pulse or repetitive plus or minus transitions with each state being maintained for no less than 1 ms.

3. The load current overshoot or undershoot shall be minimized to the greatest level possible.

4. Measurement of the dynamic load regulation and 2nd half cycle ringing shall be to the mid-point of any ripple occurring on the output voltage waveform.

FIGURE 32. Dynamic load regulation and ringing test definitions for table XV.

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TABLE XVI. Turn-on into capacitive load test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
R	Turn-on time into capacitive load	0	250	ms	33

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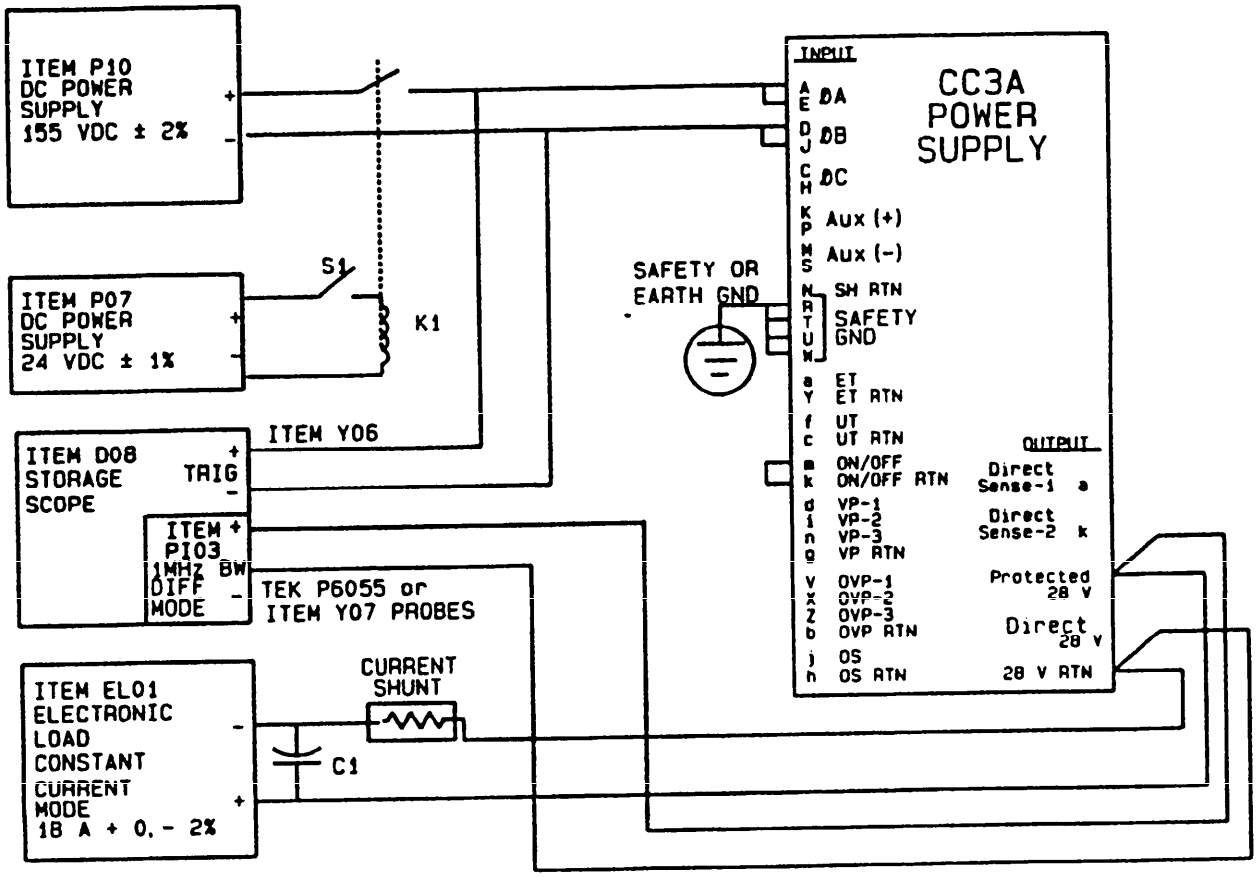


FIGURE 33. Turn-on time into capacitive load test circuit for table XVI.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. C1 = 7,200 uF, + 20, - 0 percent, 50 VDC capacitor.

6. S1 = spst switch.

7. K1 = 3pst mercury relay, Magnecraft WM35AAA-24D or equivalent.

8. At the option of the testing facility, an AC input (instead of the DC input specified) of 3-phase, 115 VRMS ± 2 percent, line-to-line, 60 ± 1 Hz voltage can be applied at the phase A, B and C inputs.

9. Prior to connecting the capacitor C1 across the output, adjust item EL01 for 20 amperes with the power supply energized. Remove the input power by opening S1 and connect capacitor C1 across the output. Close S1 and verify that the turn-on time meets the limits in table XVI.

10. Turn-on time is the time from when power is applied to the input until the output recovers to and remains within the regulation limits. The regulation limits shall be test condition C1 limits.

11. Item EL01 shall act as a constant current load while the 28V output is rising. Therefore, above a power supply output terminal voltage of 2.0 volts, the load current shall be rising to its set value at a rate two times, or greater, than the rate of the voltage.

FIGURE 33. Turn-on time into capacitive load test circuit for table XVI - Continued.

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TABLE XVII. Undervoltage and overvoltages output status tests.

Test cond	Parameter	VP contacts connected together	OVP contacts connected together	Initial limits 25 ± 5°C mounting surface temperture		Unit	Test fig
				Min	Max		
S1	Undervoltage-output status LED off	-	-	24.36	26.60	VDC	34
S2	Overvoltage-output status LED off, 28.0V 1/	-	-	29.68	31.72		
S3	Overvoltage-output status LED off, 28.5V 1/	d,g	v,b	30.21	32.29		
S4	Overvoltage-output status LED off, 29.5V 1/	d,i,g	v,x,b	31.27	33.42		
S5	Overvoltage-output status LED off, 30.5V 1/	d,i,n,g	v,x,z,b	32.33	34.22		
S6	Status LED check	----------Status LED on----------					
T1	Output signal, logic low (5mA)	-	-	0.0	0.4		
T2	Output status signal, logic high			28.0	30.5		
T3	Output status signal, undervoltage			24.36	26.60		
T4	Output status signal, overvoltage, 28.0V 1/	-	-	28.68	31.72		
T5	Output status signal, overvoltage, 28.5 1/	d,g	v,b	30.21	32.29		
T6	Output status signal, overvoltage, 29.5V 1/	d,i,g	v,x,b	31.27	33.42		
T7	Output status signal, overvoltage, 30.5 1/	d,i,n,g	v,x,z,b	32.33	34.22	VDC	
T8	Output status signal leakage current	-	-	0	4.4	uADC	34
T9	Output status signal setting time (high to low)	-	-	0	2.0	ms	35
T10	Output status signal setting time (low to high)	-	-	0	2.0	ms	35
T11	Output status signal logic low (0.5mA)	-	-	0	0.4	VDC	35

- See footnotes on the next page.

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1/ Test conditions S2-S5 and T4-T7 verify that the output status LED and output status signal detect an overvoltage condition. Additionally, the power supply vendor will verify by 100 percent (X) testing, that the output voltage is limited as specified.

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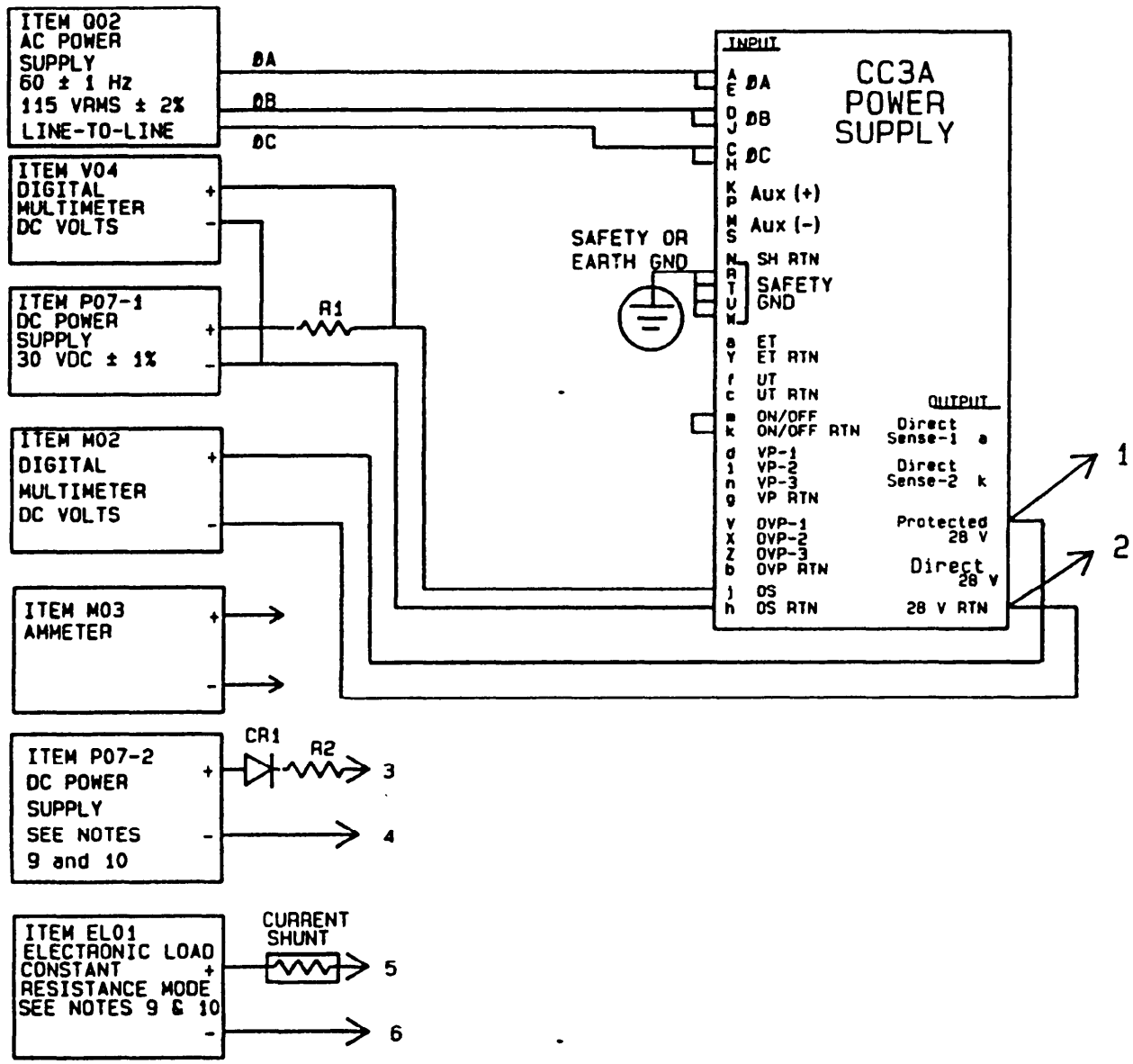


FIGURE 34. Undervoltage and overvoltage output status test circuit for table XVII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 5.9 kOhm ± 1 percent, 1/8 watt resistor.

6. R2 = 28 Ohm ± 1 percent, 28 watt resistor.

7. CR1 = 1N5616 or equivalent.

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

9. The output voltage can be reduced for undervoltage testing in one of three ways (the method to use will depend on the design of the CC3A power supply being tested):

(a) Method 1. Connect item P07-2 plus terminal to input contact d and minus terminal to input contact g (CR1 and R2 shall not be used). Beginning with item P07-2 voltage at 0 volts and item P10 ON, gradually increase the item P07-2 voltage until the output voltage is 28.0 ± 0.28 volts. Further increase of the item P07-2 voltage will cause the output voltage to drop as required. Item E101 is not required.

(b) Method 2. Prior to connecting an external power supply to the output, reduce the input voltage (as applied with item P10) until the CC3A output voltage drops to less than 24.36 volts, but not low enough for the internal housekeeping supply to stop working. Verify that the output status LED is OFF and that the output status signal is high before proceeding with the test. Connect point 1 to point 3 and point 2 to point 4. Adjust item P07-2 until the item M02 voltage is 28.0 ± 0.28 volts prior to performing each test. This external supply is used to decrease the output voltage for testing. Item EL01 is not required.

FIGURE 34. Undervoltage and overvoltage output status test circuit for table XVII - Continued.

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NOTES: - Continued.

(c) Method 3. Connect point 1 to point 5 and point 2 to point 6. For each test, begin slowly increasing the load current from 20 amperes and the output voltage shall gradually drop as required for testing. Do not exceed 26 amperes load current. Items P07-2, CR1 and R2 are not required.

10. The output voltage is increased for overvoltage testing by connecting point 1 to point 3 and point 2 to point 4. Adjust item P07-2 to 28 ± 0.28 volts prior to performing each test. This external supply is used to increase the output voltage for testing. Item EL01 is not required.

11. Perform test conditions S6 and T1 before performing the undervoltage and overvoltage tests. For S6 and T1, connect point 1 to point 5 and point 2 to point 6. Adjust the electronic load to a current of 20 + 0, -0.4 amperes. Using item M02, ensure that the output is within regulation limits for the applicable temperature (test condition C2). For test condition S6, verify that the output status LED is on. For test condition T1, the item V04 reading shall be within the table XVII limits.

12. For test condition S1, slowly decrease the output voltage from nominal 28 volts until the output status LED goes off. The item M02 voltage at which that occurs shall meet the table XVII limits.

13. For test conditions S2-S5, increase the output voltage from nominal 28.0 volts until the output status LED turns off. The item M02 voltage at which that occurs shall be within table XVII limits. Do not raise the voltage at points 1 and 2 (as measured with item M02) above 36 volts.

14. Perform test conditions T2 and T3 as follows:

(a) For test condition T2, decrease the output voltage from nominal 28.0 volts until the output status signal, as monitored by item V04, goes high (29.0 to 30.5 VDC). This item V04 voltage shall meet the table XVII limits.

(b) For test condition T3, decrease the output voltage from nominal 28.0 volts until the item V04 voltage goes high (within the test condition T2 limits). The item M02 voltage when this occurs shall meet the table XVII limits.

FIGURE 34. Undervoltage and overvoltage output status test circuit for table XVII - Continued.

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NOTES: - Continued.

15. For test conditions T4-T7, increase the output voltage from nominal 28.0 volts until the output status signal goes high (item V04 voltage within the test condition V2 limits). The item M02 voltage at which that occurs shall meet the table XVII limits. Do not raise the voltage at points 1 and 2 (as measured with item M02) above 36 volts.

16. For test condition T8, connect the item M03 ammeter between resistor R1 and item P07-1. Decrease the output voltage (item M02) for nominal 28 volts until the item V04 voltage goes high (within test condition S2 limits). The item M03 current reading when this happens shall meet the table XVII limits. Disconnect item V04 while making the current reading.

17. The output may not automatically recover when the overvoltage is removed. However, it shall recover when the input voltage is taken to zero and then reapplied.

FIGURE 34. Undervoltage and overvoltage output status test circuit for table XVII - Continued.

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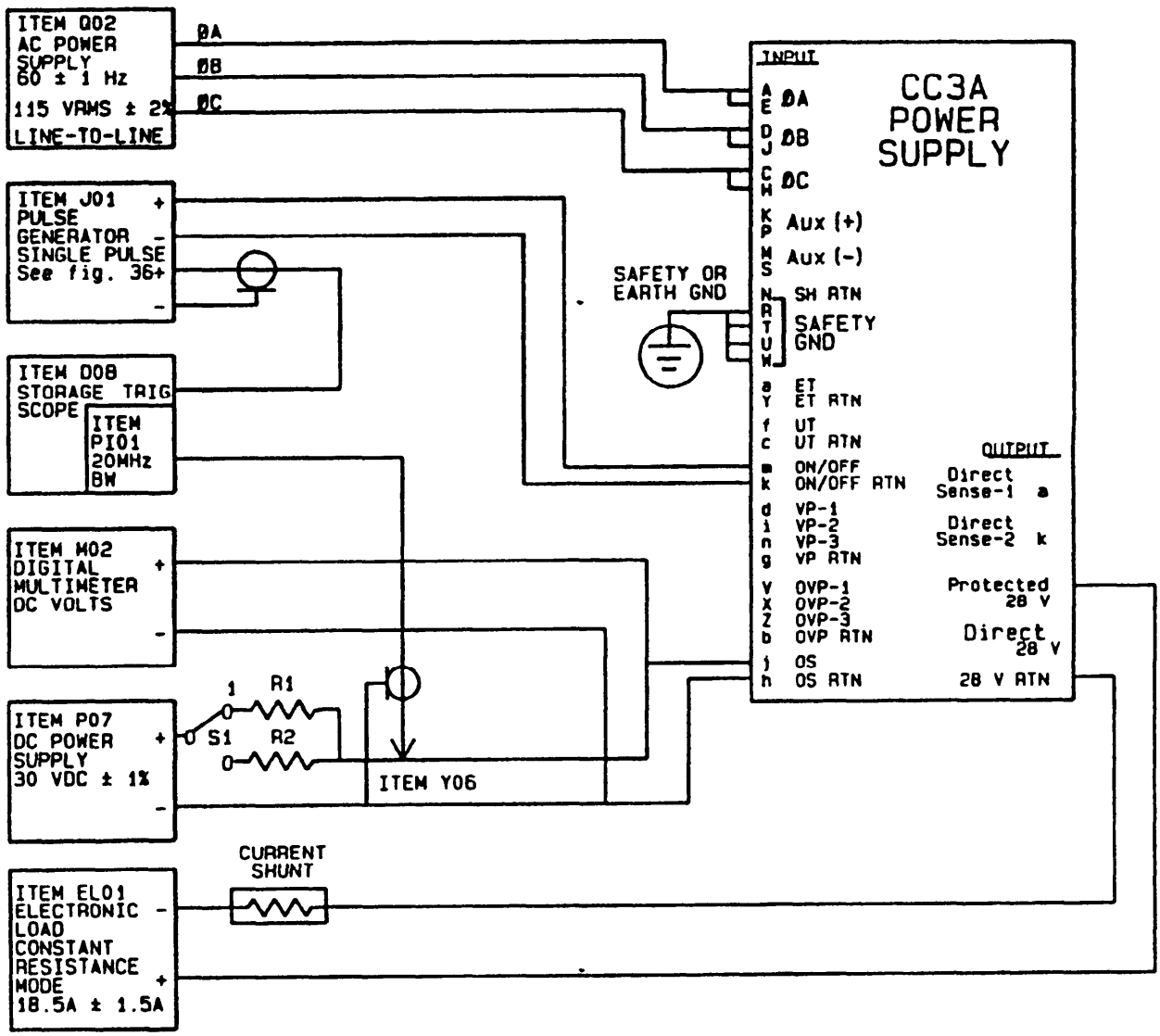


FIGURE 35. Output status signal settling time and logic level test circuit for table XVII.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 5.9 kOhm ± 1 percent, 1/4 watt resistor for test conditions T9 and T10.

6. R2 = 60.4 kOhm ± 1 percent, 1/10 watt resistor for test condition T11 (set S1 to 2).

7. S1 = spdt switch.

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

9. For test conditions T9 and T10, set up the power supply input voltage and load current with item J01 disconnected and input contacts m and k shorted (power supply on). Disconnect the short after setup. Set switch S1 to position 1.

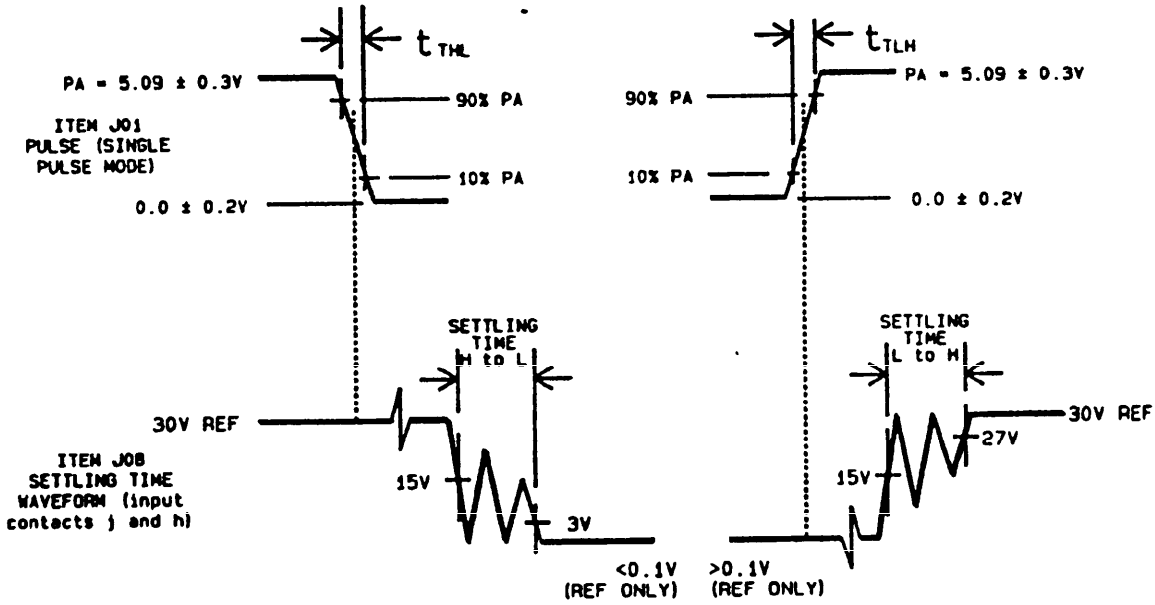
(a) For test condition T9, set up item J01 for a single high to low level pulse in accordance with figure 36. Connect item J01 as shown and trigger item J01 for a single pulse. Using the storage mode of item D08 oscilloscope, measure the settling time (see figure 36).

(b) For test condition T10, repeat (a) above, except for a single low to high level pulse from item J01.

10. For test condition T11, set switch S1 to position 2, disconnect item D08 and item J01 and short input contact m to k. Connect OS RTN (input contact h) to SAFETY GND (input contact R). Using item M02, measure the voltage across input contacts j and h.

FIGURE 35. Output status signal settling time and low level resistance test circuit for table XVII - Continued.

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NOTES:

1. Item J01 pulse parameters:
   $t_{THL} = t_{TLH} = 10 \pm 5$ us.

FIGURE 36. Output status settling time test circuit waveforms for table XVII.

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TABLE XVIII. Excessive and unsafe temperture indicatin tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperture		Unit	Test fig
		Min	Max		
U1	Excessive temperature indication (5mA current)	0	0.4	VDC	37
U2	Excessive temperature indiction (500mA current)	0	0.4		
U3	Unsafe temperature indication (5mA current)	0	0.4		
U4	Unsafe temperature indication (500mA current)	0	0.4	VDC	
U5	Unsafe-excessive differential (5mA current), 1/	1.0	25	°C	
U6	Indicatar reset (5mA current)	60	-	°C	
U7	Unsafe-excessive differential (500mA current)	1.0	25	°C	
U8	Indicator reset (500mA current)	60	-	°C	37

1/ Test conditions U5 and U7 are calculations. Subtract the trip paint temperature measured in test condition U1 (U2) from the trip point temperature measured in test condition U3 (U4).

2/ Test conditions U2 and U4 verify that the closed contact resistance is a maximum of 0.8 Ohms with a contact current of 500 milliamperes (mA). Test conditions U1 and U3 verify that the closed contact resistance is a maximum of 80 Ohms with a contact current of 5 mA.

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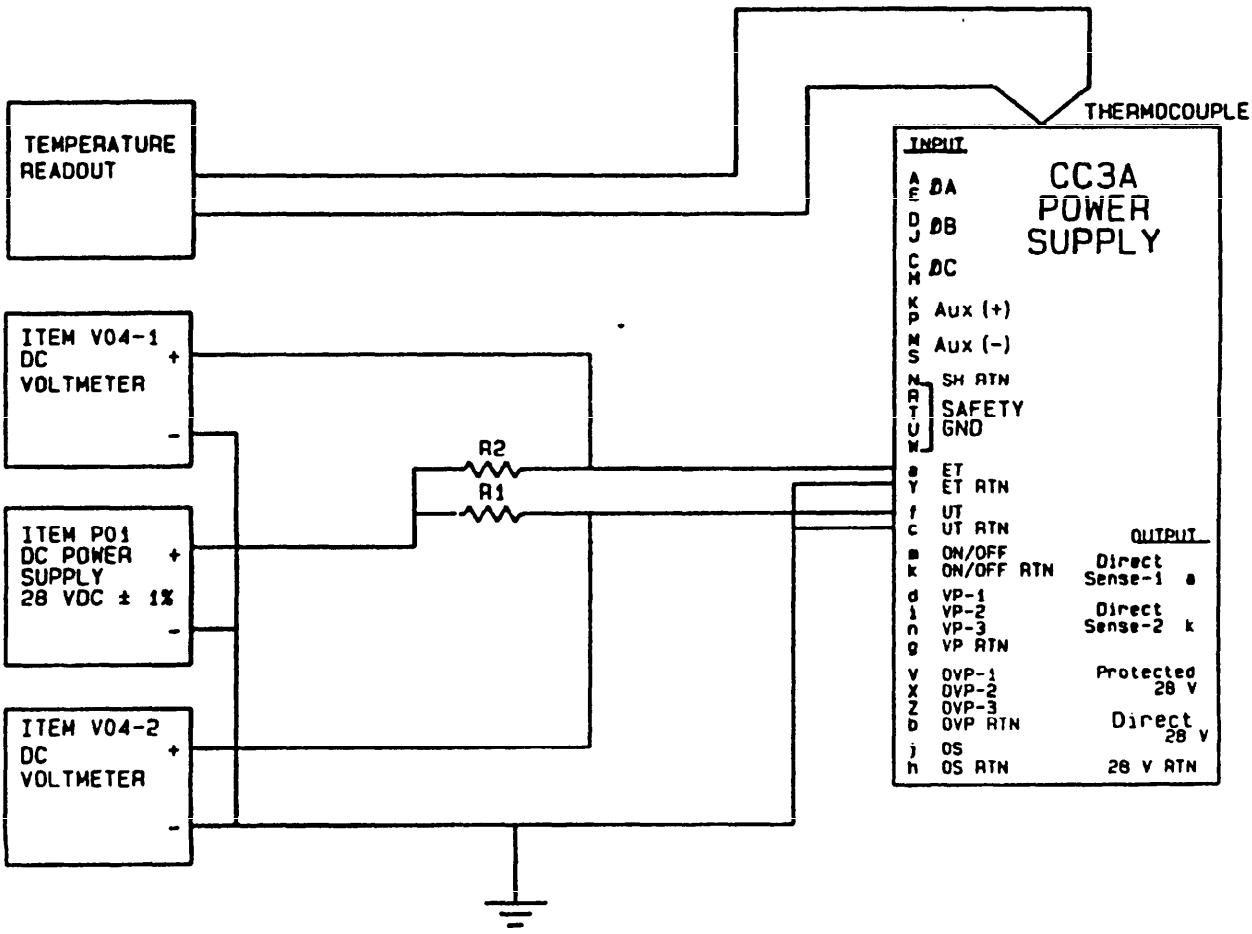


FIGURE 37. Excessive and unsafe temperature indication test circuit for table XVIII.

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NOTES:

1. Perform test conditions U1-U8 only after all other electrical and environmental tests have been performed.

2. R1 = R2 = 56 Ohm ± 1 percent, 15 watt resistor for test conditions U2, U4 and U8. R1 = R2 = 5.62 kOhm ± 1 percent, 1/4 watt resistor for test conditions U1, U3 and U6.

3. The power supply shall be unenergized during this test.

4. The temperature shall be monitored with a thermocouple connected to the power supply frame. Do not attach the thermocouple to the mounting surface.

5. Test conditions U1, U3, U5 and U6 shall be performed as follows and shall be performed prior to performing test conditions U2, U4, U7 and U8:

(a) Place the power supply in an oven and raise the temperature to 65 ± 2°C at a rate not to exceed 1°C per minute. Let the power supply soak for a minimum of 1 hour at this temperature. Verify that items V04-1 and V04-2 both read 28.0 ± 0.5 volts (thermal switches open).

(b) Begin increasing the oven temperature at a rate not to exceed 1°C per 15 minutes, until V04-2 voltage switches to a low level (excessive temperature thermal switch closed). This low level voltage shall meet test condition U1 limits. Record the power supply frame temperature at which the thermal switch closed for use in test condition U5 calculation. Do not exceed 140°C.

(c) Continue increasing the oven temperature, as in (b) above, until item V04-1 voltage switches to a low level (unsafe temperature thermal switch closed). This low level voltage shall meet test condition U3 limits. Record the power supply frame temperature at which the thermal switch closed for use in test condition U5 calculation. Do not exceed 140°C.

FIGURE 37. Excessive and unsafe temperature indication test circuit for table XVIII - Continued.

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NOTES: - Continued.

(d) Begin decreasing the oven temperature at a rate not to exceed 1°C per 15 minutes, until V04-1 and V04-2 both read 28.0 ± 0.5 volts (thermal switches open). The power supply frame temperature at which both thermal switches open shall meet the limits of test condition U6.

6. Repeat note 5 procedure for test conditions U2, U4, U7 and U8.

FIGURE 37. Excessive and unsafe temperature indication test circuit for table XVIII - Continued.

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TABLE XIX. Output voltage hold-up time test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
V	Output voltage hold-up time, 108 VDC input	50	-	us	38

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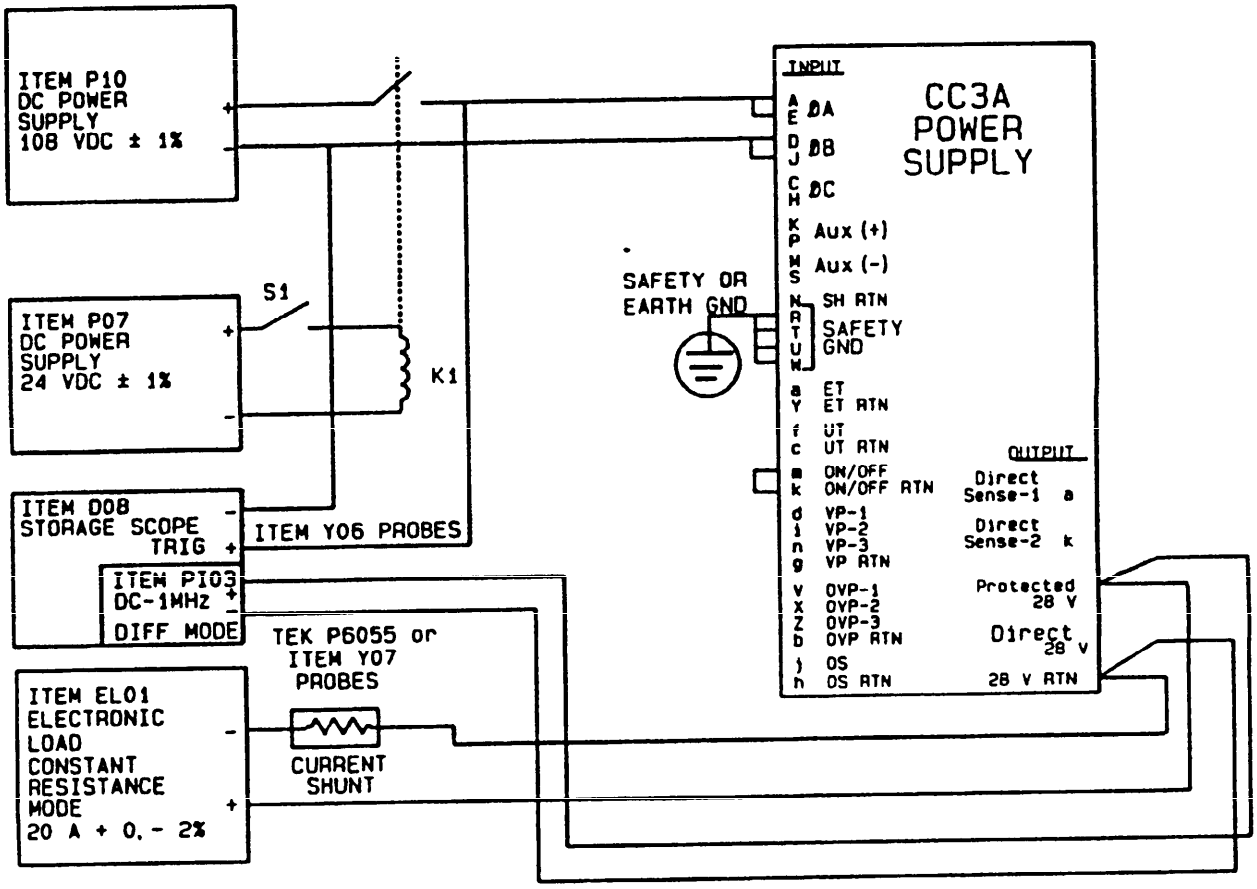


FIGURE 38. Output voltage hold-up time test circuit for table XIX.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

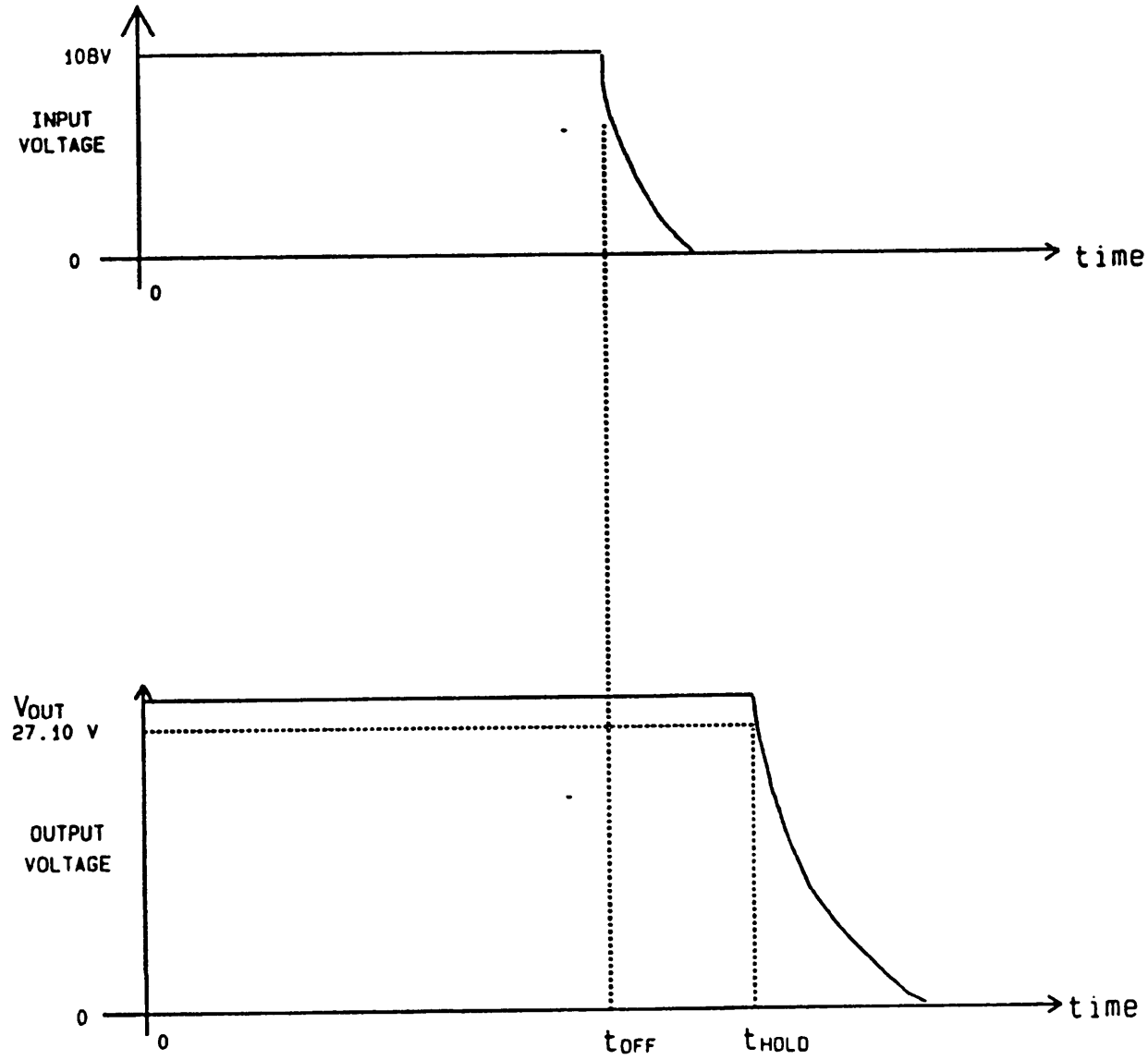
5. S1 = spst switch.

6. K1 = spst mercury relay, Magnecraft WM35A-24D or equivalent.

7. Connect the circuit as shown with switch S1 closed (input power applied). Open switch S1 (disconnecting input power) and measure the hold-up time as defined on figure 39.

FIGURE 38. Output voltage hold-up time test circuit for table XIX - Continued.

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NOTE:

1. Hold-up time is defined as $t_{HOLD} - t_{OFF}$.

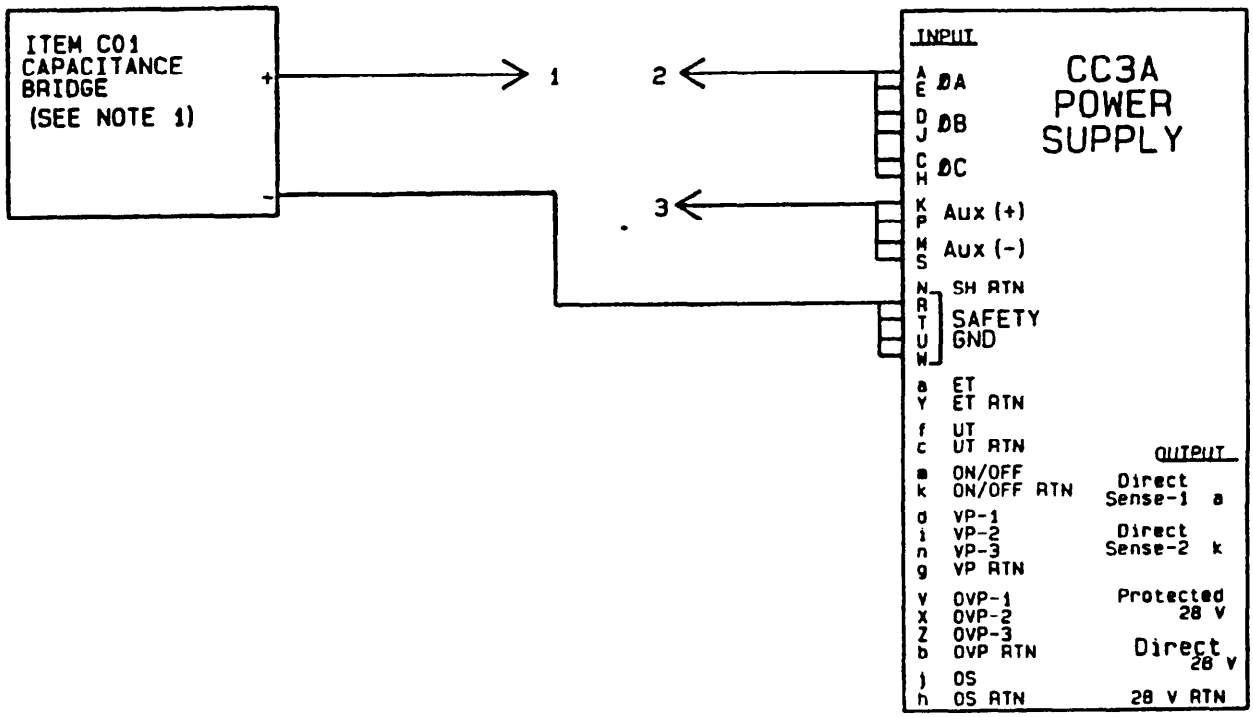
FIGURE 39. Output voltage hold-up time definition for table XIX.

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TABLE XX. Input to chassis capacitance tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
W1	AC input to chassis capacitance	0.0	0.03	uF	40
W2	Auxiliary input to chassis capacitance	0.0	0.05	uF	40

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NOTES:

1. Item C01 set-up: frequency = 1 kHz ± 5 percent, voltage = 0.3 volts peak maximum.

2. For test condition W1, connect point 1 to point 2. The item C01 capacitance reading shall meet the test condition W1 limits.

3. For test condition W2, connect point 1 to point 3. The item C01 capacitance reading shall meet the test condition W2 limits.

FIGURE 40. Input to chassis capacitance test circuit for table XX.

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TABLE XXI. Output impedance test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
X	Output impedance 1/	0	52.2	mV	41

1/ This test verifies a maximum output impedance of 300 mOhm in accordance with the relationships on figure 41, note 7(d). The mV limits in this table apply only while power supply load resistance is within the figure 41 specified value at all frequencies.

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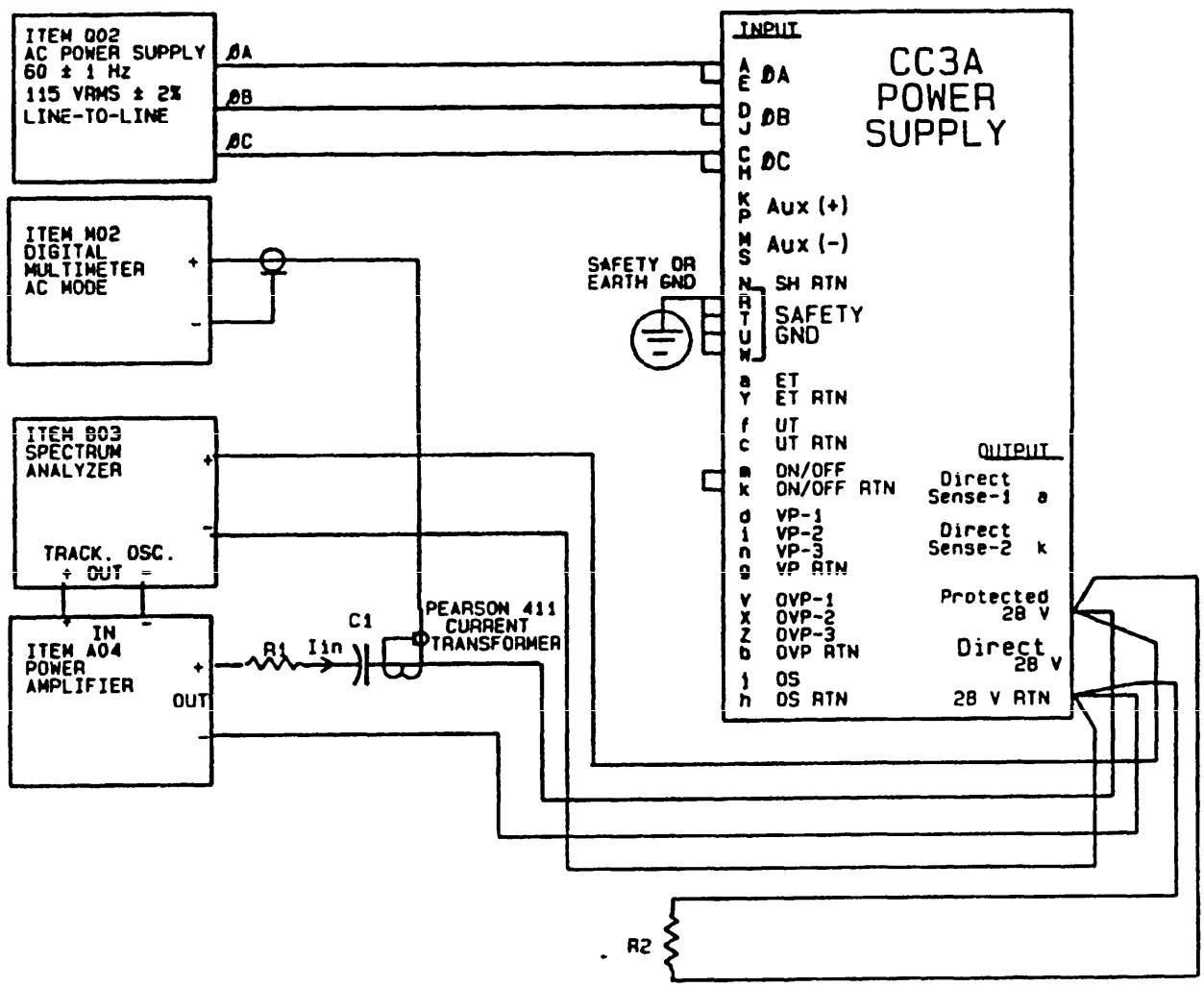


FIGURE 41. Output impedance test circuit for table XXI.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. R1 = 10 Ohm ± 5 percent, non-inductive, 0.5 watt resistor.

4. R2 = 2 Ohm ± 5 percent, 500 watt resistor, non-inductive.

5. C1 = 6000 uF + 75 percent, -10 percent, 25 volts capacitor.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. The output impedance test shall be performed as follows:

(a) Make the required connections for test condition X, with the item A04 amplitude turned down to 0 volts and apply input power to the power supply. The DC load current on the power supply will be 14 amperes ± 10 percent for test condition X.

(b) AC current adjustment: Set the spectrum analyzer to the manual mode and at a fixed frequency between 100 Hz and 1 kHz. Use the model 411 current transformer and item M02 to monitor the AC current (conversion factor = 0.1 volts per ampere). Use the item B03 tracking oscillator level control and the item A04 amplitude control to increase the AC current into the power supply output to 200 milliamperes (mA) ± 3 percent.

(c) Set the item B03 spectrum analyzer to the sweep mode and find maximum Vo (as measured by item B03) over the frequency range of 30 Hz to 25 kHz. Sweep the frequency over a minimum of two bands:

30 Hz to 50 Hz, resolution bandwidth = 3 Hz.
50 Hz to 5 kHz, resolution bandwidth = 10 Hz.
5 kHz to 25 kHz, resolution bandwidth = 30 Hz.

The AC current shall be maintained at a constant amplitude of 200 mA over the sweep range. The frequency of maximum Vo is the frequency of maximum output impedance. Read the maximum value displayed on item B03. This shall be within the limits in table XXI. Record the frequency of maximum Vo.

FIGURE 41. Output impedance test circuit for table XXI - Continued.

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NOTES: - Continued.

(d) Calculate the maximum output impedance from the maximum Vo;

$$Zo = \frac{\text{item B03 voltage}}{Iin - \frac{\text{item B03 voltage}}{R2}} \cdot = \frac{\text{item B03 voltage}}{0.2 \text{ amperes} - \frac{\text{item B03 voltage}}{R2}}$$

Where R2 includes all impedance (may be frequency dependent) from directly at the power supply positive output terminal, through the current shunt, through the resistor and to the negative terminal.

FIGURE 41. Output impedance test circuit for table XXI - Continued.

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TABLE XXII. Spike voltage protection tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
Y1	Spike voltage protection, AC input, initial transient (during first 20 us)	-1.40	1.40	V	42, 43
Y2	Spike voltage protection, AC input, secondary transient	-0.70	0.70		
Y3	Spike voltage protection, auxiliary input, initial transient (during first 20 us)	-1.40	1.40		
Y4	Spike voltage protection, auxiliary input, secondary transient	-0.70	0.70	V	42, 43

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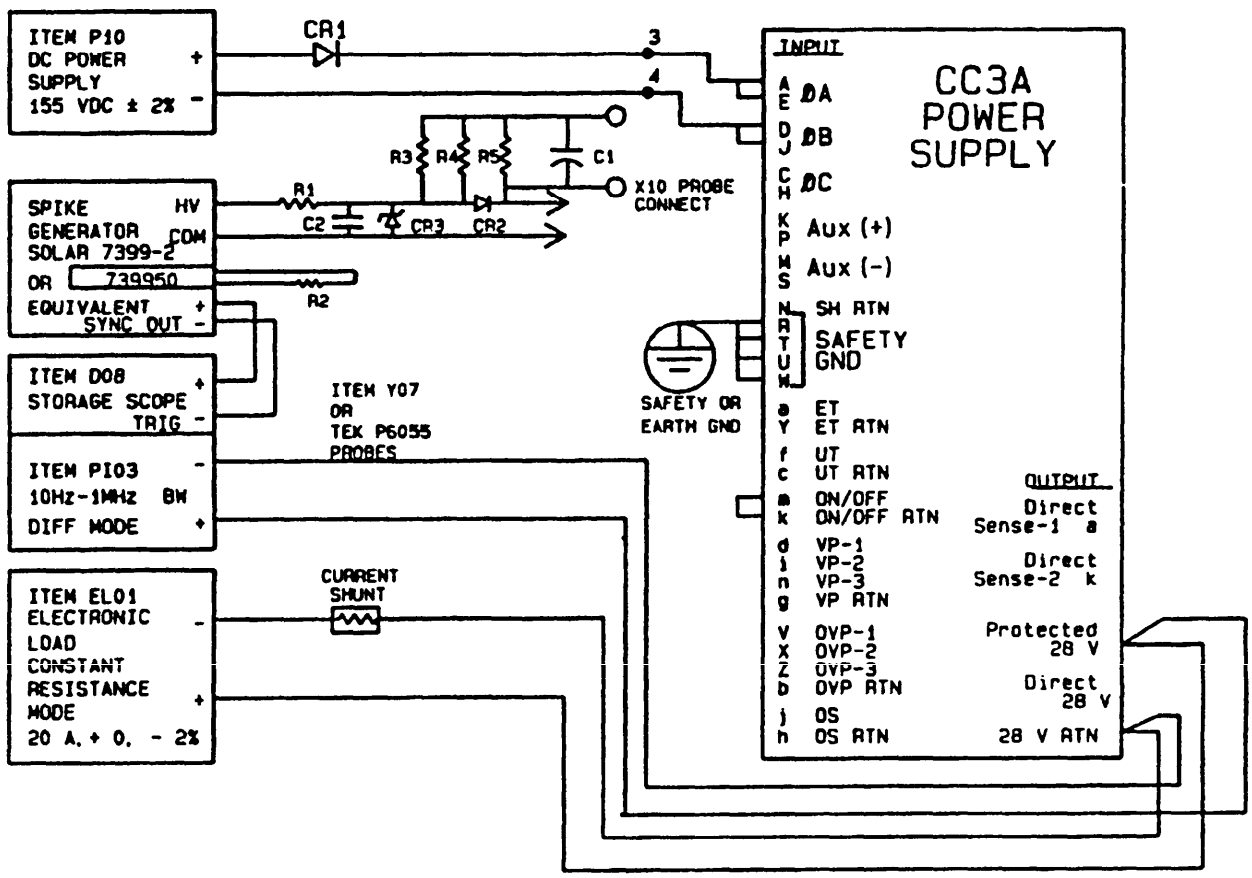


FIGURE 42. Spike voltage protection test circuit for table XXII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. CR1 = CR2 = International Rectifier, 12FL100S05 or equivalent.

6. CR3 = Transorb, General Semiconductor PIP120 or equivalent.

7. R1 = 7 Ohm ± 1 percent, 10 watt resistor.

8. R2 = 0.5 Ohm ± 5 percent, 250 watt resistor.

9. R3 = R4 = 180 kOhm ± 5 percent, 1 watt carbon composition resistor. R5 = 10 kOhm ± 5 percent, 1/2 watt carbon composition resistor. C1 = 27 picofarad (pF) ± 10 percent, 200 volts capacitor.

Select the resistors so:

$$\frac{R3 \;//\; R4}{R5} = \frac{90\ \text{kOhm}}{10\ \text{kOhm}} = 9.0 \pm 1\ \text{percent}$$

A X100 probe may be used directly in lieu of these components.

10. C2 = Selected (start at approximately 1 uF), 400 volts (see note 12(b)).

11. WARNING: High voltage is present during this test. Exercise caution and observe the procedures given in the spike generator instruction book. The spike generator shall be in the single pulse mode. Allow a minimum of 30 seconds between spikes while adjusting the waveform.

FIGURE 42. Spike voltage protection test circuit for table XXII - Continued.

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NOTES: - Continued.

12. The input voltage waveform shown on figure 43 shall be obtained at points 1 and 2 (as measured with an oscilloscope connected to the X10 PROBE CONNECT points) prior to connection to the power supply under test.

(a) Adjust the spike generator to charge its internal capacitors to approximately 1500 volts prior to discharging to produce the spike.

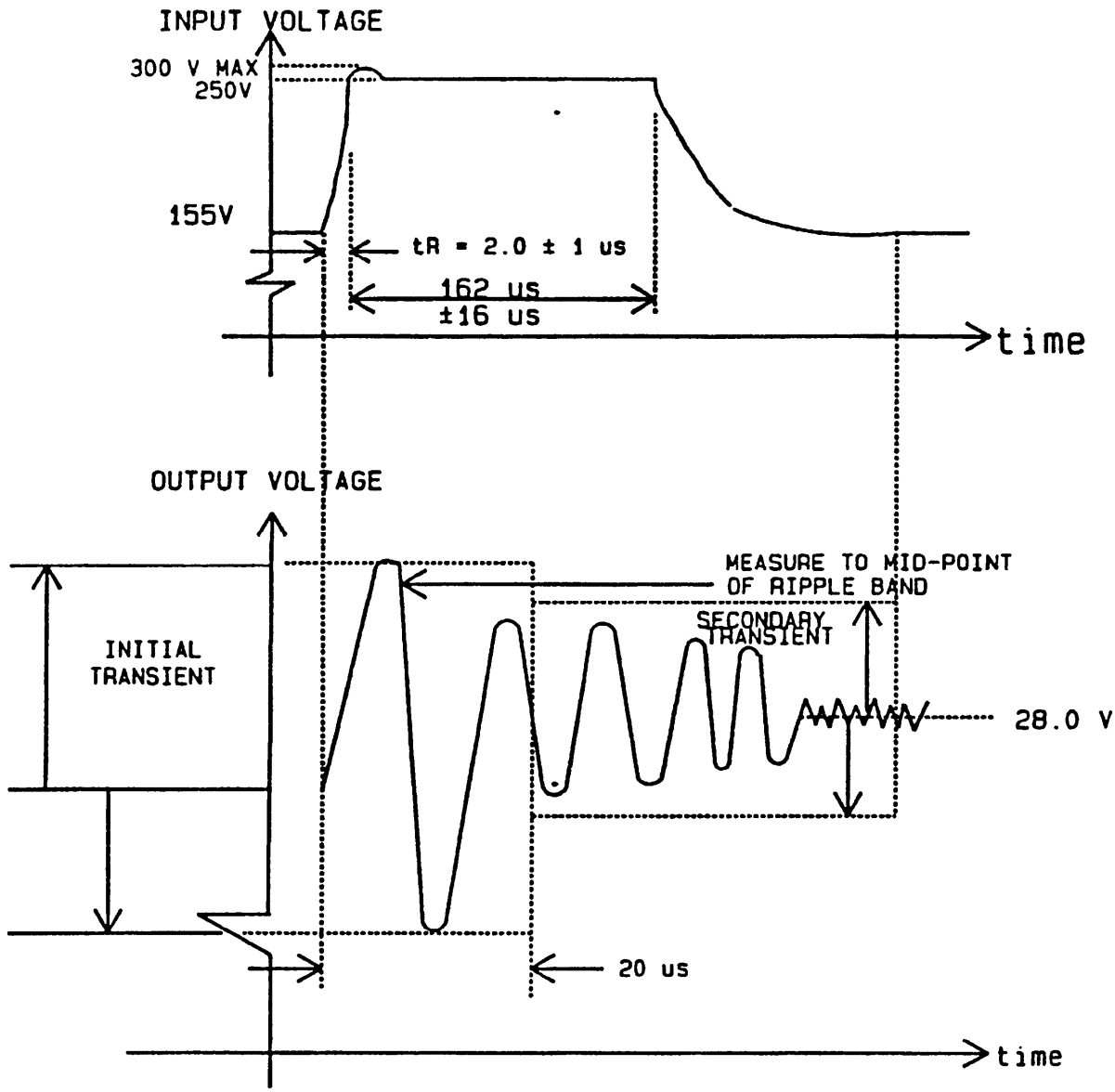
(b) Trigger the spike generator and observe that the input voltage waveform as shown on figure 43 (except the waveform will start at 0 volts and decay to 0 volts) is obtained. Select C2 to give the specified rise time. Increasing or decreasing resistor R2 will increase or decrease the pulse width. Selection of CR3 (or paralleling of more than one Transorb) will set the 250 volts peak level.

13. Perform test conditions Y1 and Y2 (after the input waveform has been adjusted according to note 12) as follows: Connect point 1 to point 3 and point 2 to point 4. Apply a single voltage spike to the power supply under test and observe that the output waveform meets the test limits. Do not reapply the input spike voltage to the power supply at intervals of less than one minute (the spike voltage and current limiting resistor R1 have been scaled down.

14. Perform test conditions Y3 and Y4 (after the input waveform has been adjusted according to note 12) as follows: Connect the circuit as shown except point 3 shall be connected to input contacts K and P and point 4 shall be connected to input contacts M and S. Repeat the procedure of note 13.

FIGURE 42. Spike voltage protection test circuit for table XXII - Continued.

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NOTES:

1. The initial transient shall be defined as the highest output transient (plus or minus) occurring during the first 20 uS after application of the input spike voltage.

2. Secondary transient shall be defined as the highest output transient (plus or minus) occurring between 20 us after the spike application and the end of the input spike.

FIGURE 43. Spike voltage definition for table XXII.

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TABLE XXIII. Reverse voltage protection tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
Z	Reverse voltage protection, auxiliary input	27.72	28.28	VDC	44
AA1	Reverse voltage protection, output at 50 ADC	0	1.6	VDC	45
AA2	Reverse voltage protection, output at 200 Apk	27.72	28.28	VDC	46

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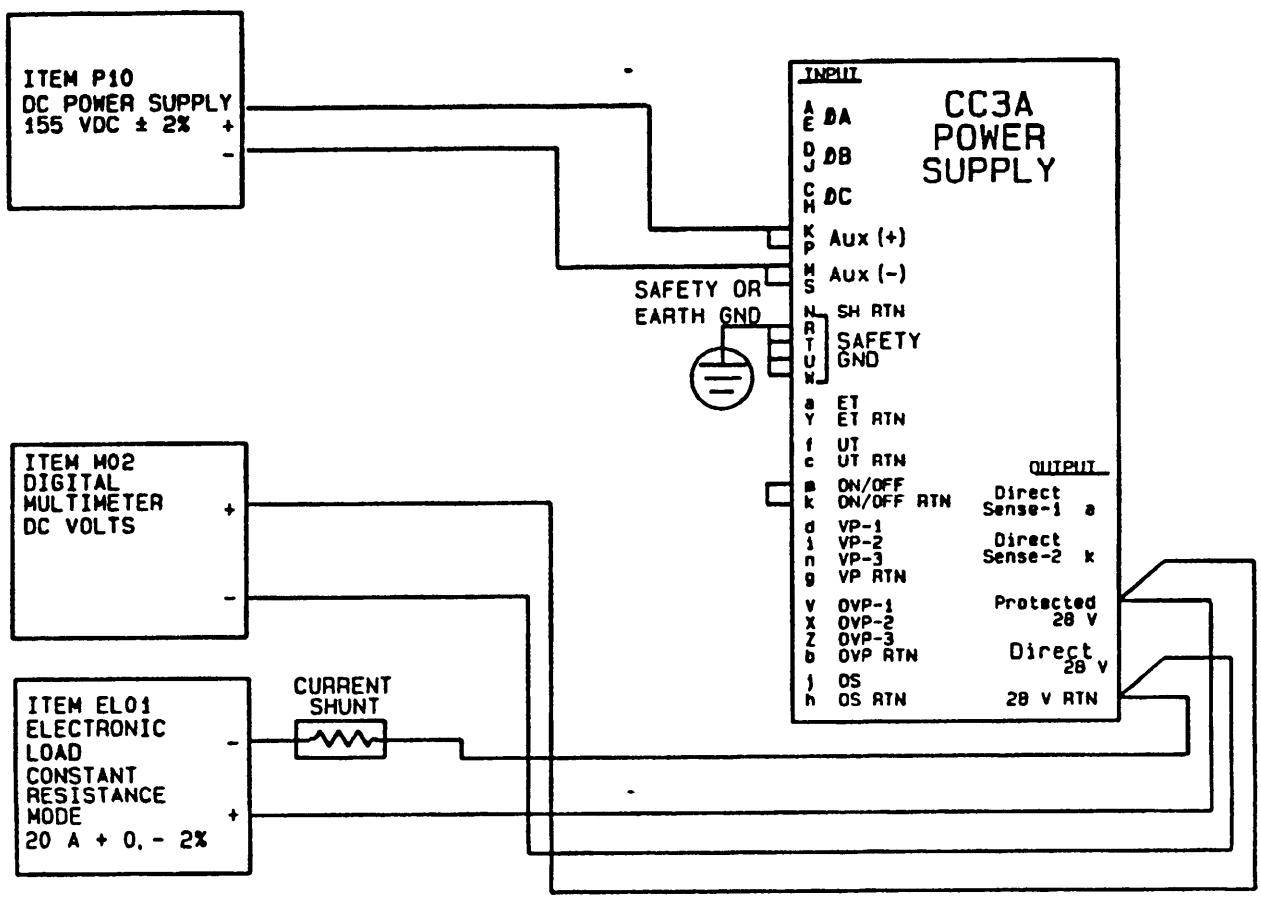


FIGURE 44. Reverse voltage protection on auxiliary input test circuit for table XXIII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. For test condition Z, connect as shown and verify that the item M02 multimeter reading 28 ± 0.28 volts. Reverse the input polarity connection for a minimum of 5 seconds. Then reconnect the circuit as shown. The voltage on the item M02 multimeter shall meet table XXIII limits. Item P10 shall be turned off when reversing the input leads and then turned back on.

FIGURE 44. Reverse voltage protection on auxiliary input test circuit for table XXIII - Continued.

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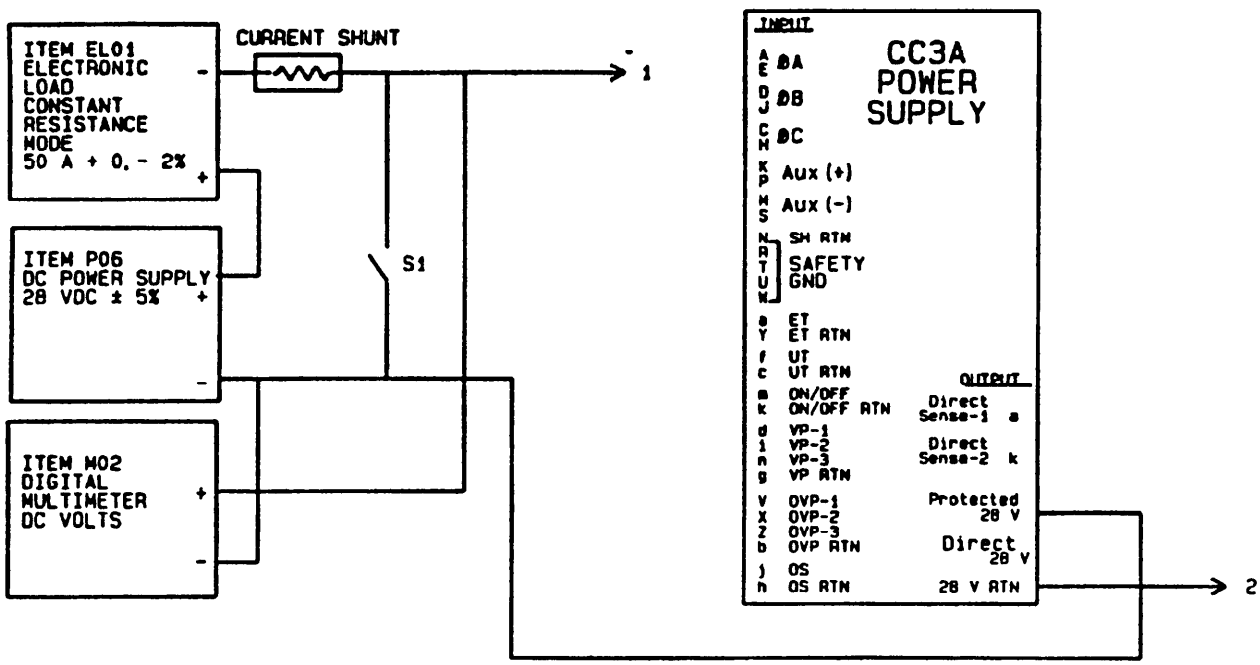


FIGURE 45. Reverse voltage protection (50 ADC) test circuit for table XXIII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst, 50 ADC switch.

6. Before connecting point 1 to point 2, close switch S1 and set the current through the current shunt at 50 ampere by varying item EL01 load. Then connect point 1 to point 2 and open switch S1. Readjust the current to 50 amperes ± 1 percent. Item M02 voltage shall be within the limits in table XXIII.

FIGURE 45. Reverse voltage protection (50 ADC) test circuit for table XXIII - Continued.

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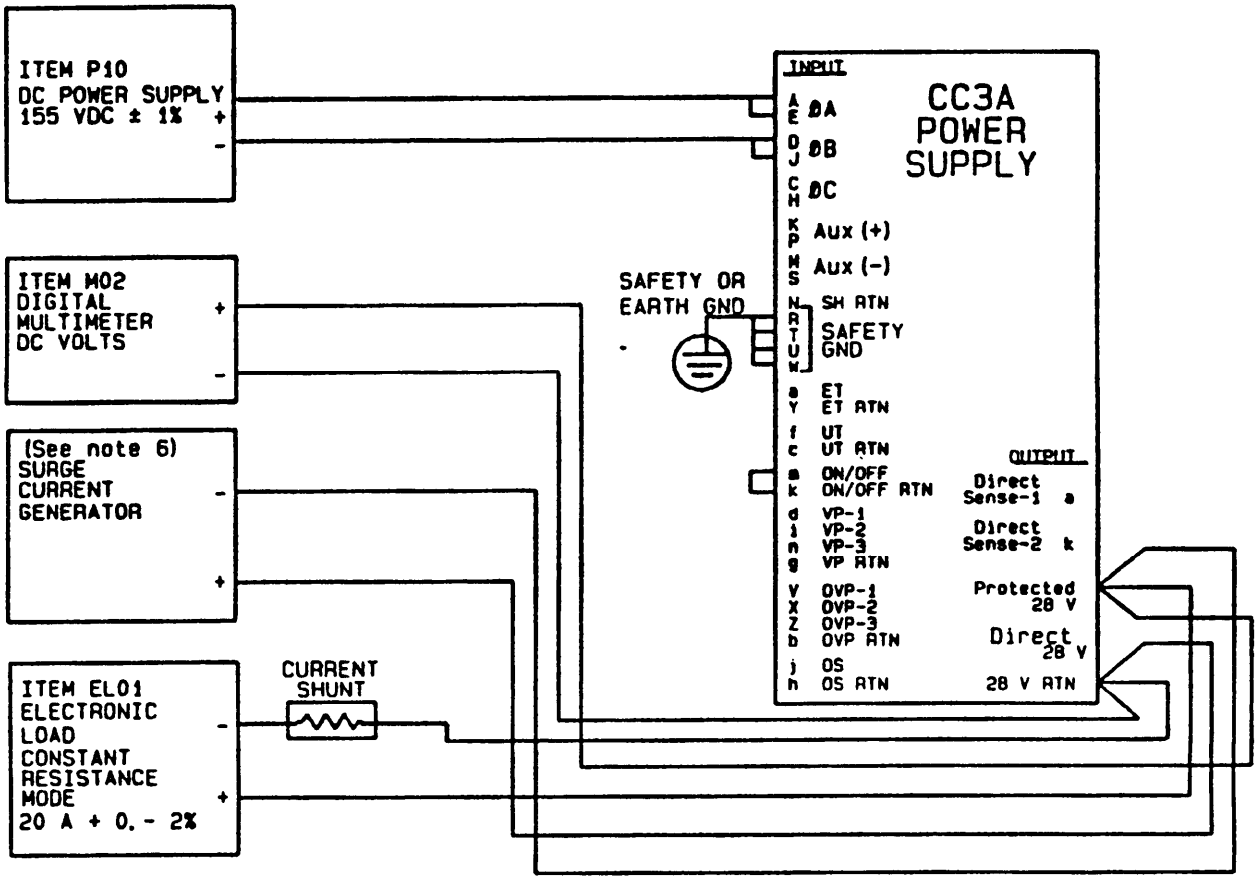


FIGURE 46. Reverse voltage protection (200Apk) test circuit for table XXIII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. At the option of the testing facility, an AC input (instead of the DC input specified) of 3-phase, 115 VRMS ± 2 percent line-to-line, 60 ± 1 Hz can be applied at the phase A, B and C inputs.

6. The surge current generator shall be capable of supplying a peak current of 200 amperes. The generator will supply 6 pulses of surge current at a 60 Hz rate.

7. Test condition AA2 is performed as follows:

(a) Connect only the surge current generator, and generate the pulses described above.

(b) Disconnect the surge current generator and connect the rest of the circuit as shown. The power supply shall operate in regulation and the item M02 voltage shall meet the limits in table XXIII.

FIGURE 46. Reverse voltage protection (200 Apk) test circuit for table XXIII - Continued.

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TABLE XXIV. Forward voltage protection and parallel output tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
BB1	Forward voltage protection on output at 300 V	0	20	mADC	47
BB2	ON/OFF parallel outputs, pff, direct	27.72	28.28	VDC	
BB3	ON/OFF parallel outputs, OS high	2.4	5.5	VDC	
BB4	ON/OFF parallel outputs, OS LED off	Output status LED to off			
BB5	ON/ON parallel outputs protected	28.91	30.09	VDC	
BB6	ON/ON parallel outputs, protected, OS low	0.0	0.4	VDC	
BB7	Positive pre-bias test	27.44	28.56	VDC	47

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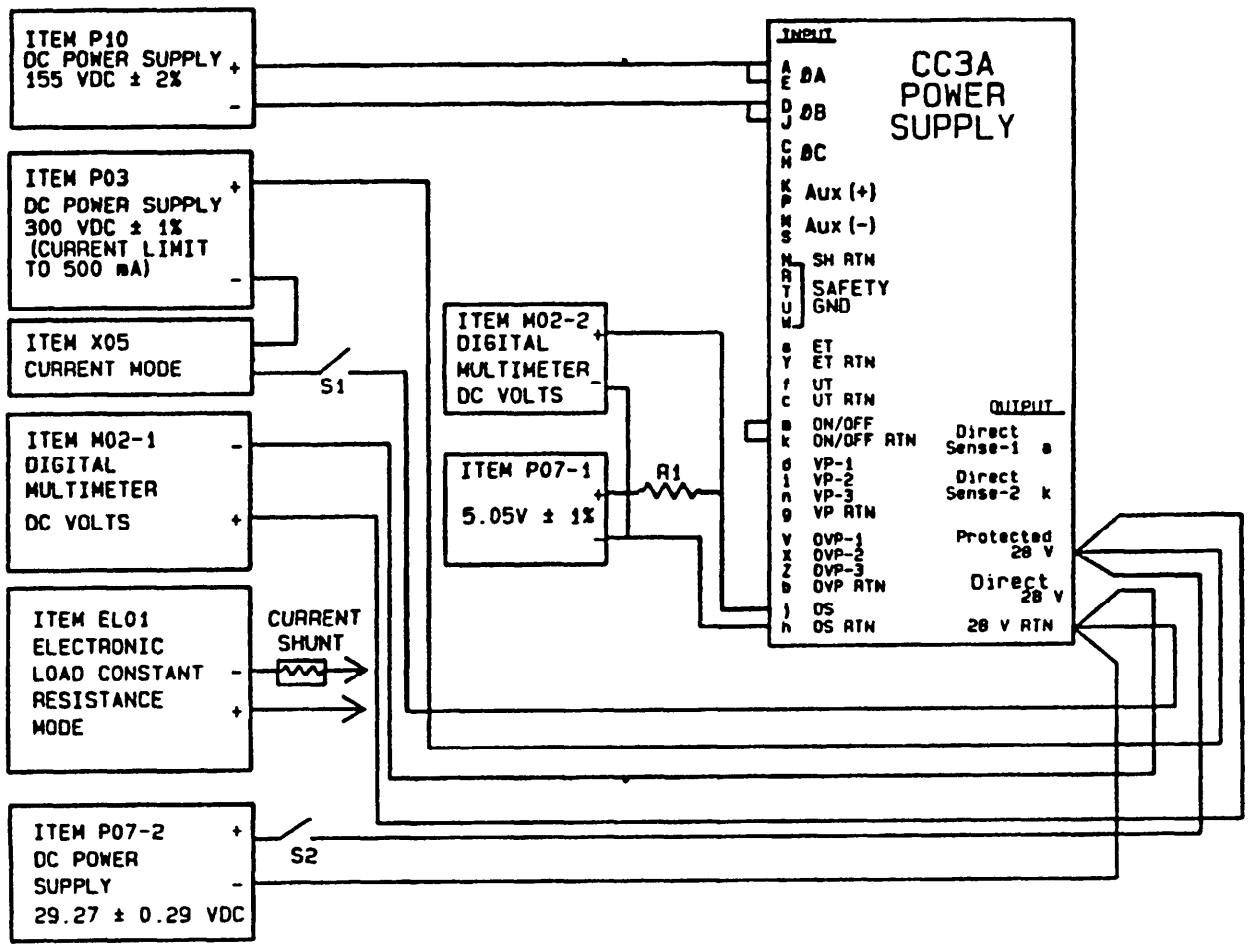


FIGURE 47. Forward voltage protection and parallel output test circuit for table XXIV.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. R1 = 1.69 kOhm ± 1 percent, 1/8 watt resistor.

5. S1 = S2 = spst switch.

6. At the option of the testing facility, an AC input (instead of the DC input specified) of 3-phase, 115 VRMS ± 2 percent line- to-line, 60 ± 1 Hz can be applied at the phase A, B and C inputs.

7. For test condition BB1, connect the circuit as shown except leave item P10 disconnected. Close S1 (for a minimum of 5 seconds) and read the item X05 current. That current shall fall within the limits in table XXIV. For test condition BB2, open input contacts m and k, and set item P03 voltage to 30.5 ± 0.05 volts. Close switch S1 for a minimum of five seconds, then open switch. Repeat four times. On the fourth closing of switch S1, item M02-2 voltage shall be within test condition BB3 limits and the output status LED shall be OFF per test condition BB4. Then open switch S1 and short input contacts m and k together. Item M02-1 voltage shall be within test condition BB2 limits.

8. For test condition BB5, close S1, short input contacts d to i to n to g, and V to X to Z to b. Open input contacts m & k. Set item P03 voltage to 30.5 ± 0.05 volts, and short input contacts m and k. Item M02-2 voltage shall be within test condition BB6 limits. Then remove item P03 from the circuit by opening S1. Item M02-1 voltage shall be within test condition BB5 limits.

9. For test condition BB7, connect as shown. Begin with item P10 turned off. Close switch S2. Turn on item P10. Open input contacts m and k (remote ON/OFF contacts) for a minimum of 5 seconds. Short input contacts m and k. Open switch S2. Item M02-1 voltage shall meet test condition BB7 limits.

FIGURE 47. Forward voltage protection and parallel output test circuit for table XXIV - Continued.

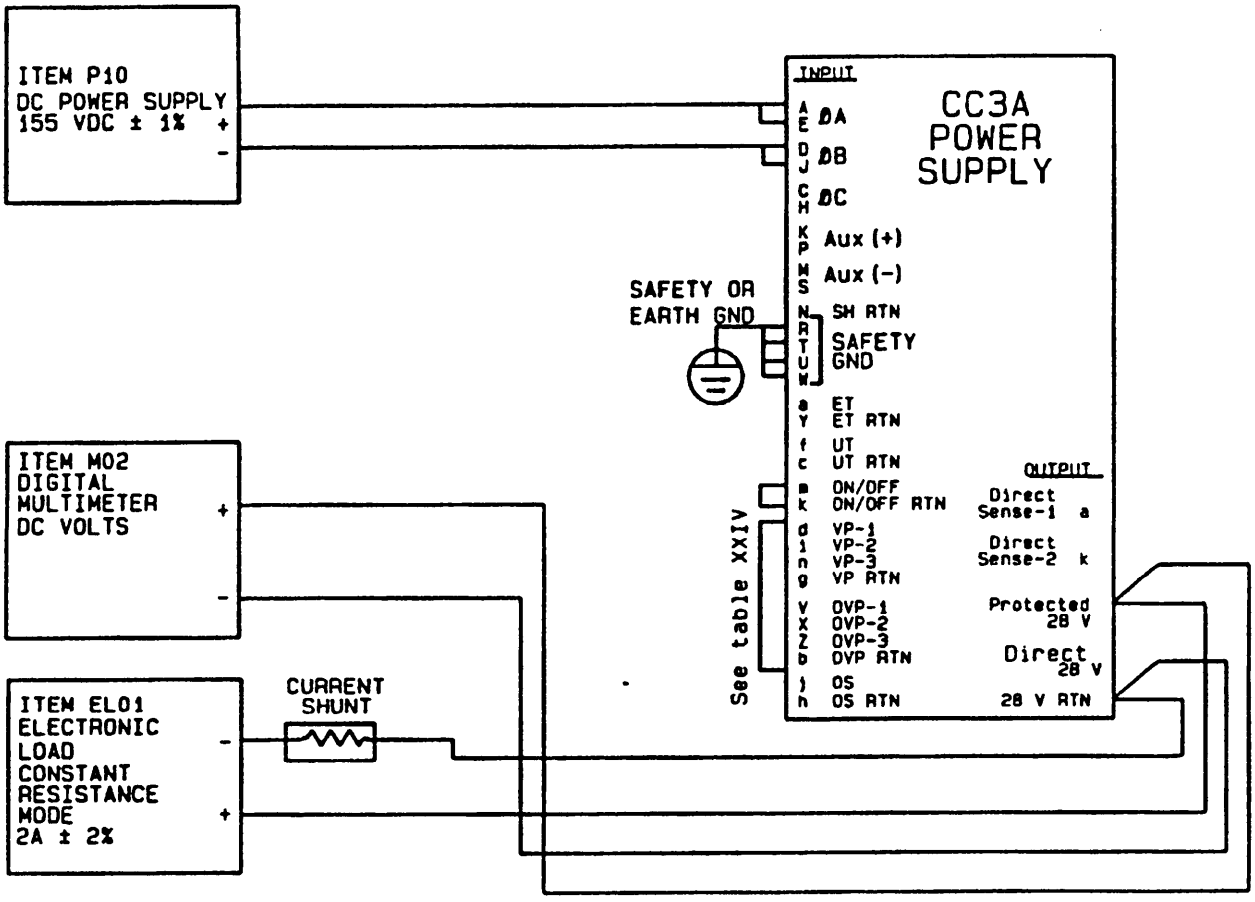
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TABLE XXV. Output voltage tests.

Test cond	Parameter	VP contacts connected together	OVP contacts connected together	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
				Min	Max		
CC1	Output voltage, 2A, 28.5V	d,g	V,b	28.21	28.79	VDC	48
CC2	Output voltage, 2A, 29.5V	d,i,g	V,X,b	29.20	29.80	VDC	48

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connector contact assignment; ... table XX for output



NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

FIGURE 48. Output voltage test circuit for table XXV.

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TABLE XXVI. Open sense load test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
DD	Open sense load voltage, direct output, 28 V	26.60	29.40	VDC	49

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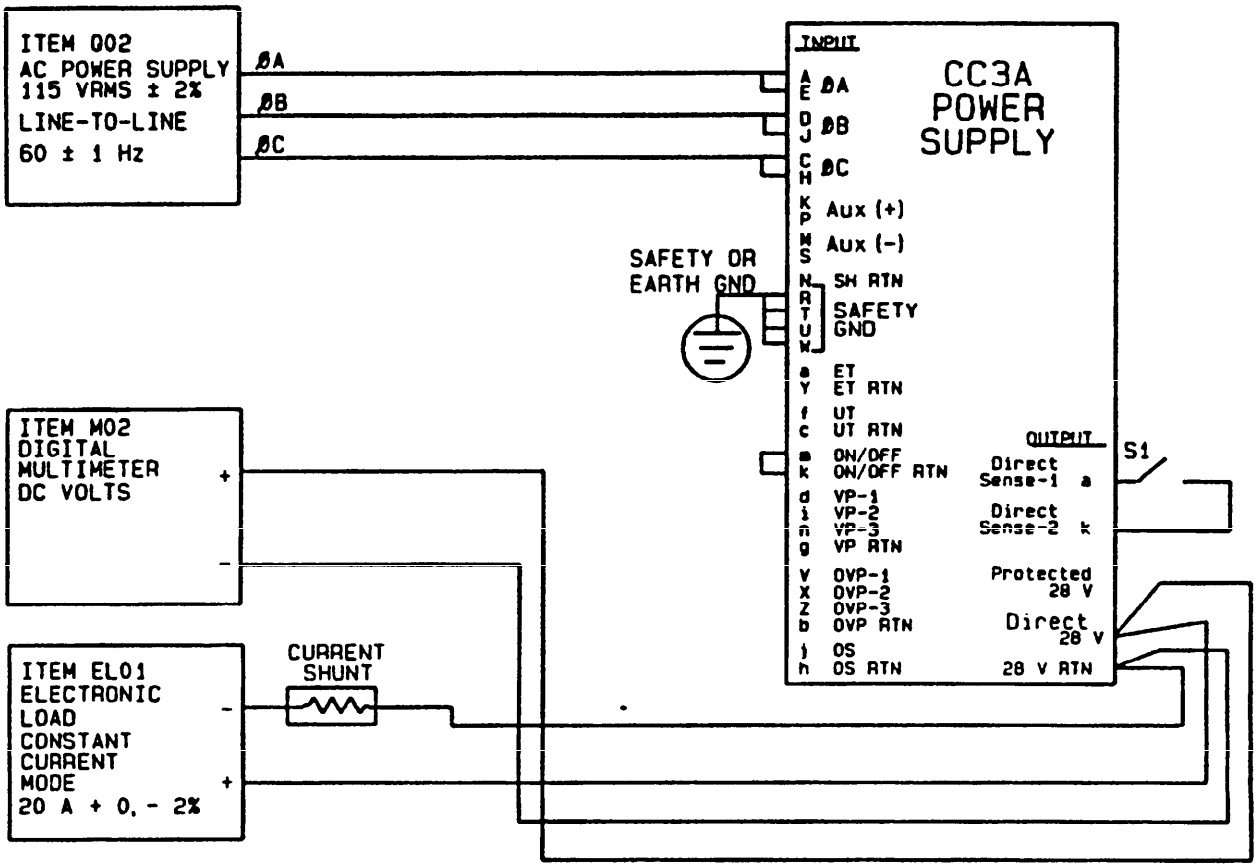


FIGURE 49. Open sense lead test circuit for table XXVI.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst switch.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. For test condition DD, connect the circuit shown and open switch S1. The output voltage, as indicated on item M02, shall be within table XXVI limits.

FIGURE 49. Open sense lead test circuit for table XXVI - Continued.

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D. Meter Action - See NOTES in Sound Level Data Sec

TABLE XXVII. Common mode output current tests.

Test cond	Parameter	Connect point 1 to point;	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
EE1	Common mode current broadband, 28V, 20A	3	0	200	mAp-p	50
EE2	Common mode current broadband, 28V, 10A	3	0	200	mAp-p	
EE3	Common mode current spectral line, 28V, 50Hz - 10kHz 1/,2/	2	-INF	-80	dBV / $\sqrt{Hz}$	
EE4	Common mode current spectral line, 28V, 10kHz - 10MHz 1/,2/	2	-INF	-56.5	dBV / $\sqrt{Hz}$	
EE5	Common mode current spectral line, 28V, largest amplitude signal 1/,2/	2	-INF	-46	dBV / $\sqrt{Hz}$	50

1/ Test conditions EE3-EE5 shall be run using the setup from test conditions EE1 or EE2 which gives the largest magnitude broadband current.

2/ Do not apply bandwidth correction to the EE3-EE5 limits.

3/ Common mode current (dBA) = (item B05 reading in dBV) - 20 log (Z) = (item B05 reading in dBV + 20 dB),

where Z = current probe transfer function of mV/mA (shown calculated above for Z = 0.1), dBV = decibel (ref 1 volt). The table XXVII limits for EE3, EE4 and EE5 will change in accordance with the above relationship if Z is not equal to 0.1. Refer to table I for the dBA/$\sqrt{Hz}$ limits.

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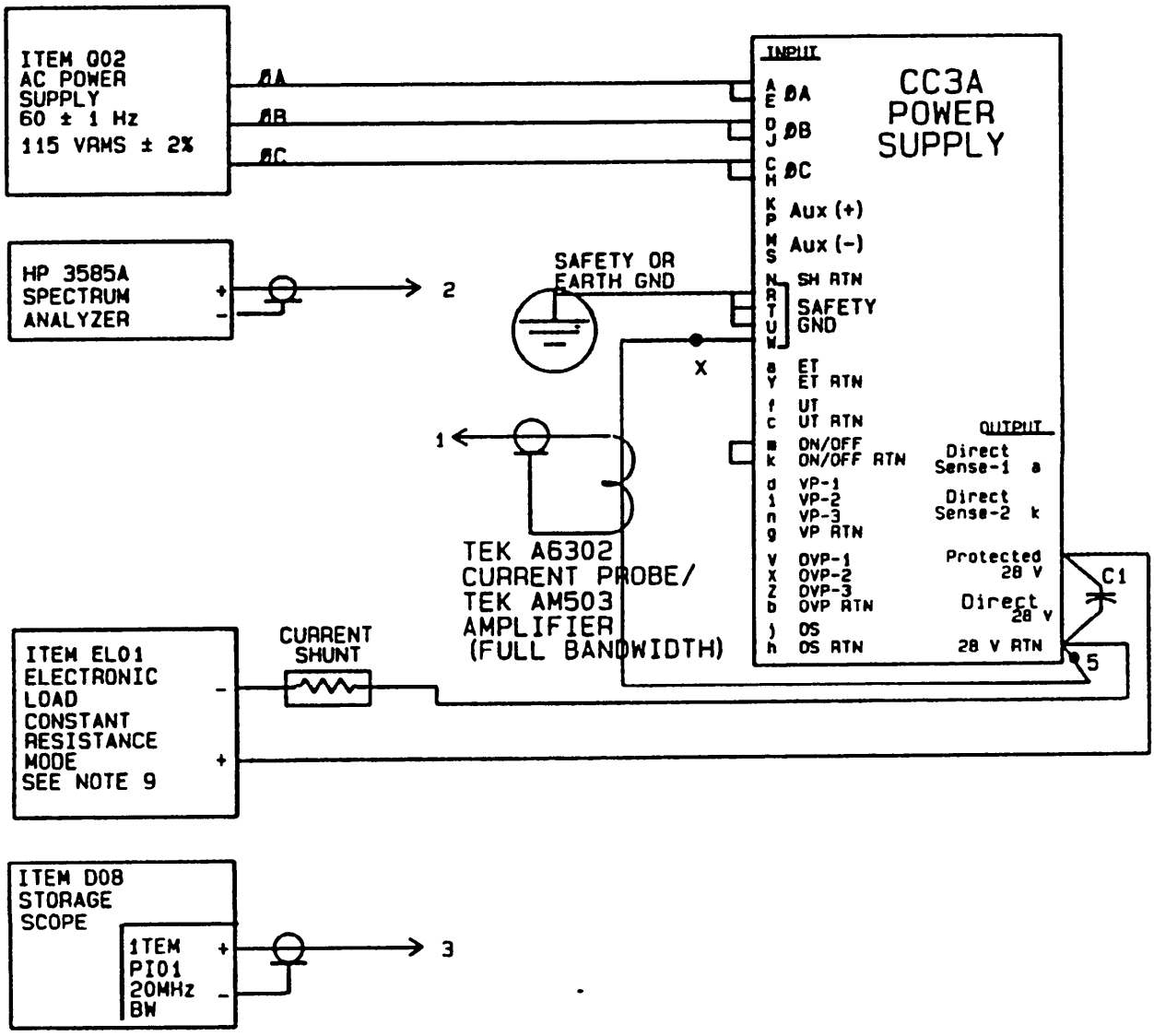


FIGURE 50. Common mode output current test circuit for table XXVII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. C1 = 10 uF ± 5 percent, 50 VDC, MIL-C-83421/01-359.

6. For test conditions EE1 and EE2, place the current probe around the wire going from point 4 to point 5. Measure the total peak-to-peak current observed on item D08 oscilloscope (including all spikes, hash, periodic and aperiodic components). Use the single sweep storage mode.

7. For test condition EE3, sweep the spectrum analyzer frequency between 50 Hz and 10 kHz. Use a 10 Hz bandwidth to make final measurements verifying the power supply meets the specified limits. The setup from test condition EE1-EE2 which gave the largest magnitude current shall be used.

8. For test conditions EE4 and EE5, sweep the spectrum analyzer frequency between 10 kHz and 10 MHz. Use a 30 Hz bandwidth to make final measurements verifying that the power supply meets the specified limits. For test condition EE5, measure the spectral line of the largest amplitude observed on the spectrum analyzer. For test condition EE4, measure the spectral line of the second largest amplitude observed on the spectrum analyzer.

9. Item EL01 shall be set to 20 amperes +0, -2 percent for test condition EE1 and to 10 amperes ± 1 percent for test condition EE2. (The load used, if different than EL01, shall have its + and - terminals isolated from safety or chassis ground so the common mode current will flow to point 4).

10. The connection from point 4 (safety gnd) to point 5 (output-1 RTN) shall be made using a 4 ± 0.125 inch piece of copper, stranded number 16 gauge wire (length measurements shall be made from the mating connector solder cups). The connection from the power supply output contacts to the C1 capacitor shall be kept as short as possible. The wires from input connector

FIGURE 50. Common mode output current test circuit for table XXVII - Continued.

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NOTES: - Continued.

contacts R, T and U to the common tie point shall each be 4.0 ± 0.5 inches of number 16 gauge wire.

11. When performing the tests in table XXVII, ensure that the current being measured is generated by the power supply under test and is not spurious noise from external sources.

FIGURE 50. Common mode output current test circuit for table XXVII - Continued.

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TABLE XXVIII. Voltage programming contact signal rejection tests.

Test cond	Parameter	Connect 1 to input contact	Spectrum analyzer frequency ±1%	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
				Min	Max		
FF1	Rejection at 100 Hz, VP	i	100 Hz	6	-	dBV	51
FF2	Rejection at 5 kHz, VP	i	5 kHz	6	-	dBV	
FF3	Rejection at 50 kHz; VP	i	50 kHz	6	-	dBV	
FF4	Rejection at 100 Hz, OVP 1/	X	100 Hz	6	-	dBV	
FF5	Rejection at 5 kHz, OVP 1/	X	5 kHz	6	-	dBV	
FF6	Rejection at 50 kHz, OVP 1/	X	50 kHz	6	-	dBV	
FF7	Rejection, output status signal	X	-	0.0	0.4	VDC	51

1/ During test conditions FF4 through FF6, the output status signal shall remain between 0.0 and 0.4 VDC.

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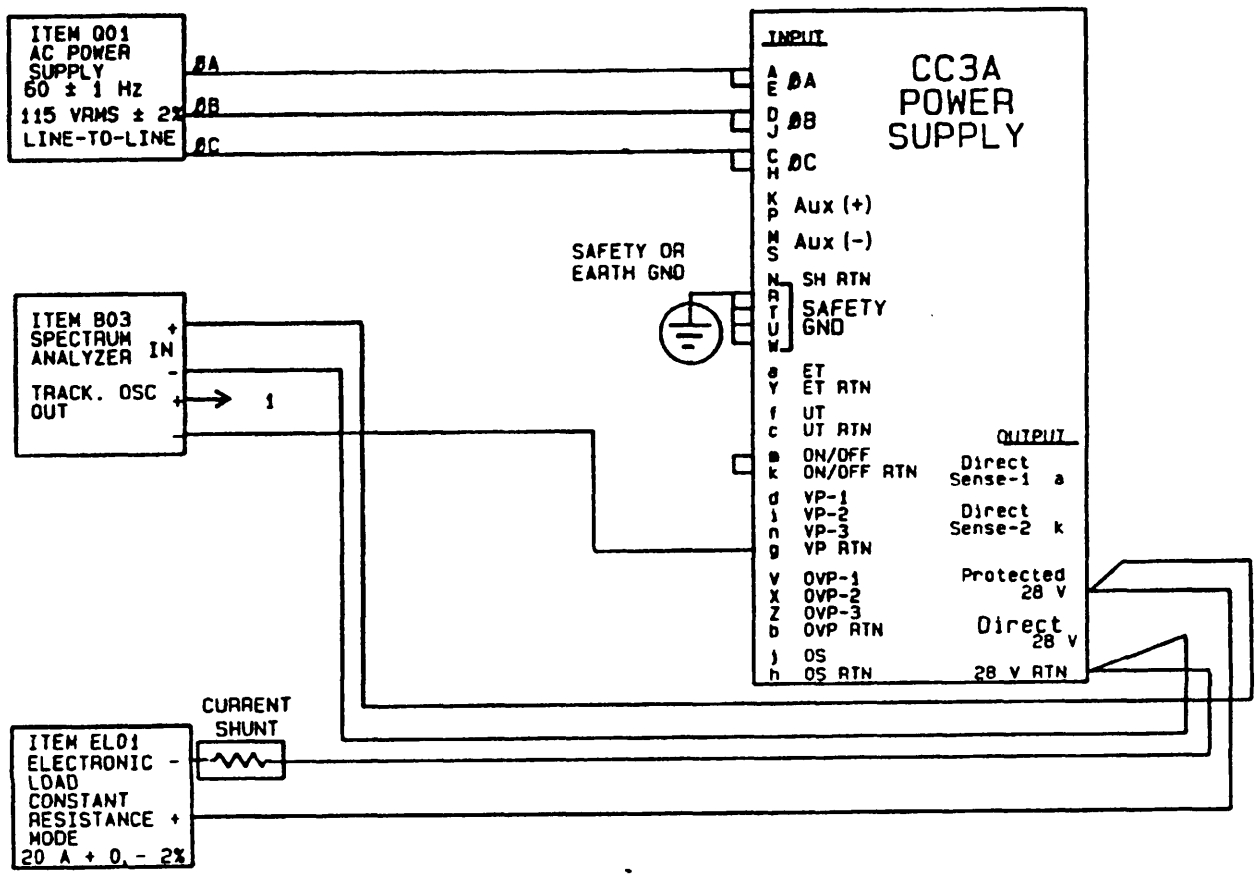


FIGURE 51. Voltage programming contact signal rejection test circuit for table XXVIII.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

6. R1 = 1.69 kOhm ± 1 percent, 1/8 watt resistor.

7. For each test condition, connect as indicated in table XXVIII, set the tracking oscillator output to 0.1 VRMS ± 5 percent (-20 ± 0.5 dBV), between point 1 and input contact g. Measure the AC voltage on the power supply output with the spectrum analyzer. The spectrum analyzer measurement shall be 15 dB or greater below the -20 dB input signal (-20 dBV minus measurement level > 15 dBV). (If the tracking oscillator output cannot be adjusted as high as 0.1 VRMS, set it to the highest value obtainable. The spectrum analyzer measurement shall still be 15 dB or greater below the input signal.)

FIGURE 51. Voltage programming contact signal rejection test circuit for table XXVIII - Continued.

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TABLE XXIX. Input voltage and current characteristics tests.

Test cond	Parameter	Initial limits 25 ± 25°C mounting surface temperature		Units	Test fig
		Min	Max		
GG1	Input voltage turn-on threshold	100	135	VDC	52,53
GG2	Input voltage turn-off threshold	--	100	VDC	
GG3	Input voltage turn-on/turn-off hysteresis 1/	18	--	VDC	
GG4	Input voltage turn-on threshold, auxiliary input	100	135	VDC	
GG5	Input voltage turn-off threshold, auxiliary input	--	100	VDC	
GG6	Input voltage turn-on/turn-off hysteresis, auxiliary input 1/	18	--	VDC	
GG7	Maximum steady-state input current	--	13.5	ADC	
GG8	Slowly rising input peak current	--	13.5	Apk	
GG9	Slowly rising input rate of current rise	--	8.4	A/ms	
GG10	Input peak current, remote turn-on	--	13.5	Apk	
GG11	Input rate of current rise, remote turn-on	--	8.4	A/ms	52,53

1/ Test conditions GG3 and GG6 are calculations:

Hysteresis = turn-on threshold minus turn-off threshold.

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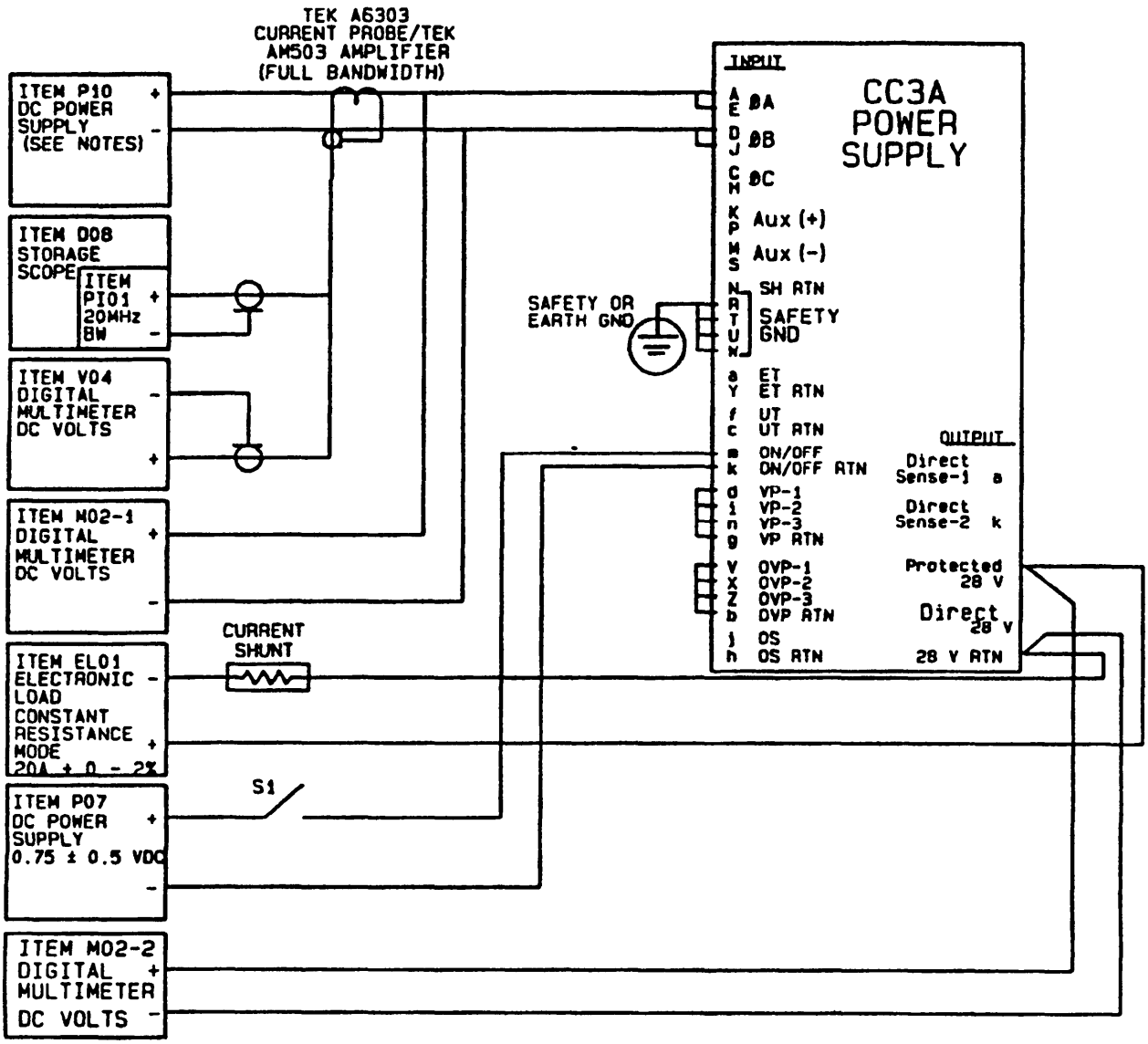


FIGURE 52. Input voltage and current characteristics test circuit for table XXIX.

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NOTES:

1. See table II for output connector contact assignments.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst, 225 VDC, 15 ampere switch.

6. Switch S1 shall be closed except as indicated in note 12 for test conditions GG10 and GG11.

7. For test conditions GG1, GG2, and GG7 through GG11, connect item P10 power supply to the AC inputs as shown and close switch S1. For test conditions GG4 and GG5, connect item P10 and item M02-1 to the auxiliary inputs: + to input contacts K and P, - to input contacts M and S and close switch S1. Ensure that item P10 is turned off or adjusted to 0 volts before attemping to make these connections.

8. Test conditions GG1 and GG4 determine the AC/DC input and auxiliary input voltage threshold for power supply turn-on. Perform the test by slowly increasing the item P10 voltage (1 volt per second or less) from 0 volts until the output voltage, as measured with the item M02-2, just equals or exceeds 29.0 volts. The item M02-1 voltage at this point shall meet the test condition GG1 or GG4 limits.

9. Test conditions GG2 and GG5 determine the AC/DC input and auxiliary input voltage threshold for power supply turn-off. Perform the tests by slowly decreasing the item P10 voltage (1 volt per second or less) from 155 volts until the output voltage, as measured with the item M02-2, just equals or drops below 29.0 volts. The item M02-1 voltage at this point shall meet the test condition GG2 or GG5 limits.

10. For test condition GG7, slowly (1 volt per second or less) sweep the item P10 voltage from 0 to 202 volts and back to 0 volts. The maximum steady state input current observed as measured with the current probe and item V04 shall meet the GG7 limits.

FIGURE 52. Input voltage and current characteristics test circuit for table XXIX - Continued.

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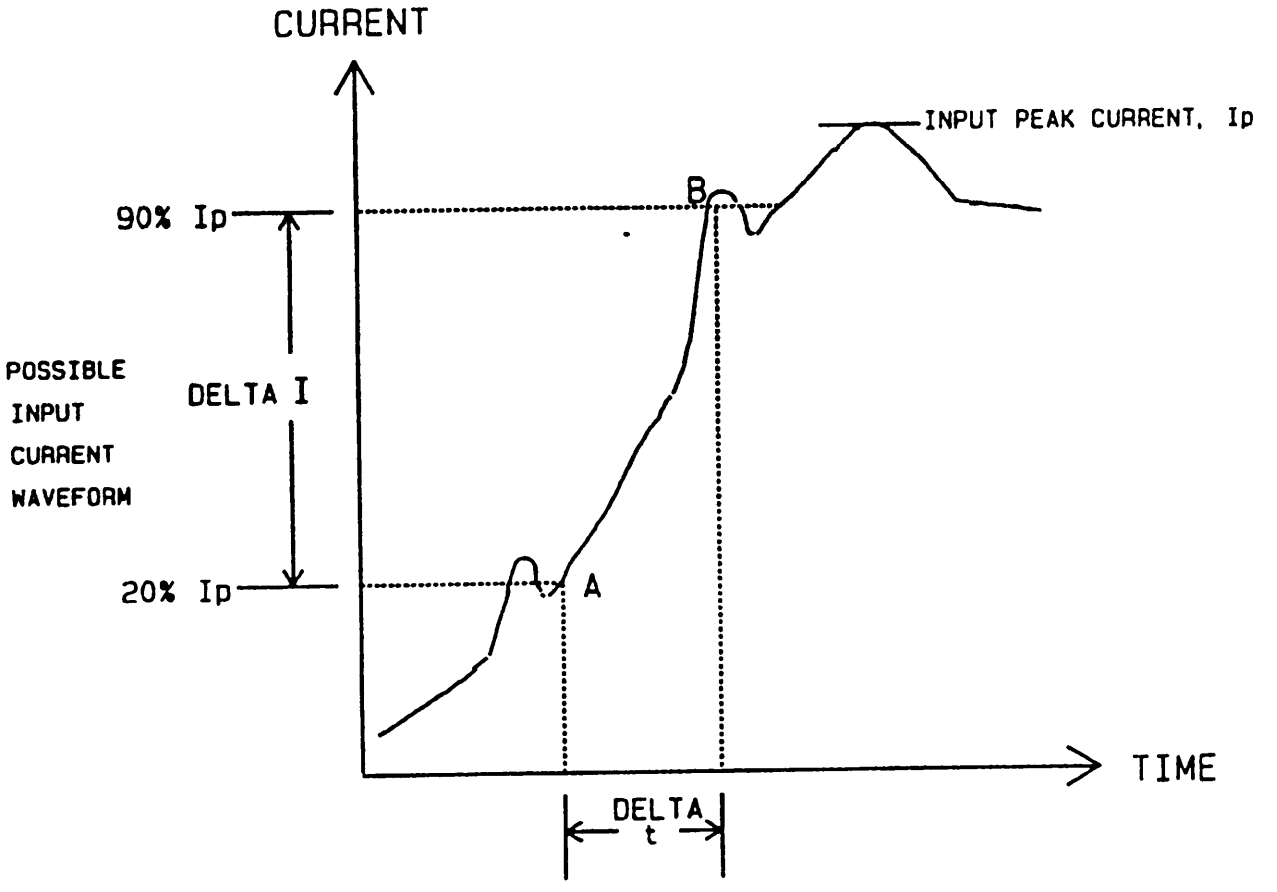
NOTES: - Continued.

11. For test conditions GG8 and GG9, increase the item P10 voltage, from 0 to 140 volts (above the turn-on threshold) at a rate of 1 volt per millisecond or less. Use the current probe and item D08 to store the input current waveform as the power supply turns on. The input current observed shall meet the GG8 and GG9 limits (see figure 53). The test shall be performed several times and the highest values recorded.

12. For test conditions GG10 and GG11, close switch S1 and adjust item P10 to 155 volts. Open switch S1. Use the current probe and item D08 to store the input current waveform as S1 is closed and the power supply turns on. The input current observed shall meet the GG10 and GG11 limits (see figure 53). The test shall be performed several times and the highest values recorded.

FIGURE 52. Input voltage and current characteristics test circuit for table XXIX - Continued.

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NOTE:

1. The rate of current rise to be recorded for test condition GG9 or G11 is the slope between points A and B on the rising input current vs time waveform. If the curve crosses the value of current represented by 20 percent Ip and/or 90 percent Ip more than once, the time between the last time it crosses 20 percent Ip and the first time it crosses 90 percent Ip shall be taken as the recorded delta t. The slope is defined as

rate of current rise = DELTA I in Amperes/DELTA t in milliseconds.

(The figure is for illustrative purposes only and the input current waveform may look different.)

FIGURE 53. Input current waveform for table XXIX.

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TABLE XXX. Emergency condition input voltage protection tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
HH1	Emergency condition input voltage protection	29.74	31.26	VDC	54
HH2	Emergency condition input voltage protection	30.20	30.80	VDC	54

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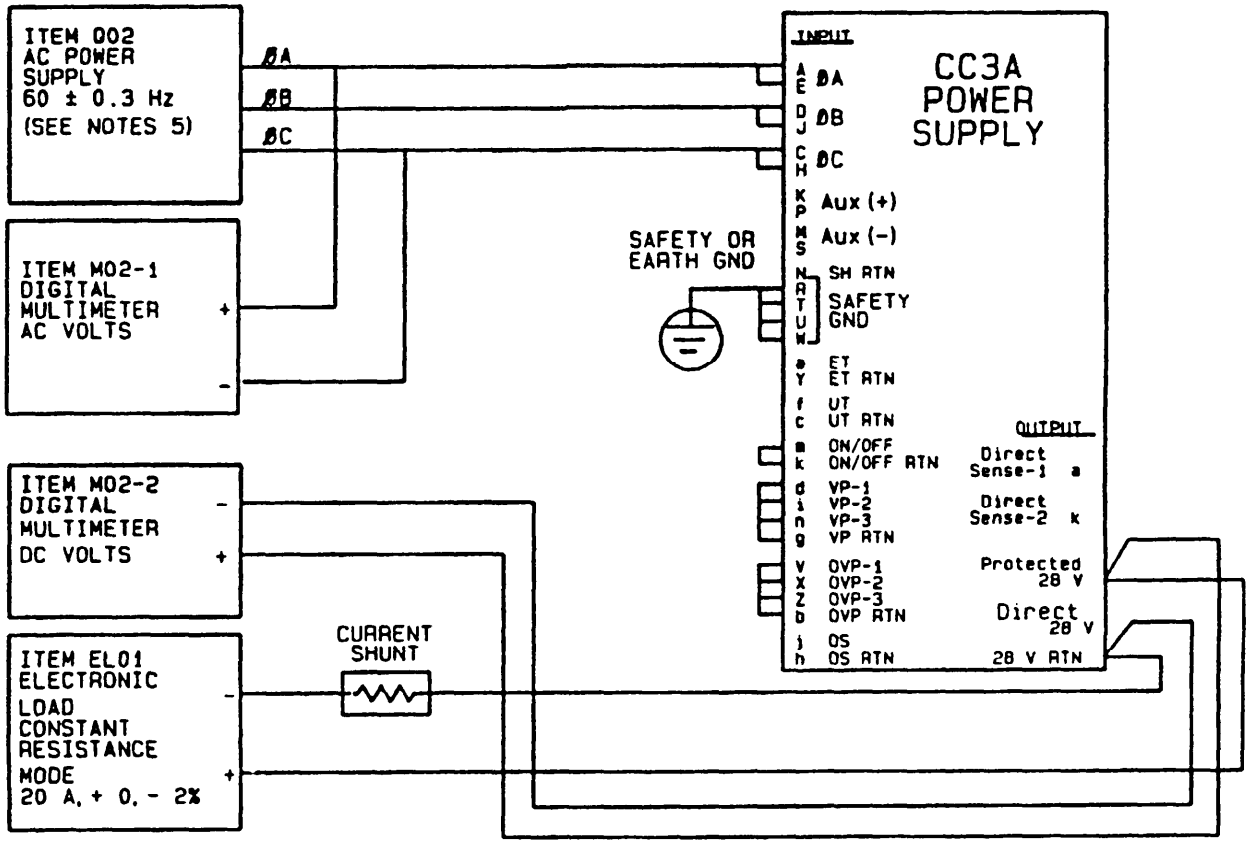


FIGURE 54. Emergency condition input voltage protection test circuit for table XXX.

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NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. The tests shall be performed as follows:

(a) Connect as shown and set item Q02 to 115 VRMS ± 2 percent line-to-line.

(b) Increase the item Q02 voltage to 155 VRMS ± 1 percent line-to-line. Leave the 155 VRMS applied for a minimum of 2 minutes and then read item M02-2 to verify that the output voltage is within the test condition HH1 limits.

(c) Reduce the item Q02 voltage to 115 VRMS ± 2 percent line-to-line. The item M02-2 voltage shall be within the test condition HH2 limits.

FIGURE 54. Emergency condition input voltage protection test circuit for table XXX - Continued.

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TABLE XXXI. Output status noise tests.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
JJ1	Output status noise, positive	0	600	mVpk	55
JJ2	Output status noise, negative	0	-600	mVpk	55

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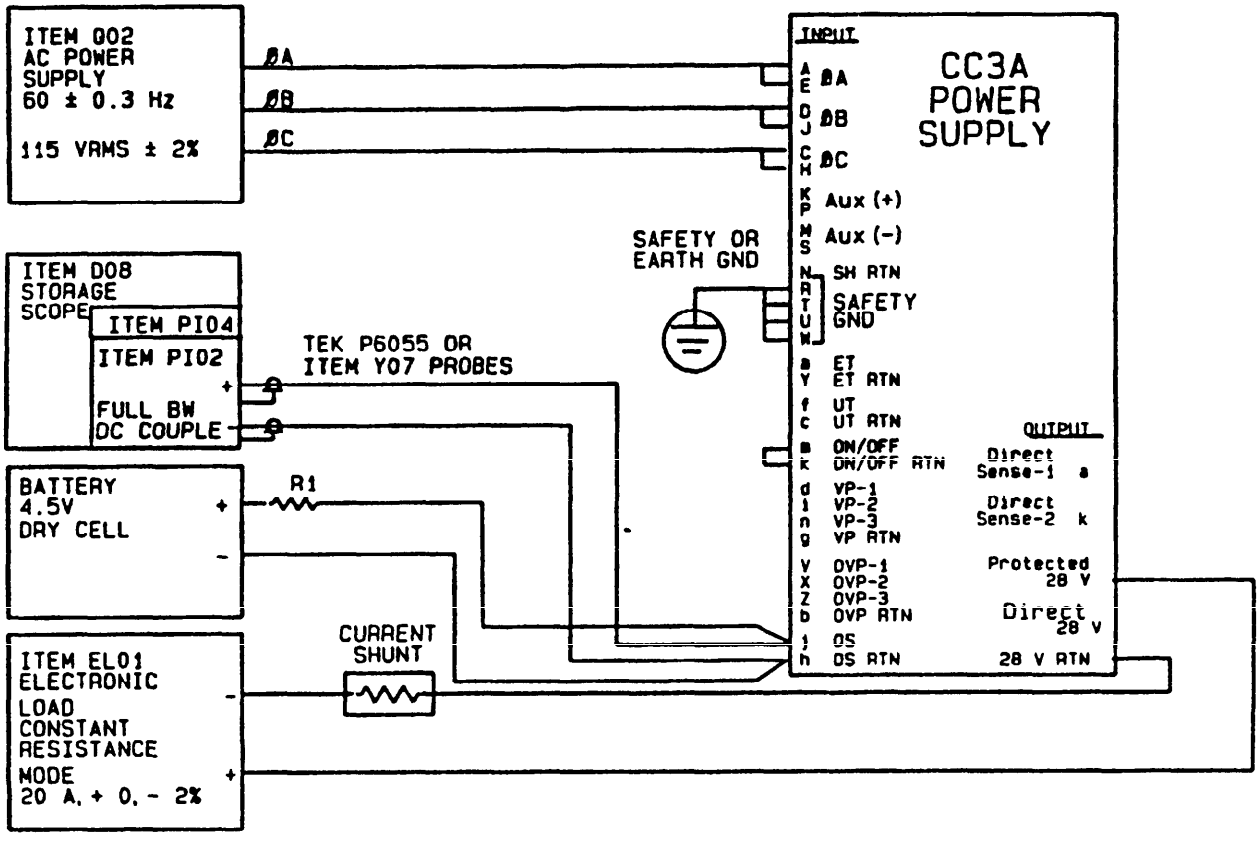


FIGURE 55. Output status noise test for table XXXI.

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NOTES:

1. See table II for output connector assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor the shunt voltage with an item V04 voltmeter to ensure that the load current is as specified.

5. R1 = 5.9 kOhm, ± 1 percent, 1/8 watt resistor.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. The leads between the battery terminals and the output status contacts of the power supply under test shall be a tightly twisted pair with a maximum length of 10 inches.

8. The mVpk measurements for test conditions JJ1 and JJ2 shall be referenced to input contact h (OS RTN). The measurement shall include all spikes, hash, periodic and aperiodic components.

FIGURE 55. Output status noise test for table XXXI - Continued.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - SH
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Navy - NW

(Project 6130-0345)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-24764/3	31 March 1993

3. DOCUMENT TITLE
Power Supplies, Shipboard, Electronic, 28 Volts, Key Code CC3A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. Larry McKibben (Code 6023) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-5107	88-482-5107
c. ADDRESS (Include Zip Code) Commander Code 6023, Bldg. 2917 (Larry McKibben) NAVSURFWARCENDIV, 300 Highway 361 Crane, Indiana 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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